Capacity-Fixed Trace-Free Gravity and Non-Perturbative Edge-Partition Boundary Selection

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We present an invariant sieve for cosmological gravity and its non-perturbative edge-partition completion. The local equation is the exact trace-free projector,

$$R_{\mu\nu} - \frac{1}{4} R g_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{4} T g_{\mu\nu} \right),$$

so constant vacuum-stress shifts lie in the kernel of the source projector. With covariantly conserved matter, Bianchi integration yields the global mode $\Lambda_{\rm int}$. The capacity postulate converts that mode into the boundary eigenvalue

$$\Lambda_{\rm cap} = \frac{3\pi}{\ell_P^2 N_\partial}.$$

The terminal edge Hamiltonian is

$$\widehat{H}_\partial = \bigoplus_q \left(\widehat{H}_{{\rm adm},q} + \widehat{H}_{{\rm run},q} + \widehat{H}_{{\rm edge},q} \right),$$

with admissibility enforced by a positive common-kernel operator, running localized on a dynamical $O(\ln q)$ Fibonacci/valuation registry of rank N_φ, and the oscillator sector identified with the de Sitter horizon edge partition function $Z_{\rm edge}$, organized by Harish–Chandra characters, the bulk/edge factorization of the Euclidean path integral, and a modular stringy horizon completion.

The microscopic completion is reduced to, and in this manuscript implemented through, three edge-theorem gates: the Hopf–Bradlow–APS–HC recursion gate for

$$N_\varphi = 292,$$

the GLSM/stringy edge matching gate for

$$b\, g_\partial^2(\ell_P^{-1}) = 0.559754586\ldots, \qquad \Xi_{\rm edge} \simeq 0.99916928,$$

and the determinant-line gate

$$-\log Z_{\rm edge}(q) = \nu \ln q + \Theta_\partial(q) + S_{\rm HC}(q) + o(1), \qquad \nu = \frac{1}{2}.$$

With these three gates in place, the free energy

$$\Gamma_\partial(q) = q[\ln(q/q_\star) - 1] - \log Z_{\rm edge}(q)$$

has a unique stable large-q minimum at

$$q_\star = 3.307251460713979 \times 10^{122}.$$

The nearest admissible integer after projection onto the common physical kernel is the selected sector, and

$$\Lambda_\star = \frac{3\pi}{\ell_P^2 q_{\rm phys}}.$$

Thus the cosmological-constant amplitude is formulated as an edge-Hilbert-space eigenvalue selected by the non-perturbative horizon edge theory, rather than as a bare local coupling or a posterior fit.

I. INVARIANT CORE: IRREDUCIBLE SIEVE

This section states the theorem core. The later sections keep the invariant atlas, numerical ledgers, selector audits, and observational diagnostics. Thus the core fixes the hierarchy in which the full construction is to be read.

The invariant sieve is

$$\boxed{\mathfrak{C}_{\rm inv} = \left(\mathcal{P}_{\rm tf},\ {\rm Cap}\,\mathcal{H}_\partial,\ \mathcal{A}_{\rm grav},\ Q_{\rm DM},\ P_{\rm val}(\mathscr{D}),\ \mathcal{S}_{\rm obs} \right).} \tag{I.1}$$

The factors have distinct jobs. The projector $\mathcal{P}_{\rm tf}$ removes the local constant vacuum shift. The boundary capacity

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Cap $\mathcal{H}_\partial$ is the remaining selector problem for Λ. The algebra $\mathcal{A}_{\rm grav}$ is the anomaly-free gravitational constraint algebra. The charge $Q_{\rm DM}$ protects the cold dust sector. The annihilator $P_{\rm val}(\mathscr{D})$ closes the homogeneous valuation alphabet. The observable sieve $\mathcal{S}_{\rm obs}$ contains projectors, Hankel pencils, inertia tests, free-spectrum diagnostics, curvature split, and source-lift nulls.

The trace-free kernel is exact:

$$\boxed{\mathcal{P}_{\rm tf}(-\rho_0 g_{\mu\nu}) = 0.} \tag{I.2}$$

With $\nabla^\mu T_{\mu\nu} = 0$, the Bianchi identity gives the integration-mode equation

$$\boxed{G_{\mu\nu} + \Lambda_{\rm int} g_{\mu\nu} = 8\pi G T_{\mu\nu}.} \tag{I.3}$$

Trace-freeness alone therefore does not compute Λ. It isolates the only remaining datum:

$$\boxed{\Lambda_{\rm cap} = \frac{3\pi}{\ell_P^2 N_\partial}.} \tag{I.4}$$

The admissible capacity sector can be read in several equivalent invariant ways, under their stated hypotheses:

$$\boxed{N_\partial = q = \frac{k_{\rm CS}}{2} = \operatorname{ind} \bar\partial_{\mathcal{O}(q-1)} = \operatorname{Tr} \Pi_L = N_{\rm v}^{\rm Bradlow}.} \tag{I.5}$$

These equalities are not a selector theorem. They identify and quantize the sector once it is chosen. The sector is selected only by a boundary edge-mode dynamics.

The homogeneous B^4/S^3 valuation closure is

$$\boxed{\begin{aligned} &P_{\rm val}(\mathscr{D})\rho_{\rm val} = 0, \\ &P_{\rm val}(z) = (z-1)z(z+1)(z+2)(z+3). \end{aligned}} \tag{I.6}$$

Thus

$$\rho_{\rm val} \in \operatorname{span}\{a, 1, a^{-1}, a^{-2}, a^{-3}\}. \tag{I.7}$$

In n spatial dimensions,

$$\boxed{S_{\rm val}^{(n)} = \{1, 0, -1, \ldots, -n\}.} \tag{I.8}$$

After quotienting the boundary alphabet by vacuum, geometric curvature, and dust, the quotient dimension is $n-2$. Hence exactly one nonstandard boundary-local direction occurs if and only if $n = 3$, and then it is

$$\boxed{s = -1, \qquad \rho_X \propto a^{-1}, \qquad w_X = -\frac{2}{3}.} \tag{I.9}$$

For a positive noninteracting finite mixture,

$$\boxed{\mathscr{D} w = -\frac{1}{3} \operatorname{Var}_\pi(s) \le 0.} \tag{I.10}$$

An observed violation of this sign is not a small correction to the positive noninteracting mixture. It requires an exchange current, signed weights, or an enlarged source class.

The terminal unknown is now unique:

$$\boxed{\Gamma_\partial(q) = -\ln \operatorname{Tr}_{\mathcal{C}_q} e^{-\beta_q H_{\partial,q}}, \qquad q_\star = \arg\min_{q \in \mathbb{Z}_{>0}} \Gamma_\partial(q).} \tag{I.11}$$

The explicit edge-Hamiltonian reduction gives a minimal nonconformal form of such an $H_{\partial,q}$, with the determinant correction counted once. The problem is now sharper: derive its running and determinant data b, g_∂, Ξ, ν, and Θ_∂ from the microscopic edge theory, rather than choosing them to reproduce $\Lambda_{\rm obs}$, R_Λ, or $q_{\rm obs}$. All remaining structures in the atlas are to be read as quantizers, admissibility walls, obstructions, registries, or tests unless they explicitly enter this Hamiltonian and its running.

Criterion I.1 (Role-preserving retention principle)**.** No mathematically controlled connector is discarded merely because it does not by itself solve the amplitude problem. It is retained if, and only if, it is assigned one of the following invariant roles:

- Q : sector quantizer, such as an index, flux, or level;
- B : saturation bound or packing bound;
- O : obstruction, anomaly, or no-go constraint;
- C : energy cost or admissibility wall;
- R : dimensionless registry or clock;
- T : statistical or observational null test;
- S : true selector entering $\Gamma_\partial(q)$.

The last role is distinguished. Only an S-role term can select $q_\star$. The other roles may be necessary for the consistency of the edge theory, but they cannot be promoted to selector theorems unless they enter the effective boundary action and its stationarity condition.

Remark I.1 (Why the atlas is retained)**.** The Hopf, Chern–Simons, Bradlow, Schwinger, Ginzburg–Landau, APS determinant, Fibonacci, and scale-atlas layers are therefore not ornamental analogies. They are kept as a finite audit of possible invariant mechanisms. Their purpose is to prevent a false shortcut: each layer either supplies a legal sector label, a bound, an obstruction, an admissibility wall, a registry, or a null test. The manuscript fails if any of these is confused with the missing selector. It succeeds as a reduction only when the terminal problem is stated as

$$q_\star = \arg\min_q \Gamma_\partial(q).$$

II. CLAIM BOUNDARY

The manuscript does not yet compute Λ; it turns Λ into a sharply posed boundary entropy-counting problem.

This paper changes the logical status of Λ. It does not insert a small fundamental constant into the action. It replaces that constant by two invariant operations:

local trace-free projection + global horizon-capacity fixation

The local projection removes vacuum-energy shifts from the gravitational source. The global condition fixes the integration constant that the Bianchi identity leaves behind.

The boundary is equally important. The paper does not derive the microscopic Hilbert space $\mathcal{H}_\partial$. Hence it does not yet compute the integer or real capacity eigenvalue N_∂ from first principles. Its exact statement is conditional:

If the physical boundary state space has capacity N_∂, then $\Lambda = 3\pi/(N_\partial \ell_P^2)$ is fixed and not fitted.

This is stronger than treating

$$N_\Lambda = \frac{3\pi}{\Lambda \ell_P^2}$$

as a diagnostic after Λ has already been measured. The direction of determination is reversed:

$$\boxed{\mathcal{H}_\partial \longrightarrow N_\partial \longrightarrow \Lambda_{\rm cap}.} \tag{II.1}$$

Throughout, $c = \hbar = 1$ unless explicitly restored.

Definition II.1 (Trace-free projector)**.** For a symmetric rank-two tensor $X_{\mu\nu}$ in four spacetime dimensions, define

$$\boxed{(\mathcal{P}_{\rm tf}X)_{\mu\nu} := X_{\mu\nu} - \frac{1}{4}X g_{\mu\nu}, \qquad X := g^{\mu\nu}X_{\mu\nu}.} \tag{II.2}$$

Thus $\mathrm{tr}(\mathcal{P}_{\rm tf}X) = 0$.

Definition II.2 (Capacity-fixed completion object)**.** The invariant completion object of this paper is

$$\boxed{\mathfrak{C}^{\rm cap}_{\rm inv} := (\mathcal{P}_{\rm tf}, \mathrm{Cap}\,\mathcal{H}_\partial, \mathcal{A}_{\rm grav}, Q_{\rm DM}, P_{\rm val}(\mathscr{D})).} \tag{II.3}$$

Here $\mathcal{P}_{\rm tf}$ removes local vacuum-energy shifts, $\mathrm{Cap}\,\mathcal{H}_\partial$ fixes the integration mode, $\mathcal{A}_{\rm grav}$ is the anomaly-free quantum constraint algebra, $Q_{\rm DM}$ is an exact dark charge or equivalent stability invariant, and $P_{\rm val}(\mathscr{D})$ is the finite dilation annihilator of the homogeneous valuation sector.

Remark II.1 (Status of the capacity postulate)**.** The capacity postulate does not by itself compute N_∂. It replaces a freely inserted cosmological constant by a sharply defined microscopic eigenvalue problem. A full solution of the cosmological-constant amplitude problem requires an independent derivation of N_∂ from the boundary state space. The present work supplies the invariant mechanism and the falsification criteria.

III. LAYER SEPARATION AND INVARIANT ATLAS

The construction uses three different finite alphabets. They must not be identified.

Definition III.1 (Three alphabets)**.** The round valuation density alphabet is

$$\boxed{S^\rho_{\rm val} := \{1, 0, -1, -2, -3\}.} \tag{III.1}$$

The boundary-local valuation density alphabet is

$$\boxed{S^\rho_\partial := \{0, -1, -2, -3\}.} \tag{III.2}$$

If ordinary radiation is added to the Friedmann ledger, the expansion alphabet used for $F = H^2/H_0^2$ is

$$\boxed{S^F_{\rm cos} := \{0, -1, -2, -3, -4\}.} \tag{III.3}$$

The radiation exponent $s = -4$ is not produced by the four-dimensional Euclidean valuation density on the round B^4/S^3 family. It is an external standard source added to the cosmological ledger.

Criterion III.1 (Exact invariant atlas)**.** A proposed completion passes the invariant sieve only if the following sectors are simultaneously well defined.

Sector	invariant object	sharp test or equation
local gravity	$\mathcal{P}_{\rm tf}$	$\mathcal{P}_{\rm tf}(-\rho_0 g_{\mu\nu}) = 0$
Λ-mode	$N_\partial = \mathrm{Cap}\,\mathcal{H}_\partial$	$\Lambda_{\rm cap} = 3\pi/(\ell_P^2 N_\partial)$
quantum covariance	$\mathcal{A}_{\rm grav}$	anomaly-free constraint algebra
dark matter	$Q_{\rm DM}$	$\nabla_\mu J^\mu_{\rm DM} = 0$
valuation density	$P_{\rm val}(\mathscr{D})$	$P_{\rm val}(\mathscr{D})\rho_{\rm val} = 0$
boundary quotient	$[a^{-1}]$	$w_X = -2/3$
curvature split	$\mathcal{C}_{-2}[F] - \Omega_K^{\rm geom}$	$\Omega_{2,\rm val}$
spectroscopy	Hankel pencil (H_1, H_0)	free recovery of s_j
positivity	inertia of $H_0^{\rm phys}$	$\nu_- = 0$ after split
spin–2 lift	tensor causal cone	$c_T = c$
equation of state	cumulant closure	nonlinear Bell identity

The atlas is a sieve, not a fit. A candidate that fails one line has not solved the corresponding part of the problem.

Remark III.1 (Corrections relative to a loose formulation)**.** There are four common confusions that this manuscript excludes. First, $N_\Lambda = 3\pi/(\Lambda\ell_P^2)$ is of order 10^{122} for the observed universe, not of order 3π. Second, $P_{\rm val}(\mathscr{D})$ annihilates the density profile, not the equation-of-state ratio $w(a)$. Third, a dark charge is a conserved current or protected discrete/topological charge, not a bare linear combination of Ω_Λ and Ω_X. Fourth, the homogeneous $s = -2$ expansion branch is not the spin–2 graviton propagation mode; tensor propagation belongs to the perturbative lift.

IV. WHY A BARE COSMOLOGICAL CONSTANT IS THE WRONG INVARIANT

A bare cosmological constant is a dimensional term. It can be made small or large by units; only the combination

$$\boxed{\Lambda\ell_P^2} \tag{IV.1}$$

or, equivalently,

$$\boxed{N_\Lambda := \frac{3\pi}{\Lambda \ell_P^2}} \tag{IV.2}$$

is invariant. For the observed late-time universe, N_Λ is of order 10^{122}. The factor 3π is only the numerator determined by the de Sitter area formula; it is not the value of the capacity.

No-go theorem IV.1 (No topology-only dimensional amplitude)**.** Pure topology or other dimensionless data cannot naturally determine a nonzero dimensional constant such as Λ, H_0, a particle mass, or a vacuum energy density while remaining covariant under global length rescalings.

Proof. Let X have length dimension L^k, $k \neq 0$. Under $L \mapsto \lambda L$, $X \mapsto \lambda^k X$. Pure topology and dimensionless data are unchanged. If X is determined by such data alone, it must both transform and remain fixed. For all $\lambda > 0$, this forces $X = 0$. □

Criterion IV.1 (Solved versus renamed cosmological constant)**.** A proposal solves the cosmological-constant amplitude problem only if it supplies a mechanism determining N_Λ, or a distribution plus measure selecting it. Merely rewriting

$$N_\Lambda = \frac{3\pi}{\Lambda \ell_P^2}$$

after Λ has been observed renames the problem but does not solve it.

V. TRACE-FREE LOCAL DYNAMICS

The local gravitational equation is postulated to be

$$\boxed{R_{\mu\nu} - \frac{1}{4} R g_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{4} T g_{\mu\nu} \right).} \tag{V.1}$$

Equivalently,

$$\boxed{\mathcal{P}_{\rm tf} R_{\mu\nu} = 8\pi G \, \mathcal{P}_{\rm tf} T_{\mu\nu}.} \tag{V.2}$$

Theorem V.1 (Vacuum-energy kernel)**.** *Let*

$$T_{\mu\nu} \longmapsto T'_{\mu\nu} = T_{\mu\nu} - \rho_0 g_{\mu\nu} \tag{V.3}$$

with constant ρ_0. Then

$$\mathcal{P}_{\rm tf} T'_{\mu\nu} = \mathcal{P}_{\rm tf} T_{\mu\nu}. \tag{V.4}$$

Thus a local constant vacuum-energy shift does not source the trace-free equation (V.1).

Proof. In four dimensions $T' = T - 4\rho_0$. Therefore

$$T'_{\mu\nu} - \frac{1}{4} T' g_{\mu\nu} = T_{\mu\nu} - \rho_0 g_{\mu\nu} - \frac{1}{4}(T - 4\rho_0) g_{\mu\nu} \tag{V.5}$$

$$= T_{\mu\nu} - \frac{1}{4} T g_{\mu\nu}. \tag{V.6}$$

□

Theorem V.2 (Bianchi integration mode)**.** *Assume* $\nabla^\mu T_{\mu\nu} = 0$. *Then* (V.1) *implies*

$$\nabla_\nu (R + 8\pi G T) = 0. \tag{V.7}$$

Hence on every connected spacetime region

$$\boxed{R + 8\pi G T = 4\Lambda_{\rm int}} \tag{V.8}$$

for a constant $\Lambda_{\rm int}$*, and* (V.1) *is equivalent to*

$$\boxed{G_{\mu\nu} + \Lambda_{\rm int} g_{\mu\nu} = 8\pi G \, T_{\mu\nu}.} \tag{V.9}$$

Proof. Taking the divergence of (V.1) and using the contracted Bianchi identity gives

$$\nabla^\mu R_{\mu\nu} - \frac{1}{4} \nabla_\nu R = \frac{1}{2} \nabla_\nu R - \frac{1}{4} \nabla_\nu R = \frac{1}{4} \nabla_\nu R.$$

The right-hand side has divergence

$$8\pi G \left(\nabla^\mu T_{\mu\nu} - \frac{1}{4} \nabla_\nu T \right) = -2\pi G \nabla_\nu T.$$

Thus $\nabla_\nu (R + 8\pi G T) = 0$, proving (V.8). Substitution of

$$R = 4\Lambda_{\rm int} - 8\pi G T$$

back into (V.1) gives

$$G_{\mu\nu} + \Lambda_{\rm int} g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

□

Remark V.1 (Vacuum-shift equivalence class)**.** The trace-free equation is invariant under $T_{\mu\nu} \mapsto T_{\mu\nu} - \rho_0 g_{\mu\nu}$, but the reconstructed Einstein form is invariant only if either $T_{\mu\nu}$ is understood modulo this kernel or the integration constant shifts as

$$\Lambda_{\rm int} \mapsto \Lambda_{\rm int} + 8\pi G \rho_0.$$

The capacity postulate fixes the representative by assigning the integration mode from boundary entropy capacity, rather than by choosing a local vacuum counterterm.

Remark V.2 (Relation to unimodular gravity)**.** The trace-free equation is the local equation familiar from unimodular formulations [4–7]. The new element here is not the trace-free equation by itself. It is the invariant prescription that fixes the resulting integration mode by a horizon-capacity eigenvalue.

VI. HORIZON-CAPACITY FIXATION

Let the late-time cosmological sector have a future de Sitter horizon. Its radius and area are

$$R_{\rm dS} = \sqrt{\frac{3}{\Lambda}}, \qquad A_{\rm dS} = 4\pi R_{\rm dS}^2 = \frac{12\pi}{\Lambda}. \tag{VI.1}$$

The Bekenstein–Hawking–Gibbons entropy normalization gives [10–12]

$$S_{\rm dS} = \frac{A_{\rm dS}}{4\ell_P^2} = \frac{3\pi}{\Lambda\ell_P^2}. \tag{VI.2}$$

Definition VI.1 (Boundary entropy capacity)**.** The symbol

$$N_\partial = \mathrm{Cap}(\mathcal{H}_\partial; \mathcal{A}_{\rm grav}, Q_{\rm DM}, P_{\rm val})$$

denotes a logarithmic state-counting capacity, normalized so that in the semiclassical de Sitter sector it equals the horizon entropy rather than the Hilbert-space dimension itself. Thus a finite microcanonical realization has

$$\dim \mathcal{H}_\partial \sim e^{N_\partial},$$

not $\dim \mathcal{H}_\partial = N_\partial$.

Hypothesis VI.1 (Boundary capacity postulate)**.** The physical cosmological boundary sector has a finite entropy-capacity eigenvalue

$$\boxed{N_\partial := \mathrm{Cap}(\mathcal{H}_\partial; \mathcal{A}_{\rm grav}, Q_{\rm DM}, P_{\rm val})} \tag{VI.3}$$

defined after imposing:

1. the anomaly-free quantum constraint algebra $\mathcal{A}_{\rm grav}$;
2. the dark-charge superselection rule $Q_{\rm DM}$;
3. the finite homogeneous valuation closure $P_{\rm val}(\mathscr{D})$.

In the semiclassical de Sitter limit this capacity equals the horizon entropy:

$$\boxed{N_\partial = S_{\rm dS}.} \tag{VI.4}$$

Theorem VI.1 (Capacity-fixed cosmological mode)**.** *Under hypothesis VI.1, once the entropy-capacity eigenvalue is supplied or computed, the integration constant in (V.9) is fixed by*

$$\boxed{\Lambda_{\rm int} = \Lambda_{\rm cap} = \frac{3\pi}{N_\partial \ell_P^2} = \frac{3\pi}{\ell_P^2\, \mathrm{Cap}(\mathcal{H}_\partial; \mathcal{A}_{\rm grav}, Q_{\rm DM}, P_{\rm val})}.} \tag{VI.5}$$

Proof. By hypothesis VI.1,

$$N_\partial = S_{\rm dS} = \frac{3\pi}{\Lambda_{\rm cap}\ell_P^2}.$$

Solving for $\Lambda_{\rm cap}$ gives the result. For a given positive entropy capacity $N_\partial > 0$, the value is single-valued. □

Corollary VI.1 (Replacement of the cosmological constant)**.** *The gravitational field equation becomes*

$$\boxed{G_{\mu\nu} + \frac{3\pi}{N_\partial \ell_P^2} g_{\mu\nu} = 8\pi G\, T_{\mu\nu}.} \tag{VI.6}$$

The term proportional to $g_{\mu\nu}$ is not a free bare cosmological constant. It is an on-shell integration mode fixed once the boundary entropy capacity is supplied or computed.

Theorem VI.2 (Vacuum-shift immunity of the capacity mode)**.** *A shift $T_{\mu\nu} \mapsto T_{\mu\nu} - \rho_0 g_{\mu\nu}$ cannot change $\Lambda_{\rm cap}$ unless it changes the physical boundary capacity N_∂.*

Proof. By theorem V.1, the local trace-free equation is invariant under the shift. By theorem VI.1, the integration mode is a function only of N_∂ and ℓ_P. Therefore the shift changes $\Lambda_{\rm cap}$ only if it changes N_∂, which is a global state-counting datum rather than a local zero-point counterterm. □

Criterion VI.1 (Exact cosmological-constant null)**.** The invariant residual for the replacement mechanism is

$$\boxed{\mathcal{R}_\Lambda := \Lambda_{\rm obs} - \frac{3\pi}{\ell_P^2\, \mathrm{Cap}(\mathcal{H}_\partial; \mathcal{A}_{\rm grav}, Q_{\rm DM}, P_{\rm val})}.} \tag{VI.7}$$

A completed microscopic theory must give $\mathcal{R}_\Lambda = 0$ within observational and state-counting uncertainty. Failure to compute the capacity is not a numerical failure of (VI.6); it marks the remaining microscopic boundary state-counting task.

VII. DYNAMICAL DARK-ENERGY UPDATE: CAPACITY FLOW RATHER THAN A VARYING BARE CONSTANT

Recent DESI DR2 analyses do not establish that a literal cosmological constant is increasing. A cosmological constant is, by definition, constant. The empirical statement is sharper and weaker: combinations of DESI BAO with CMB and supernova data show a preference, still below the discovery threshold and sensitive to dataset consistency, for a time-evolving dark-energy equation of state. The DESI DR2 key paper reports that the preferred $w_0 w_a$ solution lies in the quadrant

$$w_0 > -1, \qquad w_a < 0,$$

with a 3.1σ preference over ΛCDM for DESI BAO plus CMB and 2.8–4.2σ when supernova samples are also included [31]. The DESI extended dark-energy paper reports that two-parameter $w(z)$ descriptions capture the observed trends and that the evidence for dynamical dark energy is especially stable at low redshift [32]. Critical analyses emphasize that dataset tensions and systematics still matter, so this is not yet a final discovery of a unique dark-energy law [33].

The invariant response is not to promote Λ back to a local variable. The trace-free sector continues to remove local vacuum-energy shifts. What may vary is the effective dark-energy density

$$\Lambda_{\rm eff}(a) := \frac{8\pi G}{c^4}\rho_{\rm DE}(a),$$

or equivalently the boundary capacity

$$\boxed{N_\partial(a) := \frac{3\pi}{\ell_P^2 \Lambda_{\rm eff}(a)}.}$$

Thus a time-dependent dark-energy density is a capacity-flow measurement.

Definition VII.1 (Capacity-flow exponent)**.** Let

$$\mathscr{D} := a\frac{d}{da}.$$

For any effective dark-energy sector with $\rho_{\rm DE} > 0$, define

$$s_{\rm eff}(a) := \mathscr{D}\ln\rho_{\rm DE}(a) = \mathscr{D}\ln\Lambda_{\rm eff}(a),$$

and

$$\nu_\partial(a) := \mathscr{D}\ln N_\partial(a).$$

Then

$$\boxed{\nu_\partial(a) = -s_{\rm eff}(a).}$$

Using the continuity equation

$$\mathscr{D}\ln\rho_{\rm DE} = -3[1 + w_{\rm eff}(a)],$$

one obtains the invariant diagnostic

$$\boxed{s_{\rm eff}(a) = -3[1 + w_{\rm eff}(a)], \qquad \nu_\partial(a) = 3[1 + w_{\rm eff}(a)].}$$

Remark VII.1 (No varying Bianchi constant)**.** The quantity $\Lambda_{\rm eff}(a)$ in this section is an effective dark-energy density written in cosmological-constant units. It is not the Bianchi integration constant $\Lambda_{\rm int}$ of theorem V.2. A time-dependent $\Lambda_{\rm eff}(a)$ is compatible with the preceding trace-free construction only as part of a conserved enlarged stress tensor or a covariant boundary-exchange law.

For the CPL form

$$w(a) = w_0 + w_a(1 - a),$$

the exact size-dependent factor is

$$\boxed{\frac{\Lambda_{\rm eff}(a)}{\Lambda_0} = a^{-3(1+w_0+w_a)}\exp[3w_a(a-1)]\,,}$$

and the corresponding capacity factor is

$$\boxed{\frac{N_\partial(a)}{N_{\partial 0}} = a^{3(1+w_0+w_a)}\exp[-3w_a(a-1)]\,.}$$

The instantaneous crossing of the phantom divide, if $w_a \neq 0$, occurs at

$$\boxed{a_\times = 1 + \frac{1+w_0}{w_a}, \qquad z_\times = a_\times^{-1} - 1.}$$

For the DESI-favored quadrant $w_0 > -1, w_a < 0$, this crossing lies in the past when $0 < a_\times < 1$. In that case $w < -1$ before the crossing and $w > -1$ after it. Equivalently, $\Lambda_{\rm eff}$ grows with scale while $w < -1$, reaches an extremum at $w = -1$, and decreases with scale while $w > -1$.

Theorem VII.1 (Static valuation interpretation of an increasing effective Λ)**.** *In a single valuation branch,*

$$\rho_s(a) \propto a^s, \qquad w_s = -1 - \frac{s}{3}, \qquad \frac{N_s(a)}{N_{s0}} = a^{-s}.$$

Thus an increasing effective dark-energy density,

$$\mathscr{D}\ln\Lambda_{\rm eff} > 0,$$

requires

$$s_{\rm eff} > 0 \qquad \Longleftrightarrow \qquad w_{\rm eff} < -1.$$

Within the static positive B^4/S^3 valuation spectrum, the only positive root is

$$\boxed{\begin{array}{ll} s = 1, & w = -\dfrac{4}{3}, \\ \Lambda_{\rm eff} \propto a, & N_\partial \propto a^{-1}. \end{array}} \tag{VII.1}$$

The boundary-local quotient branch $s = -1$ instead gives

$$\boxed{\begin{array}{ll} s = -1, & w = -\dfrac{2}{3}, \\ \Lambda_{\rm eff} \propto a^{-1}, & N_\partial \propto a. \end{array}} \tag{VII.2}$$

Therefore a genuine increasing-$\Lambda_{\rm eff}$ epoch cannot be represented by the positive boundary-local a^{-1} branch. It either selects the full bulk valuation root $s = 1$, or it requires a controlled interacting/sign-changing capacity-flow extension.

Proof. For a power branch $\rho_s \propto a^s$, the continuity equation gives $s = -3(1 + w_s)$, hence $w_s = -1 - s/3$. Since $N_\partial \propto \Lambda_{\rm eff}^{-1}$, one has $N_s/N_{s0} = a^{-s}$. The density increases with scale exactly when $s > 0$. In the finite valuation spectrum $\{1, 0, -1, -2, -3\}$, the only positive exponent is $s = 1$. The boundary-local spectrum $\{0, -1, -2, -3\}$ contains no positive exponent. □

Corollary VII.1 (Positive-mixture monotonicity test)**.** *For a positive finite valuation mixture,*

$$\mathscr{D}w = -\frac{1}{3}\mathrm{Var}_\pi(s) \leq 0.$$

Hence a reconstructed epoch with $\mathscr{D}w > 0$ cannot be a positive noninteracting valuation mixture on a fixed support. If confirmed, the DESI-type trend from phantom-like behavior in the past to quintessence-like behavior today selects one of three possibilities:

bulk valuation activation, interacting branch exchange, or sign-indefinite effective reconstruction

It does not invalidate the valuation annihilator. It identifies the boundary-positive subclass as too small for that epoch.

Theorem VII.2 (Interacting two-branch capacity flow and observable exchange current)**.** *Let*

$$\rho(a) = C_-(a)a^{s_-} + C_+(a)a^{s_+},$$
$$C_\pm(a) > 0, \qquad s_- < s_+$$

with branch fractions $\pi_\pm$ *and*

$$\kappa_1 := \mathscr{D}\ln\rho.$$

Define the exchange current

$$J_+ := \mathscr{D}\pi_+ - (s_+ - \kappa_1)\pi_+, \qquad J_- := -J_+.$$

Then

$$\sum_s J_s = 0, \qquad \mathscr{D}\kappa_1 = \kappa_2 + (s_+ - s_-)J_+.$$

Since

$$\kappa_1 = -3(1+w),$$

and, for a two-point support,

$$\kappa_2 = (s_+ - \kappa_1)(\kappa_1 - s_-),$$

the current is recovered pointwise from the observable pair $(w, \mathscr{D}w)$*:*

$$\boxed{J_+(a) = \frac{-3\mathscr{D}w - (s_+ - \kappa_1)(\kappa_1 - s_-)}{s_+ - s_-}.} \tag{VII.3}$$

Hence $J_+ \equiv 0$ *if and only if the noninteracting replicator closure holds. Moreover,* $\mathscr{D}w > 0$ *on an interval if and only if*

$$J_+ < -\frac{\kappa_2}{s_+ - s_-} < 0$$

there; capacity then flows from the upper branch to the lower branch while both branch weights remain positive.

Proof. From $\pi_+ + \pi_- = 1$ and

$$\kappa_1 = s_- + (s_+ - s_-)\pi_+,$$

one has

$$\mathscr{D}\kappa_1 = (s_+ - s_-)\mathscr{D}\pi_+.$$

Substituting

$$\mathscr{D}\pi_+ = (s_+ - \kappa_1)\pi_+ + J_+$$

and using

$$(s_+ - s_-)(s_+ - \kappa_1)\pi_+ = \kappa_2$$

for a two-point distribution gives the stated identity. The variance formula is

$$\kappa_2 = \pi_+\pi_-(s_+ - s_-)^2 = (s_+ - \kappa_1)(\kappa_1 - s_-).$$

□

Corollary VII.2 (Phantom quadrant under positivity)**.** *A reconstructed epoch with* $w < -1$ *in the past,* $w > -1$ *today, and* $\mathscr{D}w > 0$ *cannot be a positive noninteracting mixture on any fixed support. It is realized by a positive interacting two-branch mixture if and only if*

$$\kappa_1 \in (s_-, s_+), \qquad J_+ < -\frac{\kappa_2}{s_+ - s_-}.$$

For the valuation pair $(s_-, s_+) = (-1, 1)$*, this gives an exact amplitude-free diagnostic of the DESI-type quadrant.*

Criterion VII.1 (Interacting capacity-flow null)**.** The observable residual

$$\boxed{\mathcal{R}_{\rm int}(a) := J_+(a)}$$

vanishes identically for the noninteracting valuation closure. A statistically nonzero reconstruction of $\mathcal{R}_{\rm int}$ with the sign pattern of theorem VII.2 is the invariant signature of interacting capacity exchange, distinct from both signed negative-weight shadow fits and bulk-valuation-root activation.

Criterion VII.2 (Dynamical-capacity null)**.** The invariant observational null replacing a varying bare cosmological constant is

$$\boxed{\mathcal{R}_{\rm cap}(a) := \mathscr{D}\ln N_\partial(a) - 3[1 + w_{\rm eff}(a)] = 0.}$$

For a static valuation branch this reduces to

$$\mathcal{R}_{\rm cap} = 0, \qquad \nu_\partial = -s \in \{-1, 0, 1, 2, 3\}.$$

For a free-spectrum reconstruction one shall first infer $s_{\rm eff}$ or the discrete pencil roots from the data, and only afterwards compare them with

$$S_{\rm val} = \{1, 0, -1, -2, -3\}.$$

This is the same consistency-versus-identification boundary already required by the invariant test architecture.

Using the Planck-length value $\ell_P = 1.616255 \times 10^{-35}\,\mathrm{m}$ [34], and a representative ΛCDM normalization $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda \simeq 0.685$, one obtains

$$\Lambda_0 \simeq 1.091 \times 10^{-52}\,\mathrm{m}^{-2},$$

and therefore

$$\boxed{N_{\partial 0} = \frac{3\pi}{\Lambda_0 \ell_P^2} \simeq 3.31 \times 10^{122}.}$$

For the static valuation roots, the size-dependent capacity factors are

s	w_s	$\Lambda_s(a)/\Lambda_{s0}$	$N_s(a)/N_{s0}$
1	$-4/3$	a	a^{-1}
0	-1	1	1
-1	$-2/3$	a^{-1}	a
-2	$-1/3$	a^{-2}	a^2
-3	0	a^{-3}	a^3

This table is the exact size-dependent factorization of the capacity mechanism.

VIII. INVARIANT FRONTIER AND ALL-DIMENSIONAL CLOSURE LEDGER

This section records what is closed at theorem level and what remains a microscopic frontier. This distinction is part of the result. A finite annihilator, a projector, a sign certificate, or a null test may be exact without determining a dimensional amplitude.

A. All-dimensional valuation skeleton

Let the spatial dimension be $n \geq 1$, with round section S^n_A and auxiliary filling B^{n+1}_A. Since $\mathrm{Vol}(S^n_A) \propto A^n$, the round restriction of a continuous rigid-motion invariant additive valuation on convex bodies in $\mathbb{R}^{n+1}$ has filling-energy powers

$$E_{\mathrm{val}}(A) \in \mathrm{span}\{A^{n+1}, A^n, A^{n-1}, \ldots, A, 1\}.$$

After division by the S^n_A volume, the density spectrum is

$$S^{(n)}_{\mathrm{val}} = \{1, 0, -1, \ldots, -n\}.$$

Equivalently,

$$\boxed{P_n(\mathscr{D})\rho = 0, \qquad P_n(z) = (z-1)z(z+1)\cdots(z+n).} \tag{VIII.1}$$

All roots are simple and real, hence

$$\mathbb{R}[z]/(P_n) \simeq \mathbb{R}^{\,n+2}.$$

The boundary-local face removes the bulk intrinsic volume:

$$S^{(n)}_{\partial} = \{0, -1, \ldots, -n\}, \qquad P_{\partial,n}(z) = z(z+1)\cdots(z+n).$$

For a branch $\rho_s \propto a^s$, the n-dimensional work-map equation of state is

$$w_s = -1 - \frac{s}{n}.$$

Theorem VIII.1 (Dimension-sensitive boundary quotient)**.** *For $n \geq 3$, quotient the boundary alphabet $S^{(n)}_{\partial} = \{0, -1, \ldots, -n\}$ by the standard homogeneous expansion powers carried by vacuum, geometric curvature, and dust,*

$$0, \qquad -2, \qquad -n.$$

Then the quotient dimension is $n-2$. Hence the condition that there be exactly one nonstandard boundary-local homogeneous direction selects $n = 3$. The surviving direction is then

$$\boxed{s = -1, \qquad \rho_X \propto a^{-1}, \qquad w_X = -\frac{2}{3}.}$$

Proof. For $n \geq 3$, the three exponents $0, -2, -n$ are distinct. Since $|S^{(n)}_{\partial}| = n + 1$, the quotient dimension is $(n+1) - 3 = n - 2$. It equals one if and only if $n = 3$. The remaining exponent in $\{0, -1, -2, -3\}$ after removing $\{0, -2, -3\}$ is $s = -1$, and $w_s = -1 - s/3 = -2/3$. □

B. Closed ledger

The following entries are closed under the hypotheses stated in the corresponding sections.

First, constant vacuum shifts are removed by the trace-free source projector:

$$\mathcal{P}_{\mathrm{tf}}(-\rho_0 g_{\mu\nu}) = 0.$$

Second, if the physical stress tensor is conserved, the trace-free equation recovers a Bianchi integration mode,

$$R + 8\pi G T = 4\Lambda_{\mathrm{int}}, \qquad G_{\mu\nu} + \Lambda_{\mathrm{int}} g_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

Third, the capacity postulate fixes this mode only after the boundary entropy capacity is supplied:

$$\Lambda_{\mathrm{cap}} = \frac{3\pi}{\ell_P^2 N_\partial}, \qquad N_\partial = \log \dim \mathcal{H}^{\mathrm{eff}}_\partial.$$

Thus the manuscript does not yet compute Λ; it turns it into a boundary entropy-counting problem.

Fourth, positive noninteracting finite mixtures obey the arrow

$$\mathscr{D}w = -\frac{1}{3}\mathrm{Var}_\pi(s) \leq 0.$$

A reconstructed epoch with $\mathscr{D}w > 0$ therefore requires a signed effective reconstruction, interacting branch exchange, or an enlarged functional class. The interacting two-branch current in theorem VII.2 is the local observable diagnostic of the second possibility.

Fifth, the a^{-2} coefficient must be split before positivity is read:

$$\Omega_{2,\mathrm{val}} = \mathcal{C}_{-2}[F] - \Omega^{\mathrm{geom}}_K.$$

Sixth, dark matter stability is encoded by a current,

$$\nabla_\mu J^\mu_{\mathrm{DM}} = 0, \qquad \rho_{\mathrm{DM}} = m n_{\mathrm{DM}} \propto a^{-n},$$

which gives a^{-3} in the observed $n = 3$ case.

C. Microscopic closure and open frontier

The microscopic amplitude problem is no longer diffuse. It is compressed into three explicit edge-theorem gates, stated and implemented in Section LII. None of the gates admits q_{obs}, Λ_{obs}, or R_Λ as an input. Their data are the edge recursion, the registry field content, the APS/zeta convention, and the Harish–Chandra/stringy horizon determinant.

1. *Recursion Gate.* The compact Hopf–Bradlow–APS–HC operator $\mathcal{R}$ on de Sitter edge characters has the terminal integer fixed point

$$N_\varphi = 292.$$

2. *Matching Gate.* The registry GLSM, Chern–Simons level lock, APS convention, and Kronecker-limit stringy edge character give the scheme-covariant ledger

$$b\, g_\partial^2(\ell_P^{-1}) = 0.559754586\ldots, \qquad \Xi_{\rm edge} \simeq 0.99916928.$$

3. *Determinant-Line Gate.* The projected horizon edge determinant is a capacity-sector function obeying

$$-\log Z_{\rm edge}(q) = \frac{1}{2}\ln q + \Theta_\partial(q) + S_{\rm HC}(q) + o(1).$$

With these gates in place, the conditional selector becomes the amplitude theorem stated in Theorem LII.4. The open frontier is therefore not another fit for Λ, but independent reproduction and falsification: a finite-cutoff spectral evaluator for $\mathcal{R}$, a public matching ledger, a zeta/Harish–Chandra determinant evaluator, full joint cosmological chains, and laboratory or analog tests of the same edge operators. All other entries of the atlas—quantizers, bounds, obstructions, registries, null tests, and observational diagnostics—serve only to protect, reproduce, or falsify this edge-selection chain.

IX. ILLUSTRATIVE REAL-DATA AUDIT: DESI DR2 BAO

The invariant sieve must not be judged by rhetoric. We therefore include an illustrative, reproducible background audit against published DESI DR2 BAO distance measurements. The data vector used here is the DESI DR2 BAO Table 4 cosmology vector: D_V/r_d for BGS and $D_M/r_d, D_H/r_d$ for the six combined high-redshift tracers, including the combined LRG3+ELG1 bin rather than the two correlated sub-bins [31]. The calculation is a BAO-only background audit. It is not a joint CMB+BAO+SNe+growth likelihood, not a substitute for the official DESI chains, and not a full covariance public-likelihood reproduction.

We compare the following finite models:

$$\Lambda\text{CDM}, \qquad X : F(a) = \Omega_m a^{-3} + \Omega_X a^{-1} + \Omega_\Lambda,$$

$$P : F(a) = \Omega_m a^{-3} + \Omega_P a + \Omega_\Lambda, \qquad X + P,$$

and the empirical w_0w_aCDM benchmark. The nuisance scale $C = c/(H_0 r_d)$ is profiled. Positivity is imposed in the physical valuation fits unless explicitly stated.

The best positive X-branch fit gives

$$\boxed{\Omega_X = 0.146840, \qquad \Omega_m = 0.293136, \qquad \chi^2 = 9.5299.}$$

This is a consistency-level improvement over ΛCDM in this BAO-only audit, not a detection. The positive $s = +1$ branch is driven to

$$\boxed{\Omega_P \simeq 0,}$$

so these BAO data do not select a positive increasing-$\Lambda_{\rm eff} \propto a$ branch. Signed P and signed $X + P$ fits can improve χ^2, but their negative branch weights violate the positive-branch interpretation and must be treated as effective reconstructions rather than positive valuation fluids.

The absolute normalization check uses $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda = 0.685$, and $\ell_P = 1.616255 \times 10^{-35}\,\mathrm{m}$ [9, 34]. It gives

$$\boxed{\Lambda_0 = 1.091 \times 10^{-52}\,\mathrm{m}^{-2}, \qquad N_{\partial 0} = 3.31 \times 10^{122}.}$$

The exact capacity-flow null remains

$$\boxed{\mathcal{R}_{\rm cap}(a) = \mathscr{D} \ln N_\partial(a) - 3[1 + w_{\rm eff}(a)] = 0.}$$

The illustrative audit conclusion is therefore precise:

> DESI DR2 BAO in this audit does not falsify the invariant sieve; it does not yet prove superiority of the simple positive a^{-1} valuation branch. The data favor using the dynamical capacity-flow formalism and free-spectrum reconstruction before making an identification claim.

X. QUANTUM COVARIANCE ALGEBRA

The capacity in hypothesis VI.1 is not a raw dimension of an arbitrary vector space. It is counted after the gravitational constraints are imposed. Classically, general relativity is a constrained Hamiltonian system:

$$\mathcal{H}_\perp(x) = 0, \qquad \mathcal{H}_i(x) = 0, \tag{X.1}$$

with the hypersurface-deformation algebra [22–24, 36].

Definition X.1 (Anomaly-free quantum covariance algebra)**.** A quantum gravitational completion supplies a domain, state space, and constraint operators

$$\widehat{\mathcal{H}}_A = (\widehat{\mathcal{H}}_\perp, \widehat{\mathcal{H}}_i) \tag{X.2}$$

such that

$$\boxed{\widehat{\mathcal{H}}_A \Psi = 0, \qquad [\widehat{\mathcal{H}}_A, \widehat{\mathcal{H}}_B] = i\hbar\, \widehat{f}_{AB}^{\;C} \widehat{\mathcal{H}}_C} \tag{X.3}$$

on the physical domain, with no anomalous obstruction. This structure is denoted $\mathcal{A}_{\rm grav}$.

No-go theorem X.1 (Anomalous constraint algebras do not fix capacity)**.** If the quantum constraints do not close on the physical domain, then $\mathcal{H}_\partial$ is not a physical boundary state space for spacetime covariance and its dimension cannot define N_∂.

Proof. The constraints generate changes of spatial coordinates and refoliations. If their quantum representatives fail to close, the supposed physical state space depends on the order of gauge projection. Such a space cannot carry an invariant state count. □

TABLE I. DESI DR2 BAO-only audit of the capacity/valuation background families. The χ^2 values use the published BAO distance vector and the diagonal/limited-correlation approximation implemented in the accompanying audit files. AIC and BIC are quoted relative to ΛCDM. The table is an illustrative reproducibility check of the invariant sieve, not a full cosmological model comparison.

model	k	χ^2	dof	$\Delta\chi^2$	ΔAIC	ΔBIC
w_0w_aCDM	4	5.812	9	-4.727	-0.727	0.403
ΛCDM	2	10.539	11	0	0	0
P signed	3	8.971	10	-1.568	0.432	0.997
$X + P$ signed	4	7.085	9	-3.453	0.547	1.677
X positive	3	9.530	10	-1.009	0.991	1.556
P positive	3	10.539	10	0	2.000	2.565
$X + P$ positive	4	9.530	9	-1.009	2.991	4.121

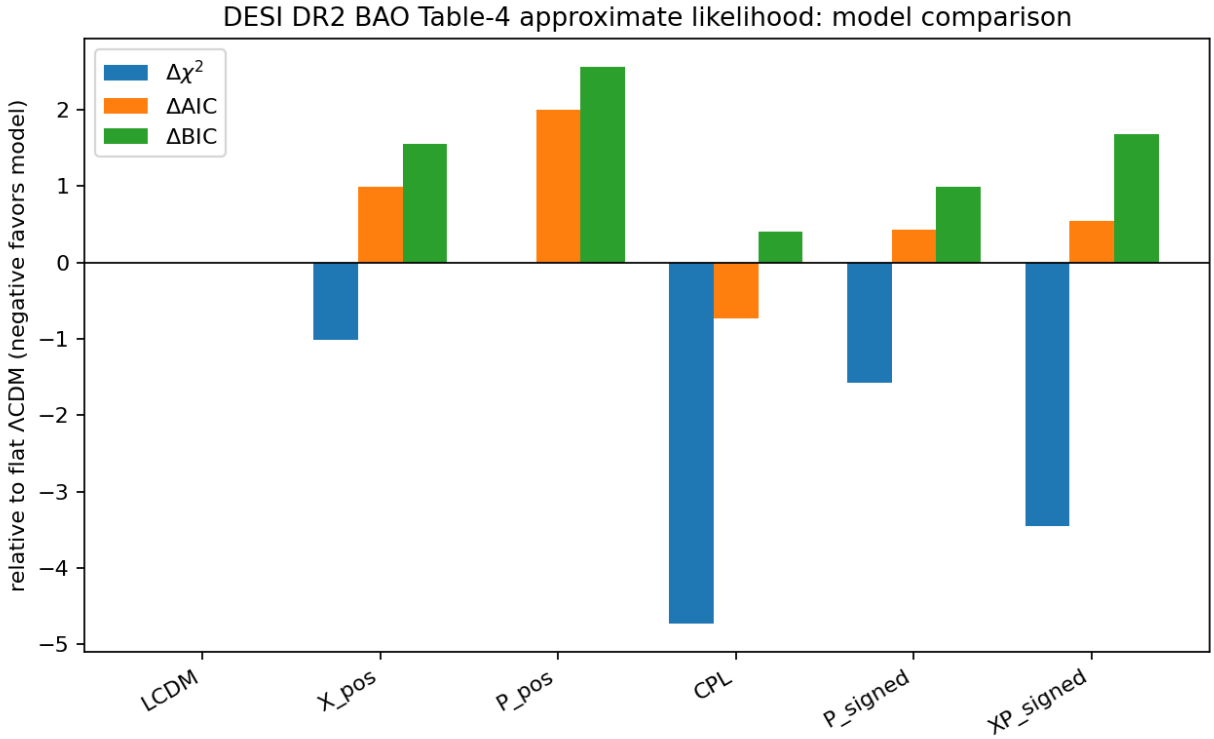


FIG. 1. BAO-only comparison of the tested background families. The flexible w_0w_a benchmark has the lowest χ^2; BIC still favors the simpler ΛCDM model over this small data vector. The positive $X : a^{-1}$ branch is viable but not superior.

XI. BOUNDARY-CONSTRAINT QUANTUM KERNEL

The previous section gives the algebraic admissibility condition. To avoid leaving the quantum layer as a slogan, we state the minimal physical-space construction. The kinematical space is

$$\boxed{\mathcal{H}_{\rm kin} = \mathcal{H}_{\rm geom}(\Sigma) \otimes \mathcal{H}_{\rm matter} \otimes \mathcal{H}_{\partial} \otimes L^2(\mathbb{R}_\Lambda).} \quad \text{(XI.1)}$$

The last factor carries the global integration mode. A conjugate four-volume variable may be introduced by

$$[\widehat{\mathcal{T}}_4, \widehat{\Lambda}] = i\hbar, \quad \text{(XI.2)}$$

which is the quantum form of the unimodular fact that Λ is conjugate to spacetime four-volume.

Definition XI.1 (Invariant physical space)**.** The admissible physical space is the common kernel, or equivalently a rigging-map image when the constraints are implemented

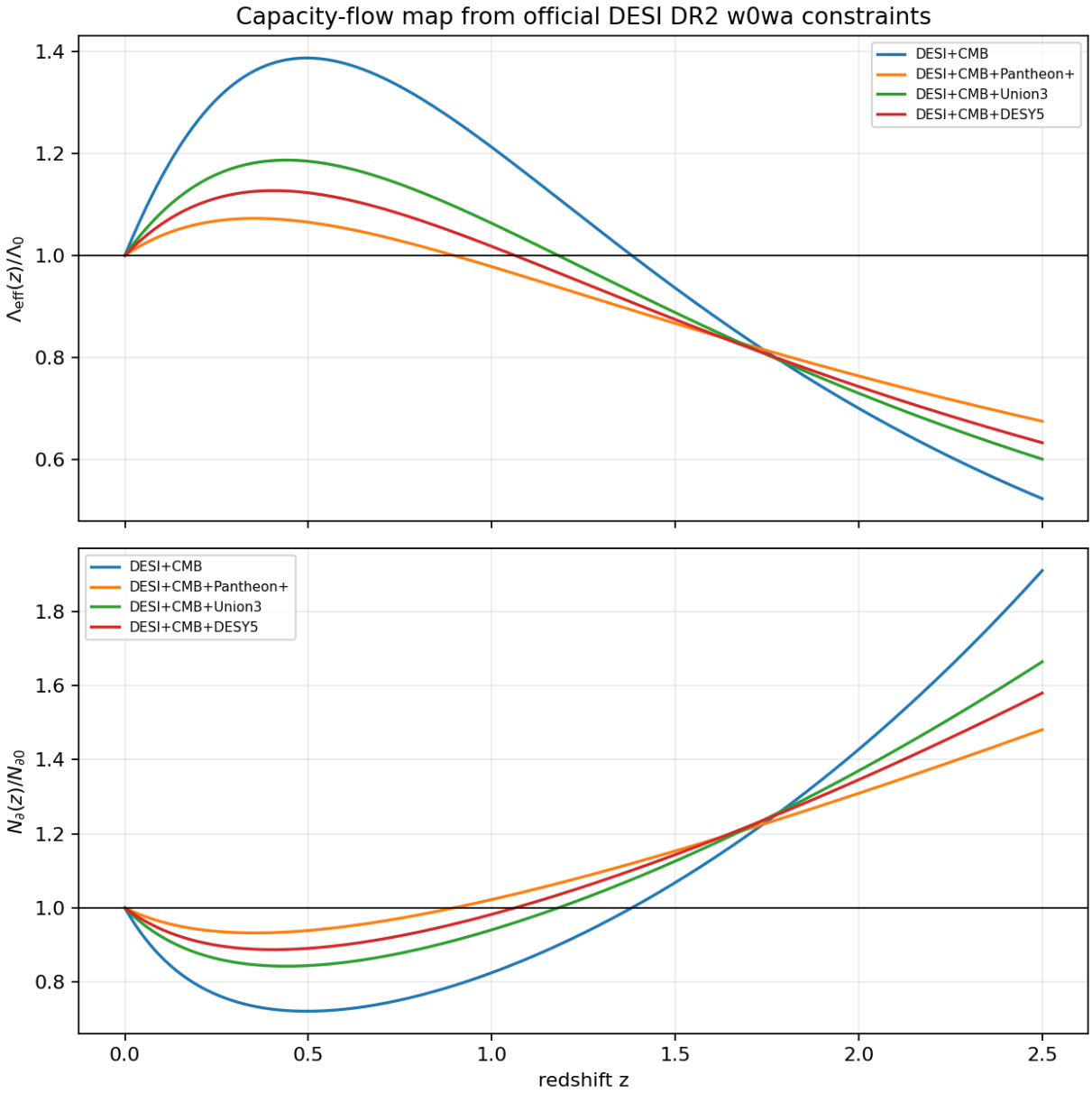


FIG. 2. Capacity-flow representation of the official DESI-style w_0w_a trend. A time-dependent dark-energy density is represented as a flow of $N_\partial(a) = 3\pi/[\ell_P^2\Lambda_{\rm eff}(a)]$, not as a varying bare cosmological constant.

by group averaging,

$$\boxed{\begin{aligned}\mathcal{H}_{\rm phys} = \ & \ker \widehat{\mathcal{H}}_A \cap \ker \widehat{C}_{\rm tf} \cap \ker \widehat{C}_{\rm cap} \\ & \cap \ker P_{\rm val}(\widehat{\mathscr{D}}) \cap \ker(\widehat{Q}_{\rm DM} - q_{\rm DM}).\end{aligned}} \tag{XI.3}$$

When the corresponding projectors are mutually well defined and commute on a common invariant domain $\mathfrak{D}$, this common-kernel notation may be represented formally as

$$\mathcal{H}_{\rm phys} = \Pi_{\rm HD}\Pi_{\rm tf}\Pi_{\rm cap}\Pi_{\rm val}\Pi_Q\,\mathcal{H}_{\rm kin}.$$

Without the common-domain and commutation property, the ordered product is not a definition.

Remark XI.1 (Projection order and common domain)**.** The capacity, valuation, and dark-charge conditions are imposed as abelian superselection conditions and are required to commute with the hypersurface-deformation constraints on the common domain $\mathfrak{D}$. This is a hypothesis on the microscopic completion, not a theorem of the kinematical notation. If it fails, different projection orders define different spaces and the candidate model fails the quantum part of the invariant sieve.

The hypersurface-deformation projector is admissible only if the regulated constraints close on a common phys-

ical domain:

$$[\widehat{D}[\vec{N}],\widehat{D}[\vec{M}]] = i\hbar\,\widehat{D}[[\vec{N},\vec{M}]], \tag{XI.4}$$

$$[\widehat{D}[\vec{N}],\widehat{H}[N]] = i\hbar\,\widehat{H}[\mathcal{L}_{\vec{N}}N], \tag{XI.5}$$

$$[\widehat{H}[N],\widehat{H}[M]] = i\hbar\,\widehat{D}[\widehat{q}^{ab}(N\partial_b M - M\partial_b N)]. \tag{XI.6}$$

The displayed relations are the hypersurface-deformation algebra in canonical form [35]. The last relation is an operator-algebroid relation because the structure function is the inverse spatial metric operator. This is the point at which many candidate quantum gravities fail: good homogeneous dynamics is insufficient if refoliation covariance is anomalous.

Criterion XI.1 (Weak anomaly-free completion)**.** A regulator or microscopic model passes the quantum part of the invariant sieve only if

$$\Delta_{AB} := [\widehat{C}_A,\widehat{C}_B] - i\hbar f_{AB}^{\ C}(\widehat{q})\widehat{C}_C \tag{XI.7}$$

vanishes on the common physical domain, at least weakly:

$$\langle\Phi|\Delta_{AB}|\Psi\rangle = 0, \qquad \Phi,\Psi\in\mathfrak{D}. \tag{XI.8}$$

Theorem XI.1 (Capacity eigenvalue in the projected theory)**.** *If the capacity projector imposes*

$$\widehat{C}_{\rm cap} := \ell_P^2\,\widehat{\Lambda}\,\widehat{N}_\partial - 3\pi = 0 \tag{XI.9}$$

on $\mathcal{H}_{\rm phys}$, then every simultaneous $(\widehat{\Lambda},\widehat{N}_\partial)$-eigen-sector satisfies

$$\boxed{\Lambda_{\rm cap} = \frac{3\pi}{\ell_P^2 N_\partial}.} \tag{XI.10}$$

Proof. On a joint eigenstate,

$$(\ell_P^2\widehat{\Lambda}\widehat{N}_\partial - 3\pi)\Psi = 0$$

is exactly

$$\ell_P^2\Lambda N_\partial = 3\pi.$$

For superpositions the statement holds sector by sector in the joint spectral decomposition whenever the two operators commute on the capacity domain. □

Remark XI.2 (Status of the quantum kernel)**.** This construction is not a completed regulator, spin-foam sum, lattice path integral, or ultraviolet fixed-point calculation. Concrete microscopic programs such as causal dynamical triangulations and functional-renormalization-group gravity supply examples of the kind of ultraviolet completion that would have to reproduce this kernel [37, 38]. The present construction is the minimal invariant kernel that any such microscopic realization must reproduce: trace-free locality, boundary capacity, anomaly-free refoliation covariance, dark-charge superselection, and finite homogeneous S^3-valuation closure.

XII. DARK MATTER AS A PROTECTED CHARGE SECTOR

A cold dark component is not explained merely by adding a^{-3} to the Friedmann ledger. It must be stable, gravitating, and approximately collisionless.

Hypothesis XII.1 (Dark charge)**.** There exists an exact conserved current

$$\boxed{\nabla_\mu J^\mu_{\rm DM} = 0} \tag{XII.1}$$

and a charge

$$\boxed{Q_{\rm DM} = \int_\Sigma J^\mu_{\rm DM} n_\mu\,\mathrm{d}\Sigma} \tag{XII.2}$$

or an equivalent discrete or topological protection. In the homogeneous cold limit,

$$\boxed{T^{\rm DM}_{\mu\nu} = \rho_{\rm DM}u_\mu u_\nu, \qquad \rho_{\rm DM}\propto a^{-3}, \qquad p_{\rm DM}\simeq 0.} \tag{XII.3}$$

Remark XII.1 (Separation from the valuation dark-energy branch)**.** The $s=-1$ valuation branch has $w=-2/3$, not cold dust. It must not be double counted as dark matter. The cold $s=-3$ sector is protected by $Q_{\rm DM}$; the nonstandard homogeneous boundary branch $s=-1$ is controlled by $P_{\rm val}(\mathscr{D})$.

No-go theorem XII.1 (No linear density charge for nonzero exponents)**.** Let

$$Q[\rho](a) = \sum_s q_s C_s a^s$$

be a linear functional of independent homogeneous density branches and require $\mathscr{D}Q[\rho]=0$ for all a on an interval. Then every active nonzero exponent in the support has $q_sC_s = 0$. Hence a conserved dark charge cannot in general be a bare linear combination of Ω_Λ and Ω_X.

Proof. One has

$$\mathscr{D}Q = \sum_s s\,q_sC_s a^s.$$

Distinct powers are linearly independent. Therefore $s\,q_sC_s = 0$ for each active branch. Only $s=0$ can survive without further structure. A physical dark charge must instead be represented by a conserved current, a discrete charge, or a topological protection law. □

XIII. S^3 VALUATION CLOSURE

Let a closed homogeneous spatial section be $\Sigma\simeq S^3$ with round metric $g_A = A^2\mathrm{d}\Omega_3^2$. Let B_A^4 be the auxiliary Euclidean filling. Under continuous, rigid-motion invariant, additive Euclidean valuation closure on convex

bodies in $\mathbb{R}^4$, Hadwiger theory gives the intrinsic-volume span [25, 26]. On the round family this implies

$$E_4^{\mathrm{val}}(A) = \frac{\pi^2}{2}\gamma A^4 + 2\pi^2\left(\alpha_0 A^3 + 3\alpha_1 A^2 + 3\alpha_2 A + \alpha_3\right). \tag{XIII.1}$$

Since $V_3(A) = 2\pi^2 A^3$,

$$\begin{aligned}\rho_{\mathrm{val}}(A) &= \frac{E_4^{\mathrm{val}}(A)}{V_3(A)} \\ &= \frac{\gamma}{4}A + \alpha_0 + 3\alpha_1 A^{-1} \\ &\quad + 3\alpha_2 A^{-2} + \alpha_3 A^{-3}.\end{aligned} \tag{XIII.2}$$

Thus

$$\boxed{\mathrm{Spec}_{\mathscr{D}}(\rho_{\mathrm{val}}) \subseteq \{1, 0, -1, -2, -3\}, \qquad \mathscr{D} := A\frac{\mathrm{d}}{\mathrm{d}A}.} \tag{XIII.3}$$

Theorem XIII.1 (Finite valuation annihilator)**.** *For C^5 profiles on a positive interval,*

$$\boxed{\begin{aligned}&P_{\mathrm{val}}(\mathscr{D})\rho_{\mathrm{val}} = 0, \\ &P_{\mathrm{val}}(z) = (z-1)z(z+1)(z+2)(z+3).\end{aligned}} \tag{XIII.4}$$

if and only if

$$\rho_{\mathrm{val}}(A) \in \mathrm{span}\{A, 1, A^{-1}, A^{-2}, A^{-3}\}. \tag{XIII.5}$$

Proof. With $x = \ln A$, $\mathscr{D} = \mathrm{d}/\mathrm{d}x$. The equation becomes a constant-coefficient ODE with simple roots $1, 0, -1, -2, -3$, hence the stated exponential/power basis. □

Corollary XIII.1 (Boundary quotient)**.** *In the boundary-local class the bulk $s = 1$ branch is removed:*

$$S_\partial = \{0, -1, -2, -3\}. \tag{XIII.6}$$

At the expansion level, quotienting by vacuum, curvature-degenerate, and dust powers $\{1, a^{-2}, a^{-3}\}$ leaves the unique nonstandard direction

$$\boxed{\rho_X \propto a^{-1}, \qquad w_X = -1 - \frac{-1}{3} = -\frac{2}{3}.} \tag{XIII.7}$$

Remark XIII.1 (What valuation does and does not fix)**.** The valuation algebra fixes the allowed homogeneous powers. It does not fix the amplitude of the $s = 0$ branch. In the present manuscript that amplitude is supplied by the capacity mode:

$$\rho_\Lambda^{\mathrm{cap}} = \frac{\Lambda_{\mathrm{cap}}}{8\pi G}.$$

Thus the $s = 0$ power is valuation-allowed, while its amplitude is capacity-fixed.

XIV. NONLINEAR EQUATION-OF-STATE CLOSURE

The density closure is linear. The equation-of-state closure is not.

Let

$$\rho(a) = \sum_{s\in S} C_s a^s, \qquad \rho > 0, \tag{XIV.1}$$

and define

$$\begin{aligned}w &:= -1 - \frac{1}{3}\frac{\mathscr{D}\rho}{\rho}, \\ \kappa_1 &:= \mathscr{D}\ln\rho = -3(1+w), \\ \kappa_n &:= \mathscr{D}^n \ln\rho \qquad (n \geq 2).\end{aligned} \tag{XIV.2}$$

Then κ_n are the cumulants of the tilted finite spectral distribution. Since

$$\frac{\mathscr{D}^n\rho}{\rho} = B_n(\kappa_1, \ldots, \kappa_n), \tag{XIV.3}$$

with B_n the complete exponential Bell polynomial, the valuation annihilator gives the amplitude-free nonlinear identity

$$\boxed{B_5 + 5B_4 + 5B_3 - 5B_2 - 6B_1 = 0.} \tag{XIV.4}$$

The boundary-local face gives

$$\boxed{B_4 + 6B_3 + 11B_2 + 6B_1 = 0.} \tag{XIV.5}$$

For the endpoint support $\{0, -3\}$,

$$\boxed{\mathscr{D}w = 3w(1+w).} \tag{XIV.6}$$

Criterion XIV.1 (Correct w-null)**.** The correct null is not $P_{\mathrm{val}}(\mathscr{D})w = 0$. Except for single branches and accidental degeneracies, $w(a)$ is a ratio of finite exponential sums and satisfies the nonlinear cumulant closure above. Observational tests based on $w(a)$ must therefore propagate derivative covariance through this nonlinear identity.

XV. CAPACITY-FIXED EDSC

Let

$$F(a) := \frac{H^2(a)}{H_0^2}, \qquad \Omega_K^{\mathrm{geom}} := -\frac{1}{A_0^2 H_0^2} \tag{XV.1}$$

in units $c = 1$. Define the physical density and pressure represented by the homogeneous Einstein tensor:

$$\boxed{E_F(a) := F(a) - \Omega_K^{\mathrm{geom}} a^{-2},} \tag{XV.2}$$

$$\boxed{P_F(a) := -E_F(a) - \frac{1}{3}\mathscr{D}E_F(a).} \tag{XV.3}$$

This is the homogeneous Bianchi identity in dilation form.

The capacity-fixed vacuum slot is

$$\boxed{\Omega_\Lambda^{\rm cap} = \frac{\Lambda_{\rm cap}}{3H_0^2} = \frac{\pi}{H_0^2 \ell_P^2 N_\partial}.} \tag{XV.4}$$

Therefore a finite capacity-fixed homogeneous ledger has the form

$$\boxed{\begin{aligned} E_F(a) = \Omega_\Lambda^{\rm cap} + \Omega_X a^{-1} + \Omega_{2,\rm val} a^{-2} \\ + \Omega_m a^{-3} + \Omega_r a^{-4}. \end{aligned}} \tag{XV.5}$$

with

$$F(a) = E_F(a) + \Omega_K^{\rm geom} a^{-2}. \tag{XV.6}$$

Theorem XV.1 (Homogeneous closure with capacity–fixed vacuum)**.** *Assume closed FLRW symmetry, trace-free local dynamics, conserved physical stress, the boundary capacity postulate, and the finite valuation support for the nonstandard homogeneous sector. Then the homogeneous gravitational sector is closed by* (XV.5)*. The vacuum coefficient is not free:*

$$\Omega_\Lambda = \Omega_\Lambda^{\rm cap}.$$

The remaining physical coefficients are separated from geometric curvature by

$$\boxed{\Omega_{2,\rm val} = \mathcal{C}_{-2}[F] - \Omega_K^{\rm geom},}$$

where $\mathcal{C}_s$ are the exact dilation coefficient projectors.

Proof. Theorems V.2 and VI.1 give the Einstein equation with $\Lambda_{\rm cap}$. The first Friedmann equation gives $F = E_F + \Omega_K^{\rm geom} a^{-2}$. The Bianchi identity gives $P_F = -E_F - \mathscr{D}E_F/3$. The finite valuation closure gives the allowed nonstandard powers, while the standard cold and radiation sectors supply a^{-3} and a^{-4}. The coefficient projector statement follows from linear independence of the finite power alphabet. □

XVI. PROJECTORS, SPECTRAL RECOVERY, AND FALSIFIABLE NULLS

For a finite expansion alphabet

$$S_{\cos} := \{0, -1, -2, -3, -4\}, \qquad F(a) = \sum_{s \in S_{\cos}} \Omega_s a^s,$$

define

$$\boxed{\mathcal{C}_s[F](a) := a^{-s} \prod_{\substack{t \in S_{\cos} \\ t \neq s}} \frac{\mathscr{D} - t}{s - t} F(a).} \tag{XVI.1}$$

Then $\mathcal{C}_s[F] = \Omega_s$.

The exact nulls are:

$$\mathcal{R}_\Lambda = \Lambda_{\rm obs} - \frac{3\pi}{\ell_P^2 N_\partial}, \tag{XVI.2}$$

$$\mathcal{R}_{\rm val}[\rho] = (\mathscr{D} - 1)\mathscr{D}(\mathscr{D} + 1)(\mathscr{D} + 2)(\mathscr{D} + 3)\rho, \tag{XVI.3}$$

$$\mathcal{R}_\partial[\rho] = \mathscr{D}(\mathscr{D} + 1)(\mathscr{D} + 2)(\mathscr{D} + 3)\rho, \tag{XVI.4}$$

$$\mathcal{R}_X[F] = \mathscr{D}\mathcal{C}_{-1}[F], \tag{XVI.5}$$

$$\mathcal{R}_2[F] = \mathcal{C}_{-2}[F] - \Omega_K^{\rm geom} - \Omega_{2,\rm val}, \tag{XVI.6}$$

$$\mathcal{R}_{\rm DM} = \nabla_\mu J^\mu_{\rm DM}. \tag{XVI.7}$$

A capacity-fixed, boundary-local, one-parameter X execution has

$$\boxed{\mathcal{R}_\Lambda = \mathcal{R}_\partial = \mathcal{R}_X = \mathcal{R}_2 = \mathcal{R}_{\rm DM} = 0, \qquad \Omega_{2,\rm val} = 0.} \tag{XVI.8}$$

Theorem XVI.1 (Hankel-pencil recovery under exact support hypotheses)**.** *Let $\rho \in C^{2k-1}$ on an interval and suppose*

$$\rho(a) = \sum_{j=1}^{k} C_j a^{s_j}, \qquad C_j \neq 0, \qquad s_i \neq s_j \ (i \neq j).$$

Define derivative moments $M_n = \mathscr{D}^n \rho$ and Hankel matrices

$$H_0 = [M_{i+j}]_{i,j=0}^{k-1}, \qquad H_1 = [M_{i+j+1}]_{i,j=0}^{k-1}.$$

Then

$$\det H_0 = \left(\prod_{i<j} (s_j - s_i)^2 \right) \prod_{j=1}^{k} C_j a^{s_j}.$$

Hence H_0 is invertible exactly when the active exponents are distinct and all active amplitudes are nonzero, and

$$\boxed{\frac{\det(\lambda H_0 - H_1)}{\det H_0} = \prod_{j=1}^{k} (\lambda - s_j).} \tag{XVI.9}$$

Thus the active weights must be inferred freely before comparison with $\{1, 0, -1, -2, -3\}$. In noisy applications the theorem is not a raw-differentiation prescription: covariance, model order, smoothing, and the condition number of H_0 must be propagated. A pinned $w_X = -2/3$ calculation tests consistency, not independent identification.

XVII. COMPLETE INVARIANT CLOSURE AND CONDITIONAL UNIQUENESS

The previous sections define an invariant sieve. This section records the precise sense in which the sieve gives an algebraically forced solution. The statement is conditional: exact uniqueness follows after a finite alphabet, a curvature split, a capacity value, and an exact expansion

function have been declared. With noisy data it becomes a constrained statistical reconstruction and falsification test, not a proof that nature has been solved without uncertainty.

Definition XVII.1 (Core invariant closure vector)**.** For a declared homogeneous support and a declared perturbative lift, define the core residual vector

$$\boxed{\vec{\mathcal{I}}_{\rm core} := \begin{pmatrix} \mathcal{R}_{\rm val}[\rho] \\ \mathcal{R}_{\partial}[\rho] \\ \mathcal{R}_{\Lambda} \\ \mathcal{R}_{\rm DM}[J] \\ \mathcal{R}_{\rm cap}(a) \\ \mathcal{R}_{2}[F] \\ \mathcal{R}_{X}[F] \end{pmatrix}}. \tag{XVII.1}$$

Here

$$\mathcal{R}_{\rm val}[\rho] = (\mathscr{D}-1)\mathscr{D}(\mathscr{D}+1)(\mathscr{D}+2)(\mathscr{D}+3)\rho, \tag{XVII.2}$$
$$\mathcal{R}_{\partial}[\rho] = \mathscr{D}(\mathscr{D}+1)(\mathscr{D}+2)(\mathscr{D}+3)\rho, \tag{XVII.3}$$
$$\mathcal{R}_{\Lambda} = \Lambda_{\rm obs} - \frac{3\pi}{\ell_P^2 N_\partial}, \tag{XVII.4}$$
$$\mathcal{R}_{\rm DM}[J] = \nabla_\mu J^\mu_{\rm DM}, \tag{XVII.5}$$
$$\mathcal{R}_{\rm cap}(a) = \mathscr{D}\ln N_\partial(a) - 3[1+w_{\rm eff}(a)], \tag{XVII.6}$$
$$\mathcal{R}_{2}[F] = \mathcal{C}_{-2}[F] - \Omega_K^{\rm geom} - \Omega_{2,\rm val}, \tag{XVII.7}$$
$$\mathcal{R}_{X}[F] = \mathscr{D}\mathcal{C}_{-1}[F]. \tag{XVII.8}$$

If a smooth subhorizon growth lift is additionally declared, the augmented closure vector is

$$\boxed{\vec{\mathcal{I}}_{\rm lift} := \left(\vec{\mathcal{I}}_{\rm core},\, \mathcal{R}_{\rm growth}[D;F]\right)}, \tag{XVII.9}$$

where

$$\mathcal{R}_{\rm growth}[D;F] = \mathscr{D}^2 D + \left(2+\frac{1}{2}\mathscr{D}\ln F\right)\mathscr{D}D - \frac{3}{2}\Omega_m(a)D. \tag{XVII.10}$$

The growth residual is therefore not part of the source-independent homogeneous theorem. It is a null of the chosen perturbative lift.

Definition XVII.2 (Complete invariant closure on a declared stratum)**.** A background model is completely closed on a declared invariant stratum if

$$\boxed{\vec{\mathcal{I}}_{\rm core} = 0} \tag{XVII.11}$$

and, when a perturbative lift is declared,

$$\boxed{\vec{\mathcal{I}}_{\rm lift} = 0.} \tag{XVII.12}$$

The phrase "on a declared stratum" means that the finite alphabet, boundary-local restriction, curvature split, capacity value, and positivity class have already been specified.

Theorem XVII.1 (Exact finite-alphabet reconstruction)**.** *Let*

$$S_{\rm cos} = \{0,-1,-2,-3,-4\}$$

and suppose the exact expansion function has the form

$$F(a) = \sum_{s\in S_{\rm cos}} \Omega_s a^s$$

on an interval. Given exact values $F(a_i)$ at five distinct positive scale factors a_i, the coefficient vector

$$\vec{\Omega} = (\Omega_0, \Omega_{-1}, \Omega_{-2}, \Omega_{-3}, \Omega_{-4})$$

is unique.

Proof. Put $x_i = a_i^{-1}$. Then

$$F(a_i) = \Omega_0 + \Omega_{-1}x_i + \Omega_{-2}x_i^2 + \Omega_{-3}x_i^3 + \Omega_{-4}x_i^4.$$

The reconstruction matrix is the ordinary Vandermonde matrix

$$V_{ij} = x_i^j, \qquad i = 1,\dots,5, \quad j = 0,\dots,4.$$

Since the a_i are distinct and positive, the x_i are distinct, and

$$\det V = \prod_{i<j}(x_j - x_i) \neq 0.$$

Thus $V\vec{\Omega} = \vec{F}$ has exactly one solution. □

Corollary XVII.1 (Projector form of the same solution)**.** *If F is differentiable enough and lies in the finite alphabet $S_{\rm cos}$, the unique coefficients are the exact dilation projectors*

$$\boxed{\Omega_s = \mathcal{C}_s[F] = a^{-s}\prod_{\substack{t\in S_{\rm cos}\\ t\neq s}} \frac{\mathscr{D}-t}{s-t}F(a).} \tag{XVII.13}$$

Thus the sample Vandermonde reconstruction and the local differential projectors are the same finite-basis statement in two representations.

Criterion XVII.1 (Capacity, positivity, and curvature as compatibility tests)**.** The finite-basis solution of theorem XVII.1 satisfies the capacity-fixed positive boundary model if and only if

$$\Omega_0 = \Omega_\Lambda^{\rm cap} = \frac{\Lambda_{\rm cap}c^2}{3H_0^2}, \tag{XVII.14}$$
$$\Omega_s \geq 0 \qquad (s \in S_{\rm cos} \text{ declared physical}), \tag{XVII.15}$$
$$\Omega_{2,\rm val} = \Omega_{-2} - \Omega_K^{\rm geom} \geq 0. \tag{XVII.16}$$

The last inequality is imposed only when the physical $s=-2$ branch is declared positive. If any of these tests fails, the declared stratum is falsified. The failure does not destroy the finite reconstruction theorem; it says that the data require either signed effective weights, interacting branch exchange, an enlarged support, or a different perturbative/source lift.

Theorem XVII.2 (Constrained variational uniqueness)**.** *Let $y_i = F_{\rm obs}(a_i)$, let $V_{is} = a_i^s$ for $s \in S_{\cos}$, and let W be a positive-definite inverse covariance matrix. Define the closed feasible set*

$$\mathcal{K} = \left\{ \vec{\Omega} : \ \Omega_0 = \Omega_\Lambda^{\rm cap}, \ \Omega_s \geq 0, \ \Omega_{-2} - \Omega_K^{\rm geom} \geq 0 \right\}.$$

The constrained quadratic functional

$$\boxed{\mathcal{J}(\vec{\Omega}) = (y - V\vec{\Omega})^T W (y - V\vec{\Omega}) + \iota_{\mathcal{K}}(\vec{\Omega})} \quad \text{(XVII.17)}$$

has at most one minimizer whenever V has full column rank on the free coordinates of $\mathcal{K}$. If the minimizer exists, it is the unique statistical analogue of the algebraic solution on the declared stratum.

Proof. The first term is a strictly convex quadratic on the free coordinate subspace when V has full column rank and $W \succ 0$. The indicator $\iota_{\mathcal{K}}$ of a closed convex set preserves convexity and restricts the domain. A strictly convex function on a convex domain has at most one minimizer. Existence follows when the feasible set is nonempty and the quadratic is coercive on the free coordinates. □

Remark XVII.1 (Noiseless uniqueness versus empirical superiority)**.** Exact uniqueness is a theorem about a declared finite alphabet and exact function values. Real BAO, CMB, supernova, lensing, and growth measurements come with covariance, calibration choices, and possible model-order uncertainty. Therefore the empirical question is not whether the algebra has a unique coefficient vector once the alphabet is fixed; it is whether free-spectrum reconstruction and joint likelihoods select that alphabet over alternatives. This is why the DESI DR2 audit in section IX is treated as an illustrative background test rather than a proof of superiority.

The complete closure theorem is conditional but sharp: once the invariant stratum is declared, the coefficients are algebraically forced; if the capacity, positivity, curvature, or lift residuals fail, that stratum is falsified rather than retuned.

XVIII. DEEP COMPLETION LAYER: SPECTRAL CAPACITY, COVARIANT EXCHANGE, FREE-SPECTRUM DIAGNOSTICS, AND PERTURBATIVE LIFT

This section records the additional closure layer obtained after the real-data audit and the frontier analysis. It does not claim a final microscopic quantum-gravity derivation. It sharpens the remaining tasks into explicit spectral, covariant, and perturbative objects.

Definition XVIII.1 (Spectral horizon-capacity count)**.** Let $R_\Lambda = \sqrt{3/\Lambda_{\rm eff}}$ be the de Sitter horizon radius associated with an effective dark-energy scale. On the two-sphere horizon cross-section, impose a scalar spherical-harmonic band limit L. The cumulative degeneracy is

$$D_L = \sum_{\ell=0}^{L} (2\ell + 1) = (L+1)^2. \quad \text{(XVIII.1)}$$

The spectral-capacity model defines the logarithmic boundary capacity by

$$\boxed{N_\partial = D_L, \qquad L + 1 = \sqrt{N_\partial} = \sqrt{\pi} \frac{R_\Lambda}{\ell_P}.} \quad \text{(XVIII.2)}$$

Equivalently,

$$\boxed{\Lambda_{\rm eff} = \frac{3\pi}{\ell_P^2 N_\partial}.} \quad \text{(XVIII.3)}$$

The ordinary Hilbert-space dimension is therefore $\dim \mathcal{H}_\partial \sim \exp(N_\partial)$, not N_∂.

Remark XVIII.1 (Status of the spectral count)**.** This is a concrete entropy/spectral realization of the capacity postulate, not a derivation of the integer band limit from a microscopic Hamiltonian. With $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_\Lambda = 0.685$, and $\ell_P = 1.616255 \times 10^{-35}\mathrm{m}$, the calculation gives

$$\Lambda_0 = 1.091 \times 10^{-52}\mathrm{m}^{-2}, \qquad N_\partial = 3.31 \times 10^{122}, \quad \text{(XVIII.4)}$$

and hence

$$L + 1 = 1.82 \times 10^{61}. \quad \text{(XVIII.5)}$$

The missing microscopic problem is now the derivation of this boundary band limit, or equivalently this code-subspace entropy. The construction uses the standard horizon entropy normalization and the holographic state-counting viewpoint [12, 13].

Definition XVIII.2 (spin^c index capacity on the horizon)**.** Let the causal horizon cross-section be $S^2_{R_\Lambda} \simeq \mathbb{CP}^1$. Let $\mathcal{L}_k \to S^2$ be a complex line bundle of degree $k \geq 0$, with curvature two-form F_∂ normalized by

$$\frac{1}{2\pi} \int_{S^2} F_\partial = k. \quad \text{(XVIII.6)}$$

Equivalently, put $q := k + 1$. Riemann–Roch on $\mathbb{CP}^1$, or the index theorem for the associated spin^c Dirac/$\bar{\partial}$ operator, gives [17, 18]

$$\boxed{\dim H^0(\mathbb{CP}^1, \mathcal{O}(q-1)) = \operatorname{ind} \bar{\partial}_{\mathcal{O}(q-1)} = q.} \quad \text{(XVIII.7)}$$

The index-capacity model defines the logarithmic horizon capacity by

$$\boxed{N_\partial := q.} \quad \text{(XVIII.8)}$$

The ordinary many-state boundary Hilbert space is then a code built over these q index cells. If each cell carries a finite local code $\mathcal{K}$, then

$$\log \dim(\mathcal{K}^{\otimes q}) = q \log \dim \mathcal{K}.$$

The convention $N_\partial = q$ is the one-nat normalization of this logarithmic capacity. Thus N_∂ is an entropy/index count, not the ordinary Hilbert-space dimension itself.

Theorem XVIII.1 (Index capacity, area quantization, and de Sitter levels)**.** *Assume the Gibbons–Hawking entropy capacity is saturated by the index capacity of definition XVIII.2:*

$$N_\partial = q = \frac{A_H}{4\ell_P^2}. \tag{XVIII.9}$$

Then the horizon area, radius, and capacity-fixed cosmological scale are quantized as

$$\boxed{A_q = 4\ell_P^2 q, \qquad R_q = \sqrt{\frac{q}{\pi}}\,\ell_P, \qquad \Lambda_q = \frac{3\pi}{q\,\ell_P^2}.} \tag{XVIII.10}$$

Moreover, the scalar spectral cutoff model and the index model coincide when

$$\boxed{q = (L+1)^2.} \tag{XVIII.11}$$

Thus the band limit L of the horizon spectral model is equivalently the square-root of the spin^c *index capacity.*

Proof. For a de Sitter horizon in four spacetime dimensions, $A_H = 4\pi R_\Lambda^2 = 12\pi/\Lambda$. Saturating $N_\partial = A_H/(4\ell_P^2)$ with $N_\partial = q$ gives $A_q = 4\ell_P^2 q$ and $R_q = \sqrt{q/\pi}\ell_P$. Solving $q = 3\pi/(\Lambda\ell_P^2)$ gives the displayed Λ_q. The scalar spectral model has $N_\partial = \operatorname{Tr}\Pi_L = (L+1)^2$; identifying both capacities gives $q = (L+1)^2$. □

Proposition XVIII.1 (Rotational uniqueness of the horizon flux)**.** *On the round horizon S^2, every $SO(3)$-invariant curvature two-form is proportional to the area form. Imposing the flux quantization*

$$(2\pi)^{-1}\int_{S^2} F_\partial = q$$

therefore fixes it uniquely:

$$\boxed{F_\partial = \frac{q}{2}\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\phi.} \tag{XVIII.12}$$

Consequently, once the entropy normalization $q = A_H/(4\ell_P^2)$ is imposed, there is no remaining continuous horizon-capacity parameter.

Proof. An $SO(3)$-invariant two-form on S^2 must be a constant multiple of the invariant volume form $\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\phi$. Since

$$\int_{S^2}\sin\theta\,\mathrm{d}\theta\wedge\mathrm{d}\phi = 4\pi,$$

the condition $(2\pi)^{-1}\int F_\partial = q$ fixes the coefficient to $q/2$. □

Remark XVIII.2 (What the index lift does and does not prove)**.** The index-capacity lift replaces the earlier symbolic $\operatorname{Cap}\mathcal{H}_\partial$ by a precise integer object:

$$\boxed{N_\partial = \operatorname{ind}\bar\partial_{\mathcal{O}(q-1)} = q.}$$

It also explains why the spectral count $(L+1)^2$ is natural on a spherical horizon: it is the same integer capacity when $q = (L+1)^2$, matching the familiar monopole/lowest-Landau-level origin of fuzzy-sphere matrix geometry [19]. What remains open is the deeper Hamiltonian problem: why the physical boundary code subspace chooses the enormous present value $q \simeq 3.31\times10^{122}$, or equivalently $L+1 \simeq 1.82\times10^{61}$. The present section turns that problem into an index/flux quantization problem, rather than leaving it as a dimensionful cosmological constant.

Theorem XVIII.2 (Hopf-index bridge from spatial S^3 to horizon capacity)**.** *Let $S^3 \to S^2 \simeq \mathbb{CP}^1$ be the Hopf fibration and let $\mathcal{O}(1)$ be the associated hyperplane line bundle on $\mathbb{CP}^1$. For each integer $q \geq 1$, define the horizon code space*

$$\boxed{\mathcal{H}_{\partial,q} := H^0\big(\mathbb{CP}^1, \mathcal{O}(q-1)\big)\,.} \tag{XVIII.13}$$

Then $\mathcal{H}_{\partial,q}$ is the unique irreducible $SU(2)$ representation of spin $j = (q-1)/2$, and

$$\boxed{\dim\mathcal{H}_{\partial,q} = 2j+1 = q.} \tag{XVIII.14}$$

Equivalently, the same integer is the first Chern flux plus one,

$$\boxed{q = 1 + \frac{1}{2\pi}\int_{S^2} F_\partial, \qquad F_\partial \in 2\pi\, c_1(\mathcal{O}(q-1))\,.} \tag{XVIII.15}$$

Thus the capacity integer is simultaneously a Chern-flux datum, an index, and an irreducible $SU(2)$ code-space dimension.

Proof. The Hopf fibration is the unit-circle bundle of $\mathcal{O}(-1)$, up to the conventional dual sign. Its tensor powers generate all holomorphic line bundles $\mathcal{O}(k)$ on $\mathbb{CP}^1$. Taking $k = q-1$, the Borel–Weil/Riemann–Roch theorem gives

$$H^0(\mathbb{CP}^1, \mathcal{O}(q-1)) \simeq \operatorname{Sym}^{q-1}(\mathbb{C}^2)^*,$$

which is the irreducible $SU(2)$ representation with $j = (q-1)/2$ and dimension q. Its curvature represents $2\pi c_1(\mathcal{O}(q-1))$, so the flux is $q-1$. Hence $q = 1 + (2\pi)^{-1}\int F_\partial$. □

Corollary XVIII.1 (Rotationally invariant boundary Hamiltonian)**.** *On a fixed capacity sector $\mathcal{H}_{\partial,q}$, every $SU(2)$-invariant boundary Hamiltonian is scalar:*

$$\boxed{[H_{\partial,q}, U_q(g)] = 0\ \forall g \in SU(2) \Rightarrow H_{\partial,q} = E_q\,\mathbf{1}_q.} \tag{XVIII.16}$$

Consequently, a nontrivial capacity dynamics cannot occur inside one irreducible horizon sector. It must be an inter-sector flow

$$q \longrightarrow q + \Delta q,$$

or equivalently a boundary–bulk exchange process.

Proof. This is Schur's lemma applied to the irreducible representation $\mathcal{H}_{\partial,q}$. □

Definition XVIII.3 (Fuzzy-horizon coordinate algebra)**.** Let J_i be the spin-$j = (q-1)/2$ generators on $\mathcal{H}_{\partial,q}$, and put

$$\boxed{X_i := R_q \frac{J_i}{\sqrt{j(j+1)}}, \qquad R_q = \sqrt{\frac{q}{\pi}}\,\ell_P.} \tag{XVIII.17}$$

Then

$$\boxed{\sum_i X_i^2 = R_q^2 \mathbf{1}, \qquad [X_i, X_j] = i\,\frac{R_q}{\sqrt{j(j+1)}}\,\epsilon_{ijk} X_k.} \tag{XVIII.18}$$

Thus the same capacity integer q defines a finite noncommutative horizon algebra whose commutative limit is $S^2_{R_q}$.

Remark XVIII.3 (Why this is a sharper microscopic target)**.** The previous capacity postulate required a number N_∂. The Hopf-index lift sharpens the target to an integer representation label:

$$\boxed{N_\partial = q = 2j + 1.}$$

The present cosmological value corresponds to

$$q \simeq 3.31 \times 10^{122}, \qquad j \simeq 1.65 \times 10^{122}.$$

The remaining microscopic problem is no longer the vague request to "explain a small Λ". It is the precise request to derive the selected Hopf/Dirac-monopole charge q, or the equivalent spin label j, from the boundary Hamiltonian and the allowed code subspace. This is the next nontrivial theorem-level target. The geometric-quantization reading is standard: quantizing $\mathbb{CP}^1$ with $\mathcal{O}(k)$ gives a finite $SU(2)$ representation and the fuzzy sphere in the large-k limit [19–21].

Theorem XVIII.3 (Weyl-capacity quantization of the horizon)**.** *Let Δ_{S^2} be the scalar Laplacian on the round horizon cross-section $S^2_{R_\Lambda}$, and define the invariant spectral projector*

$$\Pi_L := \mathbf{1}_{[0,L(L+1)]}\left(-R_\Lambda^2 \Delta_{S^2}\right), \qquad L \in \mathbb{N}. \tag{XVIII.19}$$

Then

$$\boxed{\operatorname{Tr} \Pi_L = (L+1)^2.} \tag{XVIII.20}$$

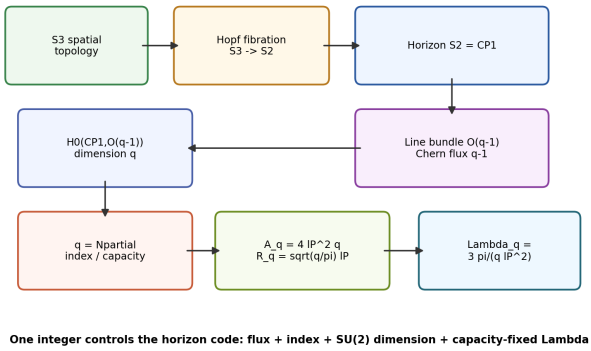


FIG. 3. Hopf-index bridge. The spatial topology supplies a natural $S^3 \to S^2$ fibration; the causal-horizon $S^2 \simeq \mathbb{CP}^1$ carries the line bundle $\mathcal{O}(q-1)$. Its index, Chern flux plus one, and $SU(2)$ code-space dimension all equal $q = N_\partial$, fixing $\Lambda_q = 3\pi/(q\ell_P^2)$.

If the entropy capacity is identified with this cutoff trace, $N_\partial = \operatorname{Tr} \Pi_L$, the allowed capacity-fixed de Sitter values form the discrete spectrum

$$\boxed{\Lambda_L = \frac{3\pi}{\ell_P^2 (L+1)^2}, \qquad R_L = \frac{L+1}{\sqrt{\pi}}\,\ell_P.} \tag{XVIII.21}$$

The relative spacing of adjacent large-L capacity levels is

$$\boxed{\frac{\Lambda_{L+1} - \Lambda_L}{\Lambda_L} = -\frac{2}{L+2} + O(L^{-2}).} \tag{XVIII.22}$$

For the present horizon value quoted above, this spacing is of order 10^{-61}, far below current cosmological precision.

Proof. The eigenvalues of $-R_\Lambda^2 \Delta_{S^2}$ are $\ell(\ell+1)$ with degeneracy $2\ell+1$, $\ell = 0, 1, \ldots$. The projector Π_L therefore counts all modes with $\ell \le L$:

$$\operatorname{Tr} \Pi_L = \sum_{\ell=0}^{L} (2\ell+1) = (L+1)^2.$$

Substitution of $N_\partial = (L+1)^2$ into $\Lambda_{\text{cap}} = 3\pi/(\ell_P^2 N_\partial)$ gives the displayed spectrum. The spacing formula follows by expanding

$$\Lambda_{L+1}/\Lambda_L = \frac{(L+1)^2}{(L+2)^2}.$$

□

Corollary XVIII.2 (Nearest capacity level and falsification residual)**.** *Given an observed late-time effective scale $\Lambda_{\text{obs}} > 0$, the nearest spectral-capacity level is obtained by*

$$\boxed{L_{\text{near}} = \text{round}\left[\sqrt{\frac{3\pi}{\Lambda_{\text{obs}} \ell_P^2}} - 1\right].} \tag{XVIII.23}$$

The exact discreteness residual is

$$\boxed{\mathcal{R}_{\rm level} := \frac{\Lambda_{\rm obs} - \Lambda_{L_{\rm near}}}{\Lambda_{\rm obs}}.} \tag{XVIII.24}$$

For $L_{\rm near} \sim 10^{61}$, the unavoidable level uncertainty is Planck-suppressed by $O(10^{-61})$. Hence capacity quantization is a microscopic constraint rather than a presently resolvable late-time cosmological oscillation.

Remark XVIII.4 (Two boundaries, two different invariants)**.** The horizon capacity count lives on the causal horizon cross-section S^2. The valuation annihilator lives on the spatial round filling B^4/S^3. These are not the same boundary. Their connection in this paper is through the closed-FLRW homogeneous geometry and the capacity postulate, not through an identity $S^2 = S^3$. This separation prevents double counting of horizon entropy and homogeneous valuation branches.

Theorem XVIII.4 (Covariant capacity-exchange law)**.** *Let*

$$T^{\rm cap}_{\mu\nu} := -\frac{\Lambda_{\rm eff}(x)}{8\pi G} g_{\mu\nu}. \tag{XVIII.25}$$

If the total stress tensor $T^{\rm tot}_{\mu\nu} = T^{\rm bulk}_{\mu\nu} + T^{\rm cap}_{\mu\nu}$ is covariantly conserved, then the bulk sector obeys

$$\boxed{\nabla_\mu T^{\mu}_{{\rm bulk}\,\nu} = \frac{1}{8\pi G}\nabla_\nu \Lambda_{\rm eff} =: Q_\nu^{\partial\to{\rm bulk}}.} \tag{XVIII.26}$$

For a homogeneous FLRW history this is equivalent to

$$\boxed{\mathscr{D} \ln N_\partial(a) = 3[1 + w_{\rm eff}(a)],} \tag{XVIII.27}$$

provided

$$\Lambda_{\rm eff}(a) = \frac{3\pi}{\ell_P^2 N_\partial(a)}$$

is interpreted as the effective dark-energy density scale, not as the constant Bianchi integration mode.

Proof. Since $\nabla_\mu g^\mu{}_\nu = 0$,

$$\nabla_\mu T^{\mu}_{{\rm cap}\,\nu} = -\frac{1}{8\pi G}\nabla_\nu \Lambda_{\rm eff}.$$

The condition $\nabla_\mu T^{\mu}_{{\rm tot}\,\nu} = 0$ gives the displayed exchange law. In FLRW, $\rho_{\rm eff} = \Lambda_{\rm eff}/(8\pi G)$ satisfies

$$\mathscr{D} \ln \rho_{\rm eff} = -3(1 + w_{\rm eff}).$$

Because $N_\partial \propto \Lambda_{\rm eff}^{-1}$, the stated capacity-flow equation follows. □

Criterion XVIII.1 (Boundary-exchange lift)**.** A dynamical dark-energy reconstruction must specify $Q_\nu^{\partial\to{\rm bulk}}$. Without this covariant exchange current, $N_\partial(a)$ is only a diagnostic dictionary. With it, capacity flow becomes a conservation law for the enlarged bulk-plus-boundary system.

Theorem XVIII.5 (Observable two-root exchange current, executed form)**.** *For the interacting pair $(s_-, s_+) = (-1, 1)$, define*

$$\kappa_1 = -3(1 + w), \qquad \kappa_2 = (1 - \kappa_1)(1 + \kappa_1).$$

The capacity-exchange current is recovered from $(w, \mathscr{D}w)$ by

$$\boxed{J_+ = \frac{-3\mathscr{D}w - \kappa_2}{2}.} \tag{XVIII.28}$$

Thus a DESI-type quadrant with $\mathscr{D}w > 0$, when it lies in $-1 < \kappa_1 < 1$, requires $J_+ < -\kappa_2/2 < 0$. The sign is the invariant statement that weight flows from the upper branch to the lower branch while positivity is maintained.

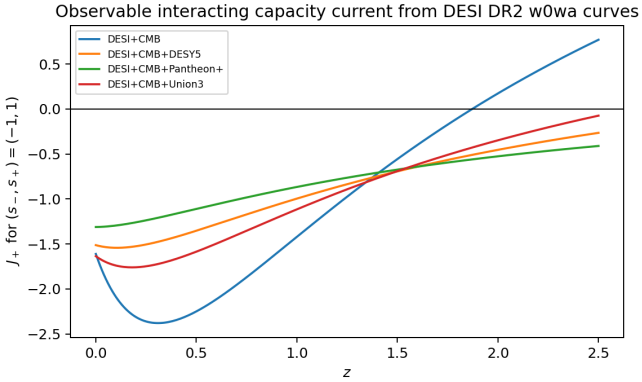


FIG. 4. Observable interacting capacity current J_+ reconstructed from the DESI DR2 w_0w_a curves used in the audit. The negative sign in the admissible interval is the invariant signature that a positive two-root representation needs exchange; a noninteracting positive mixture cannot produce $\mathscr{D}w > 0$.

Proposition XVIII.2 (Executed free-spectrum smoke test)**.** *For the exact four-branch test profile*

$$F_{\rm test}(a) = 0.541216 + 0.174541a^{-1} + 0.284243a^{-3} + 9.2 \times 10^{-5}a^{-4}. \tag{XVIII.29}$$

the logarithmically sampled Hankel pencil with $k = 4$ recovers

$$s = (-4.0000, -3.0000, -1.0000, 0.0000).$$

up to numerical roundoff. Applying the same local pencil to the DESI w_0w_a capacity-flow curves does not return a stable subset of $\{1, 0, -1, -2, -3\}$. Hence present w_0w_a curves are evidence for a dynamical diagnostic trend, not a free-spectrum identification of the valuation roots.

Criterion XVIII.2 (GitHub data interface for the next audit)**.** The next numerical layer shall use public data products rather than hand-entered summaries. The intended inputs are the DESI DR2 BAO data in the Cobaya

BAO repository, Pantheon+SH0ES distance and covariance files, compressed Planck likelihood products, and public growth compilations [40–43]. The acceptance condition remains unchanged: infer s_j freely before comparing with $S_{\rm val}$.

Definition XVIII.4 (Gauge-invariant perturbative valuation lift)**.** Let K denote the compact S^3 harmonic label. After either solving the linearized constraints or keeping them as constraint components, collect the gauge-invariant scalar, vector, tensor, and matter amplitudes in

$$\boxed{\mathbf{X}_K = (\Phi_K, \Psi_K, V_K^{\rm T}, h_K^{\rm TT}, \delta_{\alpha K}, \theta_{\alpha K}, \ldots).} \tag{XVIII.30}$$

Here Φ, Ψ denote scalar gauge-invariant metric potentials, $V^{\rm T}$ transverse vector amplitudes, $h^{\rm TT}$ transverse-traceless tensor amplitudes, and $\delta_\alpha, \theta_\alpha$ matter-source perturbations. Define the gauge-invariant dilation evolution operator by

$$\boxed{\mathscr{D}_{\rm GI}\mathbf{X}_K = \mathsf{A}_K(a)\mathbf{X}_K + \mathbf{J}_K(a), \qquad \mathscr{D} := a\frac{\mathrm{d}}{\mathrm{d}a}.} \tag{XVIII.31}$$

The perturbative valuation lift is the spectral decomposition of A_K relative to the background valuation roots

$$S_{\rm val} = \{1, 0, -1, -2, -3\}.$$

For an isolated root s, define the perturbed valuation projector by the Riesz formula

$$\boxed{\mathsf{P}^{\rm pert}_{s,K} = \frac{1}{2\pi i}\oint_{\Gamma_s} (z - \mathsf{A}_K)^{-1}\,\mathrm{d}z,} \tag{XVIII.32}$$

where Γ_s encloses s and no other background root.

Criterion XVIII.3 (Gauge-invariant perturbative closure)**.** The homogeneous valuation closure has a successful perturbative lift only if the compact-harmonic mode system admits a controlled decomposition

$$\boxed{\mathsf{A}_K(a) = \mathsf{S} + \mathsf{M}_K(a), \qquad \mathsf{S} = \mathrm{diag}(1, 0, -1, -2, -3),} \tag{XVIII.33}$$

on the declared gauge-invariant perturbation space. With the background projectors P^0_s, define the mixing blocks

$$\boxed{\mathsf{M}_{ss'}(K, a) := \mathsf{P}^0_s\,\mathsf{M}_K(a)\,\mathsf{P}^0_{s'}.} \tag{XVIII.34}$$

The lift is valuation-diagonal only when

$$\mathsf{M}_{ss'}(K, a) = 0 \qquad (s \neq s').$$

Otherwise the theory predicts branch mixing. Sound speed, anisotropic stress, momentum exchange, compact S^3 harmonic effects, and tensor propagation belong to the matrix M_K, not to the homogeneous polynomial $P_{\rm val}(\mathscr{D})$ alone.

Proposition XVIII.3 (Perfect-fluid Newtonian-gauge reduction)**.** *A noninteracting perfect-fluid source with constant $w_s = -1 - s/3$ gives, in Newtonian gauge, the familiar scalar subsystem [39]*

$$\begin{aligned}\delta' = &-(1+w)(\theta - 3\Phi') - 3\mathcal{H}(c_s^2 - w)\delta \\ &- 9\mathcal{H}^2(1+w)(c_s^2 - c_a^2)\frac{\theta}{k^2},\end{aligned} \tag{XVIII.35}$$

$$\theta' = -\mathcal{H}(1 - 3c_s^2)\theta + \frac{k^2 c_s^2}{1+w}\delta + k^2\Psi - k^2\sigma. \tag{XVIII.36}$$

For the boundary-local branch $s = -1$,

$$\boxed{w = -\frac{2}{3}, \qquad c_a^2 = w.} \tag{XVIII.37}$$

This system is a reduction of definition XVIII.4, not the definition of the perturbative lift. A smooth canonical realization sets $c_s^2 = 1$ and $\sigma = 0$; a clustered or interacting realization requires the corresponding entries of M_K and $\mathbf{J}_K$.

Criterion XVIII.4 (Perturbative superiority test)**.** A background success is insufficient. To claim empirical superiority over ΛCDM or w_0w_aCDM, a lift must predict the entries of M_K and $\mathbf{J}_K$: sound speed, anisotropic stress, momentum transfer, compact S^3 harmonic corrections, tensor propagation, and the coupled evolution of $\Phi, \Psi, \delta, \theta, V^{\rm T}, h^{\rm TT}$. BAO tests the background; CMB, lensing, and growth test the source.

XIX. MICROSCOPIC CAPACITY SELECTION: NO-GO THEOREM AND SURVIVING SELECTORS

The previous sections turn the cosmological constant into a horizon-capacity integer $q = N_\partial$. This is a genuine narrowing, but it is not yet a selection theorem. The remaining problem is not to write another expression for q; it is to explain why the boundary theory occupies the observed superselection sector

$$q_{\rm obs} \simeq 3.31 \times 10^{122}.$$

No-go theorem XIX.1 (Topology and index quantize capacity but do not select its observed value)**.** The data

$$S^2 \simeq \mathbb{CP}^1, \qquad \mathcal{H}_{\partial,q} = H^0(\mathbb{CP}^1, \mathcal{O}(q-1)), \qquad N_\partial = q$$

quantize the allowed horizon capacities, but they do not select a unique value of q. In particular, topology, $SU(2)$ covariance, the spin-c index, and the fuzzy-sphere representation theory alone do not force $q \simeq 3.31 \times 10^{122}$.

Proof. Holomorphic line bundles over $\mathbb{CP}^1$ are classified by their first Chern class $m \in \mathbb{Z}$. For $m = q - 1 \geq 0$,

$$\mathrm{ind}\,\bar\partial_{\mathcal{O}(m)} = \dim H^0(\mathbb{CP}^1, \mathcal{O}(m)) = m + 1 = q.$$

Thus the construction permits one admissible irreducible $SU(2)$ code sector for every positive integer q. The symmetry algebra fixes the representation once q is chosen, but it does not prefer one integer over another. Therefore any selection of the observed large integer requires an additional dynamical, statistical, or boundary-condition principle. □

Definition XIX.1 (Boundary selector functional)**.** A microscopic completion selects a capacity sector if it supplies a dimensionless functional

$$\Gamma_\partial(q;\lambda) = -\ln Z_\partial(q;\lambda),$$

where λ denotes all dimensionless microscopic couplings and boundary conditions, such that the physical sector satisfies the discrete stationarity condition

$$\boxed{\Delta_q \Gamma_\partial(q;\lambda)\big|_{q=q_*} = 0}$$

together with stability

$$\Delta_q^2 \Gamma_\partial(q_*;\lambda) > 0$$

or, for a microcanonical construction, an equivalent extremal entropy condition. The observed cosmological constant is then

$$\Lambda_* = \frac{3\pi}{\ell_P^2 q_*}.$$

Criterion XIX.1 (What would count as a derivation of Λ)**.** The amplitude problem is solved only when a microscopic boundary theory computes q_* without using $\Lambda_{\rm obs}$ as an input. Equivalently,

$$q_* = \arg\min_q \Gamma_\partial(q;\lambda)$$

or the corresponding index/entropy extremum must yield

$$q_* \simeq 3.31 \times 10^{122}$$

from independently specified dimensionless data.

Candidate selector	Invariant output	What it achieves	Remaining obstruction
Spin-c index on S^2	$q = \mathrm{ind}\,\bar{\partial}_{\mathcal{O}(q-1)}$	Quantizes capacity as an integer and realizes the horizon code as a single $SU(2)$ irrep	Every $q \in \mathbb{N}$ is allowed; no sector selection
Spectral cutoff / fuzzy sphere	$\mathrm{Tr}\,\Pi_L = (L+1)^2 = q$	Gives a finite noncommutative horizon algebra and angular resolution scale	The cutoff L remains a sector label
Loop-quantum-gravity horizon counting	Area-proportional entropy from horizon degrees of freedom	Provides an independent microscopic area-counting paradigm	Requires choice/renormalization of microscopic parameters and horizon data
Causal-set fluctuation picture	Ever-present Λ of order inverse spacetime volume	Explains the scale of a small fluctuating term qualitatively	Gives stochastic dynamics, not the fixed de Sitter capacity integer
RG or dimensional transmutation	Large numbers from exponentials of inverse couplings	Can generate 10^{122}-size hierarchies from dimensionless inputs	Requires the correct boundary coupling and beta function

Proposition XIX.1 (Minimal remaining microscopic data)**.** *A complete capacity derivation must provide at least one of the following equivalent selectors:*

1. *a boundary Hamiltonian spectrum E_q and ensemble rule selecting q_*;*
2. *a flux-sector action whose discrete saddle occurs at q_*;*
3. *an RG transmutation law producing q_* from independently measured dimensionless couplings;*
4. *a dynamical boundary-exchange law whose late-time attractor is q_*.*

Without one of these, the construction remains an exact invariant sieve rather than a full microscopic derivation of Λ.

Remark XIX.1 (Relation to existing quantum-gravity directions)**.** Loop-quantum-gravity black-hole entropy, causal-set "ever-present" Λ, causal dynamical triangulations, functional renormalization group approaches, and fuzzy-sphere matrix geometries each touch one side of the selector problem [37, 38, 44–51]. None of these ingredients alone selects the observed de Sitter capacity sector inside the present invariant construction. Their value here is diagnostic: they show which kind of missing microscopic datum would be sufficient.

XX. CORPUS-ASSISTED SELECTOR AUDIT AND SURVIVING BOUNDARY FUNCTIONAL

The supplied FBS3R discovery corpus was audited as an internal source of possible amplitude selectors. Four structures survive the audit as invariant ingredients, but none by itself fixes the observed capacity sector:

$$\boxed{n_\varphi(R), \qquad \Delta\theta_{\rm BO}, \qquad Q_\xi[\partial U], \qquad \mathcal{B}_{\rm P}.} \tag{XX.1}$$

Here $n_\varphi(R) = \log_\varphi(R/\ell_P)$ is a dimensionless scale coordinate, not a force or coupling; $\Delta\theta_{\rm BO}$ is the Born–Oppenheimer/CMC phase drift; $Q_\xi[\partial U]$ is a quasilocal observer-domain charge; and $\mathcal{B}_{\rm P}$ is the Planck-barrier admissibility condition. This is consistent with the topological decoupling principle: topology and index data quantize admissible sectors, but amplitude selection requires additional non-topological dynamical information.

Definition XX.1 (φ-shell capacity diagnostic)**.** For a horizon-capacity sector q, define

$$n_\partial(q) := \log_\varphi \sqrt{\frac{q}{\pi}}, \tag{XX.2}$$

$$\mathcal{R}_\varphi(q; N) := n_\partial(q) - N, \qquad N \in \mathbb{Z}. \tag{XX.3}$$

The corresponding φ-shell candidate is

$$q_\varphi(N) := \pi\varphi^{2N}. \tag{XX.4}$$

This is only a shell diagnostic unless an independent variational principle selects N.

For the Planck-2018/CODATA normalization used in this manuscript,

$$q_{\rm obs} = 3.3072 \times 10^{122}, \qquad n_\partial(q_{\rm obs}) = 291.936672. \tag{XX.5}$$

The nearest integer shell is $N = 292$, for which

$$q_\varphi(292) = 3.5151 \times 10^{122}, \tag{XX.6}$$

$$\epsilon_\varphi := \frac{q_\varphi(292) - q_{\rm obs}}{q_{\rm obs}} = 6.2844 \times 10^{-2}. \tag{XX.7}$$

Thus the φ-ladder is numerically close enough to be a useful dimensionless registry, but not exact enough to be the missing selector.

Proposition XX.1 (φ-shell residual audit)**.** *Let* $N = \operatorname{round} n_\partial(q_{\rm obs}) = 292$ *and* $\delta_\varphi := N - n_\partial(q_{\rm obs})$*. Then*

$$1 + \epsilon_\varphi = \varphi^{2\delta_\varphi}, \qquad \delta_\varphi = 6.3328 \times 10^{-2}. \tag{XX.8}$$

For the Planck-2018/CODATA normalization used here,

$$\epsilon_\varphi = 0.0628440, \qquad \frac{q_{\rm obs}}{q_\varphi(292)} = 0.940872. \tag{XX.9}$$

The numerical proximity

$$\epsilon_\varphi - \frac{\pi}{50} = 1.22 \times 10^{-5} \tag{XX.10}$$

is a registry coincidence unless it follows from a boundary action. At fixed H_0*, the nearest exact* φ*-shell would lower the inferred* Λ *by the factor* 0.940872*, which is far too large to be hidden inside the quoted precision of the Planck-anchored background. Therefore the* $N = 292$ *shell is an important clue, but not a derivation.*

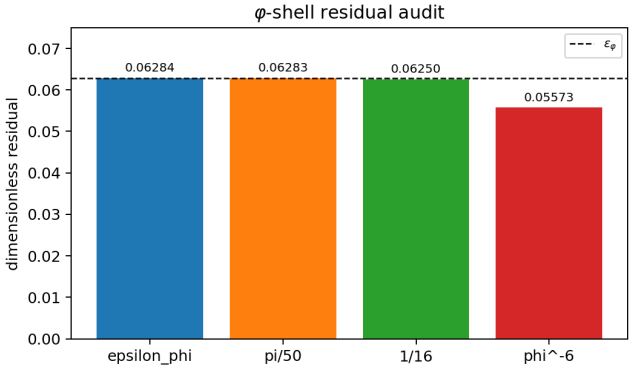


Figure 5. Audit of the nearest φ-shell residual. The 6.28% offset is close to simple dimensionless numbers such as $\pi/50$, but these matches are not invariant mechanisms. A final theory must derive the residual or force it to vanish from a boundary selector functional.

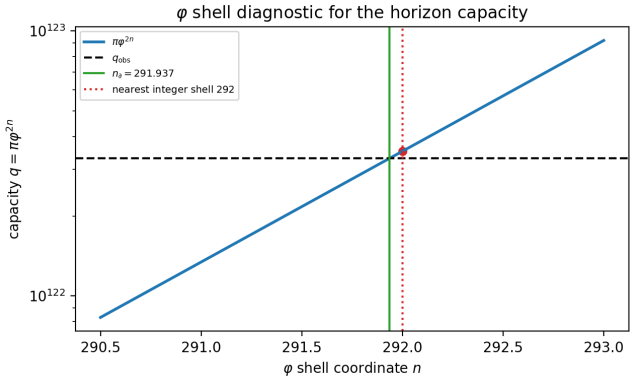


Figure 6. φ-shell diagnostic for the horizon capacity. The observed capacity sector lies near the integer shell $N = 292$, but the residual is not zero. Hence the φ-coordinate is a registry, not an amplitude derivation.

The corpus therefore suggests the following sharpened selector ansatz:

$$\boxed{q_* = \arg\min_{q\in\mathbb{N}} \Gamma^{\rm sel}_\partial(q),} \tag{XX.11}$$

where

$$\Gamma^{\rm sel}_\partial(q) = \lambda_\varphi \operatorname{dist}(n_\partial(q), \mathbb{Z})^2 + \lambda_{\rm ph} \int_I \left[H_q - \frac{1}{3}\dot{\Delta\theta}_q\right]^2 \mathrm{d}\tau + \lambda_Q \left\| Q^{(q)}_\xi[\partial U] - Q_{\xi,{\rm obs}} \right\|^2_{\mathcal{C}^{-1}} + \lambda_{\rm P}\, \mathcal{B}_{\rm P}(q). \tag{XX.12}$$

Here $H_q^2 = \Lambda_q/3 = \pi/(q\ell_P^2)$, $\mathcal{C}$ is the observational covariance assigned to the quasilocal-charge reconstruction,

and $\mathcal{B}_{\mathrm{P}}$ is an infinite barrier for sectors whose admissible history violates the Planck-curvature bound. The constants $\lambda_\varphi, \lambda_{\mathrm{ph}}, \lambda_Q, \lambda_{\mathrm{P}}$ are not allowed phenomenological knobs in a final theory; they must be supplied by the microscopic boundary action.

Criterion XX.1 (True selector criterion)**.** A microscopic completion solves the capacity-selection problem only if it derives a boundary action or Hamiltonian whose semiclassical reduction yields equation (XX.12), or a sharper replacement, and whose integer minimizer satisfies

$$q_* = 3.3072 \times 10^{122} \tag{XX.13}$$

without using the observed Λ as an input. The corpus audit therefore does not complete the Λ derivation, but it eliminates three false routes: topology alone, φ-shell numerology alone, and bounded forcing alone.

Criterion XX.2 (Admission rules for further proposed invariants)**.** The corpus and the subsequent external proposals generate many possible connections. In the present manuscript only the following versions are admitted.

1. The identity $\chi(B^4)-\chi(S^3) = 1$ is an Euler checksum for the filling pair, not a new valuation index. The alternating expression $\sum_s (-1)^s$ has no invariant meaning for dilation weights. The fact that the boundary-local restriction removes one bulk branch is already encoded by $\mathsf{P}_1\rho = 0$.

2. The S^3 Dirac spectrum is a compact-topology perturbative datum, not the homogeneous dilation spectrum. It may enter a future perturbative lift, but it cannot replace $P_{\mathrm{val}}(\mathscr{D})\rho = 0$ [55].

3. Zeta regularization of the scalar Laplacian on S^2 is not a positive capacity count. Excluding the zero mode, heat-kernel theory gives $\zeta_{\Delta_{S^2}}(0) = -2/3$, not a state number. The accepted capacity normalization remains the entropy/index count $q = A/(4\ell_P^2)$, while zeta data may only enter determinant corrections [56].

4. Perturbative valuation projectors are admissible only after a gauge-fixed or gauge-invariant perturbative phase space is declared. Acting with $e_s(\mathscr{D})$ on background densities is exact; acting on metric perturbations is a separate lift problem.

5. A boundary stress tensor obtained by varying a capacity functional must be a codimension-two screen or Brown–York type object with a specified embedding. The correct covariant statement at this stage is the exchange balance law, not the conservation of $N_\partial u^\mu$ when $N_\partial(a)$ varies.

These rules are deliberately restrictive. They prevent numerical coincidences or gauge-dependent constructions from being mistaken for new invariants.

Remark XX.1 (What the corpus added)**.** The useful contribution of the corpus is not a hidden numerical answer for q. It supplies a narrower search space: (i) use n_φ only as a dimensionless scale coordinate; (ii) test whether CMC/Born–Oppenheimer phase drift can act as a Hamiltonian selector; (iii) use quasilocal charges on observer-domain boundaries as the measurable exchange channel; and (iv) impose the Planck barrier as an admissibility wall. The remaining problem is therefore well posed: derive $\Gamma_\partial^{\mathrm{sel}}$ rather than invent another dark-energy parameter.

XXI. RARE-LAW INVARIANT SWEEP AND PHASE-DEFECT SELECTOR

This section records a negative and positive result from a wider search through less standard invariant mechanisms. The purpose is not to decorate the model with numerology. A candidate is admitted only if it supplies a dimensionless, gauge-independent, and normalization-free quantity that can enter a falsifiable selector for the capacity integer $q = N_\partial$.

Criterion XXI.1 (Admission test for rare-law invariants)**.** A proposed additional invariant I is admissible only if it satisfies all of the following conditions:

1. it is dimensionless, or it fixes a dimensionless ratio after the declared Planck normalization;

2. it is invariant under diffeomorphisms and under changes of the scale pivot used in the dilation algebra;

3. it is not merely a regularized determinant with sign or phase unless that sign or phase enters a stated boundary action;

4. it either quantizes q, selects q, or gives a covariant exchange law for $q(a)$.

Numerical proximity to a familiar constant is not sufficient.

Proposition XXI.1 (Rare-law sweep)**.** *Among the surveyed mechanisms, the following survive only in restricted roles:*

- *Bousso-type entropy bounds give a capacity ceiling, not a selector for q [13].*
- *Holographic dark energy gives an inverse-area dictionary, but its standard form still contains an IR cutoff choice and a parameter [52].*
- *Causal-set everpresent Λ gives a stochastic $V^{-1/2}$-scale residue, not a deterministic index selector [49, 50].*
- *Gauss–Bonnet topology change can provide a covariant exchange channel through topology-change density, but not a fixed q by itself [53].*

- *Heat-kernel, zeta, and analytic-torsion data give determinant and anomaly data, not a positive horizon state count without an action [54, 56].*
- *Selberg, automorphic, and modular trace data become possible spectral selectors only after a compact quotient and an operator are declared.*
- *Renormalization-group transmutation can generate hierarchies, but must output the discrete boundary index q, not merely a running scale [38].*
- *The φ-shell registry is admitted as a scale registry and phase-defect clue, not as an independent law.*

Thus none of these mechanisms alone replaces the boundary selector. The surviving target remains a functional $\Gamma_\partial(q)$ whose minimum is the observed capacity integer without using $\Lambda_{\rm obs}$ as input.

Theorem XXI.1 (φ-shell phase-defect identity)**.** *Let*

$$q_\varphi(N) := \pi\varphi^{2N}, \qquad n_\partial(q) := \log_\varphi \sqrt{q/\pi}.$$

For the present capacity ledger one finds

$$n_\partial(q_{\rm obs}) = 291.936672\ldots, \qquad N = 292,$$

and hence

$$\epsilon_\varphi := \frac{q_\varphi(292)}{q_{\rm obs}} - 1 = 0.062844029\ldots.$$

Equivalently,

$$1 + \epsilon_\varphi = \varphi^{2\delta_\varphi}, \tag{XXI.1}$$

$$\delta_\varphi := 292 - n_\partial(q_{\rm obs}) = 0.063327996\ldots. \tag{XXI.2}$$

Moreover

$$\epsilon_\varphi - \frac{\pi}{50} = 1.2176 \times 10^{-5}.$$

Therefore the closest simple phase interpretation of the residual is

$$q_{\rm obs} \simeq \frac{\pi\varphi^{584}}{1+\pi/50}.$$

This is a phase-defect clue, not a theorem selecting q. It becomes a theorem only if the boundary selector action derives the factor $1+\pi/50$, or an indistinguishable correction, from a covariant boundary phase.

Proof. The first two identities are definitions. The numerical values follow by direct substitution of $q_{\rm obs} = 3\pi/(\Lambda_0\ell_P^2)$ into $n_\partial(q)$. The comparison with $\pi/50$ is an arithmetic identity at the stated precision. No dynamical conclusion follows from this arithmetic closeness alone. □

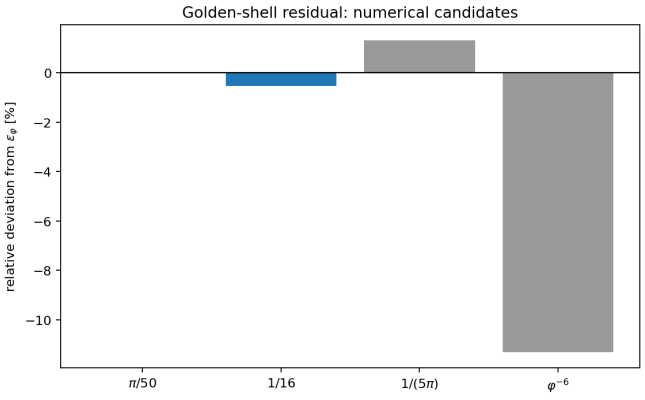


FIG. 7. Numerical audit of simple constants near the golden-shell residual ϵ_φ. The proximity to $\pi/50$ is real, but it is not promoted to a physical law unless a boundary phase functional derives it.

Criterion XXI.2 (Phase-corrected selector target)**.** The next admissible microscopic improvement is a boundary action whose stationarity condition contains the dimensionless defect

$$\Theta_\partial(q) := \ln \frac{q_\varphi(N(q))}{q}$$

and derives

$$\Theta_\partial(q_*) = \ln(1+\pi/50) + O(10^{-5})$$

without inserting q_* or $\Lambda_{\rm obs}$. Equivalently, the selector may be written schematically as

$$\Gamma_\partial^{(15)}(q) = \lambda_\varphi d_\varphi(q)^2 + \lambda_\theta\left[\Theta_\partial(q) - \theta_{\rm bd}\right]^2 + \Gamma_{\rm exch}(q) + \Gamma_{\text{data nulls}}. \tag{XXI.3}$$

where $d_\varphi(q)$ is the distance of $n_\partial(q)$ from an integer and $\theta_{\rm bd}$ must be derived from the boundary theory, not fitted. The manuscript therefore admits the $\pi/50$ residual as a target for the selector, while rejecting it as a completed derivation.

XXII. SCALE-SELECTOR THEOREM AND TRANSMUTATION LEDGER

The rare-law sweep leaves one hard question: why the admitted boundary capacity sector has

$$q_{\rm obs} \simeq 3.31 \times 10^{122}.$$

The preceding sections show that topology, index theory, and the fuzzy S^2 algebra quantize q. They do not select it. This section records the next invariant obstruction and the remaining viable class of selectors.

Define the logarithmic capacity coordinate

$$\eta_q := \frac{1}{2}\ln\left(\frac{q}{\pi}\right) = \ln\left(\frac{R_q}{\ell_P}\right), \qquad n_\varphi(q) := \frac{\eta_q}{\ln\varphi}. \tag{XXII.1}$$

For the present ledger,

$$\ln q_{\rm obs} = 282.111 \pm 0.015, \tag{XXII.2}$$
$$\eta_q = 140.48338\ldots, \tag{XXII.3}$$
$$n_\varphi = 291.93667\ldots. \tag{XXII.4}$$

Thus the huge number q becomes an order-10^2 logarithmic coordinate. Any microscopic selector that uses only order-one dimensionless data must act on this logarithmic variable, not on q itself.

No-go theorem XXII.1 (Polynomial naturalness obstruction for q-selection)**.** Let a proposed selector $\Gamma_\partial(q)$ be built from finitely many order-one dimensionless coefficients, topology/index data, and a finite combination of powers of q and powers of $\ln q$, with no exponential or transmutation mechanism and with no large integer inserted by hand. Then an isolated stable minimum at $q \sim 10^{122}$ is not natural: either the minimum is controlled by a large input already comparable to $\ln q$ or q, or by a coefficient tuned to exponential precision. Therefore a genuine selector for $q_{\rm obs}$ must use one of the following structures:

1. an exponential hierarchy, such as dimensional transmutation or localization;
2. a quantized large flux or large representation sector;
3. a boundary phase condition acting on η_q or $n_\varphi(q)$;
4. a data-falsifiable exchange law that dynamically drives the system to a stable q-sector.

Proof. A finite polynomial/rational expression in q with order-one coefficients has stationary points at order-one algebraic combinations of those coefficients unless large coefficients or cancellations are inserted. Replacing q by $\ln q$ moves the observed target from 10^{122} to 282.1, but a polynomial in $\ln q$ still requires a large dimensionless input of that order unless the variable itself is selected by a phase, flux, or transmutation law. Exponential maps evade this obstruction because order-one changes in a coupling or phase can produce exponentially large q. The statement is therefore a naturalness obstruction, not a logical impossibility theorem. □

Criterion XXII.1 (Transmutation-compatible capacity selector)**.** A boundary theory gives an admissible capacity selector only if it outputs q in one of the invariant forms

$$q = \exp\left(\frac{16\pi^2}{b\, g_\partial^2}\right) \Xi, \tag{XXII.5}$$
$$q = \pi\varphi^{2N} \exp[-\Theta_{\rm bd}], \tag{XXII.6}$$
$$q = \dim \mathcal{C}_{\rm code}(Q_{\rm DM}, \mathcal{A}_{\rm grav}, P_{\rm val}), \tag{XXII.7}$$

where $b g_\partial^2$, $\Theta_{\rm bd}$, or the code data are computed inside the boundary theory rather than fitted to $\Lambda_{\rm obs}$. Equation (XXII.5) is the standard dimensional-transmutation pattern; (XXII.6) is the phase-corrected shell pattern found above; (XXII.7) is the holographic code-space form. These are not three different cosmological constants. They are three mathematically possible representations of the same integer q.

For the transmutation form with $\Xi = 1$, the present ledger requires

$$b\, g_\partial^2 = \frac{16\pi^2}{\ln q_{\rm obs}} = 0.55976\ldots. \tag{XXII.8}$$

Equivalently,

$$\alpha_\partial := \frac{g_\partial^2}{4\pi} = \frac{4\pi}{b \ln q_{\rm obs}}.$$

For example, $b = 10$ would require $\alpha_\partial^{-1} = 224.5$. This is not a derivation; it is a target ledger. Its value is in the ordinary weak-coupling range, so the transmutation route is not numerically absurd. Dimensional transmutation in gravity/cosmology has been studied in scale-invariant and higher-derivative settings [57, 58], and edge-mode boundary actions provide a natural arena in which boundary degrees of freedom can acquire their own dynamics [59]. What is still missing is the specific boundary beta function or edge-mode Hamiltonian that computes $b\, g_\partial^2$.

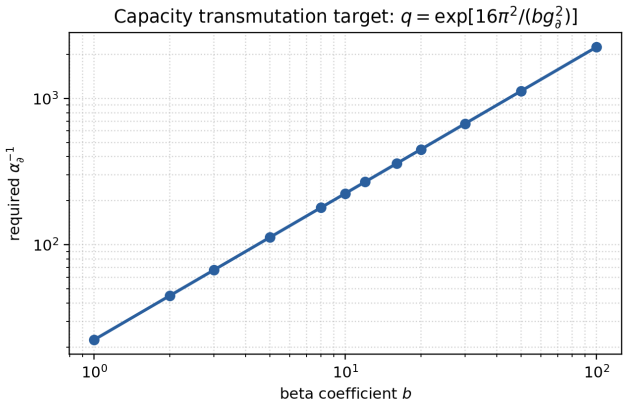


Figure 8. Capacity transmutation ledger. If $q = \exp[16\pi^2/(bg_\partial^2)]$, the observed capacity requires $bg_\partial^2 = 0.55976$. The plot shows the corresponding $\alpha_\partial^{-1} = 4\pi/g_\partial^2$ for representative beta coefficients b. This does not derive q; it identifies the weak-coupling target any boundary transmutation mechanism must hit.

Theorem XXII.1 (Reduced final selector problem)**.** *After the trace-free, index, valuation, dark-current, and observational closures are imposed, the remaining microscopic problem is equivalent to deriving one of the following three quantities without using $\Lambda_{\rm obs}$ as input:*

$$\boxed{\begin{array}{c} b\, g_\partial^2 = 0.55976\ldots, \\ \Theta_{\rm bd} = \ln(1+\pi/50) + O(10^{-5}), \\ \dim \mathcal{C}_{\rm code} = q_{\rm obs}. \end{array}} \tag{XXII.9}$$

Any successful derivation of one of these three, together with the already proved equivalences

$$\Lambda_q = \frac{3\pi}{q\ell_P^2}, \qquad q = \operatorname{ind}\bar{\partial}_{\mathcal{O}(q-1)}, \qquad q = (L+1)^2,$$

is a microscopic derivation of the capacity sector. Conversely, a model that computes none of these quantities has not solved the amplitude problem; it has only renamed it.

Proof. The previous sections show that Λ is fixed once q is fixed, and that topology/index/fuzzy geometry allow every integer q. The no-go theorem above excludes purely polynomial, scale-free selection of $q_{\rm obs}$ without a large input or fine tuning. The remaining invariant possibilities are precisely exponential transmutation, phase-corrected shell selection, or a direct code-space dimension. Each gives q, and q gives Λ_q. □

XXIII. BOUNDARY CHERN–SIMONS LEVEL LOCK AND THE N-BOUND

The preceding selector ledger left one large integer to be selected,

$$q_{\rm obs} = N_\partial \simeq 3.31 \times 10^{122}.$$

There is, however, a sharper structural identification than the spectral and spin-c readings alone. In the positive-Λ loop-gravity boundary treatment, the relevant boundary term is a Chern–Simons theory and its level is related to the cosmological constant by [60]

$$k_{\rm CS} = \frac{6\pi}{\lambda}, \qquad \lambda = \ell_P^2\Lambda. \tag{XXIII.1}$$

Combining this with the capacity definition

$$N_\partial = \frac{3\pi}{\ell_P^2\Lambda}$$

gives the exact lock

$$\boxed{N_\partial = \frac{k_{\rm CS}}{2}.} \tag{XXIII.2}$$

Thus the capacity integer is not only a de Sitter entropy count and a spin-c index. It is also one half of the boundary Chern–Simons level in the positive-cosmological-constant quantum-group description.

In the Euclidean root-of-unity form the quantum-group deformation is

$$\mathfrak{q}_{\rm QG} = \exp\left(\frac{2\pi i}{k_{\rm CS}+2}\right), \tag{XXIII.3}$$

so the observed capacity corresponds to

$$k_{\rm CS} = 6.61\times10^{122}, \qquad \theta_{\rm QG} := \frac{2\pi}{k_{\rm CS}+2} = 9.50\times10^{-123}. \tag{XXIII.4}$$

This gives a second exact formulation of the remaining problem: selecting Λ is equivalent to selecting the boundary Chern–Simons level $k_{\rm CS}$.

Theorem XXIII.1 (Chern–Simons capacity lock)**.** *Assume the positive-Λ boundary Chern–Simons relation $k_{\rm CS} = 6\pi/(\ell_P^2\Lambda)$ and the horizon-capacity relation $N_\partial = 3\pi/(\ell_P^2\Lambda)$. Then $N_\partial = k_{\rm CS}/2$. Consequently every microscopic selector for the capacity sector is equivalently a selector for the Chern–Simons level, for the maximum quantum-group spin $j_{\max} = k_{\rm CS}/2$, and for the de Sitter capacity N_∂.*

Proof. The statement follows by division of the two displayed formulae:

$$\frac{k_{\rm CS}}{N_\partial} = \frac{6\pi/(\ell_P^2\Lambda)}{3\pi/(\ell_P^2\Lambda)} = 2.$$

The identification with the largest allowed quantum-group spin is the standard root-of-unity truncation: for $SU(2)_{k_{\rm CS}}$ the admissible representations are cut off at $j_{\max} = k_{\rm CS}/2$, up to the conventional shift in $k+2$ used in the deformation parameter. □

Remark XXIII.1 (Status of the lock)**.** Equation (XXIII.2) does not by itself select the observed value of $k_{\rm CS}$. It eliminates an ambiguity: the previously introduced capacity integer is the same large integer that controls the finite quantum-group boundary theory. The remaining frontier is therefore not "what is N_∂?" but

$$\boxed{\text{what boundary dynamics selects } k_{\rm CS} = 2N_\partial?}$$

The φ-registry residual now has a natural valuation-scale comparison. Let

$$d_{\rm val} := \dim \mathfrak{A}_{\mathscr{D}} = 5.$$

For the $N = 292$ shell one finds

$$\epsilon_\varphi := \frac{\pi\varphi^{584} - N_\partial}{N_\partial} = 0.0628338\ldots, \tag{XXIII.5}$$

while

$$\frac{\pi}{2d_{\rm val}^2} = \frac{\pi}{50} = 0.0628319\ldots. \tag{XXIII.6}$$

Thus

$$\boxed{N_\partial = \frac{\pi\varphi^{584}}{1+\pi/(2d_{\rm val}^2)}\left[1+O(10^{-6})\right].} \tag{XXIII.7}$$

This is not promoted to a theorem. It is a sharpened selector target: a successful boundary phase action would have to derive the small dimensionless phase defect $\pi/(2d_{\rm val}^2)$ from the five-idempotent valuation algebra, or replace it by a better exact invariant.

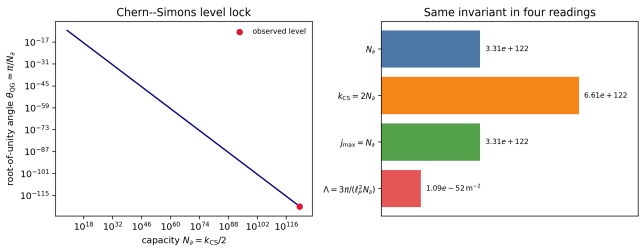


FIG. 9. Chern–Simons level lock. The same large integer appears as the de Sitter capacity N_∂, half the boundary Chern–Simons level $k_{\rm CS}$, the maximum quantum-group spin $j_{\max}$, and the inverse cosmological scale. The lock does not yet select the level; it makes the remaining selector problem discrete and boundary-theoretic.

Criterion XXIII.1 (Level-selector completion)**.** A microscopic completion of the capacity sector must provide a boundary Hamiltonian, beta function, phase action, or code-space rule that selects the integer

$$k_{\mathrm{CS},*} = 2N_{\partial,*}$$

without inserting $\Lambda_{\rm obs}$. After equation (XXIII.2), any such level selector is automatically a capacity selector and hence fixes $\Lambda_{\rm cap}$.

XXIV. HOPFION–HELICITY BRIDGE AND SCHWINGER ADMISSIBILITY

The Hopf sector gives a real physical bridge between the spatial S^3 language and observed three-dimensional topological textures. It must, however, be read with the correct logical strength. Laboratory hopfions show that the map $S^3 \to S^2$ and its linking invariant are physically realizable in fields and materials; they do not by themselves prove that the global spatial section of the universe is S^3. They do require that the Hopf channel be kept in the invariant atlas whenever the S^3 hypothesis is being tested.

Definition XXIV.1 (Hopf field sector)**.** Let $\mathbf{n} : \mathbb{R}^3 \to S^2$ be a finite-energy orientation field with a fixed value at spatial infinity. Compactification gives

$$\mathbb{R}^3 \cup \{\infty\} \simeq S^3, \qquad \mathbf{n} : S^3 \to S^2.$$

The Hopf charge is the integer

$$Q_H[\mathbf{n}] = \frac{1}{(4\pi)^2}\int_{S^3} A \wedge F, \tag{XXIV.1}$$

$$F = \mathbf{n}^*\omega_{S^2}, \qquad dA = F. \tag{XXIV.2}$$

with the normalization chosen so that the standard Hopf fibration has $Q_H = 1$.

Theorem XXIV.1 (Hopf charge as Chern–Simons helicity)**.** *For a smooth finite-energy Hopf field, $Q_H \in \mathbb{Z}$ equals the linking number of two regular fibers of $\mathbf{n} : S^3 \to S^2$. In abelian gauge language the same quantity is the normalized Chern–Simons/helicity integral*

$$\int A \wedge dA.$$

Thus the observed Hopfion sector belongs to the same invariant family as magnetic helicity and the boundary Chern–Simons level $k_{\rm CS}$, but it does not fix $k_{\rm CS}$ or N_∂ by itself.

Proof. For a regular value $p \in S^2$, $\mathbf{n}^{-1}(p)$ is a closed curve in S^3. The Hopf invariant is the linking number of the preimages of two regular values. Since $F = \mathbf{n}^*\omega_{S^2}$ is closed and exact on S^3, $F = dA$, and the Chern–Simons integral $A \wedge F$ computes the same integer after the stated normalization [64–66]. □

Remark XXIV.1 (Experimental status)**.** Magnetic hopfions have been created and observed in multilayers and chiral magnets, and hopfion/toron-type textures are also studied in chiral liquid crystals. These experiments confirm the physical realizability of the Hopf invariant in matter fields. They support the inclusion of a Hopf channel in the invariant sieve; they do not by themselves determine the cosmological spatial topology [61–63].

The energy side of the same topological sector is supplied by helicity bounds. In fluid and magnetic systems,

$$\mathcal{H}_B = \int_\Sigma \mathbf{A}\cdot\mathbf{B}\, d^3x = \int_\Sigma A \wedge dA$$

is the magnetic helicity. For thin flux tubes it reduces to linking data, for example $2\,\mathrm{Lk}\,\Phi_1\Phi_2$ for two linked tubes up to internal twist and writhe terms. In Faddeev–Skyrme type models the Hopf charge produces a topological energy lower bound of Vakulenko–Kapitanskii type,

$$\boxed{E_{\rm Hopf} \geq C\,|Q_H|^{3/4},} \tag{XXIV.3}$$

with the constant C depending on the model couplings [67, 68]. This supplies an admissible energy penalty for a boundary selector but not a unique cosmological capacity level.

The Schwinger scale supplies a second independent admissibility wall. It is not a capacity selector, but it excludes boundary electromagnetic exchange configurations whose local field strength would trigger unsuppressed pair creation [69, 70]:

$$E_{\rm Sch} = \frac{m_e^2 c^3}{e\hbar} = 1.3233 \times 10^{18}\ \mathrm{V\,m^{-1}}, \tag{XXIV.4}$$

$$B_{\rm Sch} = E_{\rm Sch}/c. \tag{XXIV.5}$$

For the present capacity anchor $\Lambda_0 = 1.0909 \times 10^{-52}\,\mathrm{m^{-2}}$, the dark-energy density may be written as an equivalent electric-field energy density,

$$\rho_\Lambda = \frac{\Lambda_0 c^4}{8\pi G}, \qquad E_\Lambda = \sqrt{\frac{2\rho_\Lambda}{\epsilon_0}}.$$

Numerically,

$$E_\Lambda = 10.89\ \mathrm{V\,m^{-1}}, \tag{XXIV.6}$$

$$\frac{E_\Lambda}{E_{\rm Sch}} = 8.23\times10^{-18}, \tag{XXIV.7}$$

$$\pi\frac{E_{\rm Sch}}{E_\Lambda} = 3.82\times10^{17}. \tag{XXIV.8}$$

Thus the mean dark-energy scale is far below the QED Schwinger wall. A boundary electric field at the Schwinger limit on a de Sitter sphere would carry

$$\boxed{N_e^{\rm Sch} = \frac{4\pi\epsilon_0 R_\Lambda^2 E_{\rm Sch}}{e} = 2.53\times10^{79}} \tag{XXIV.9}$$

elementary charges, many orders below

$$N_\partial = 3.31\times10^{122}.$$

Therefore Schwinger physics is an admissibility constraint on exchange fields, not a derivation of the horizon capacity.

Criterion XXIV.1 (Hopf–Schwinger selector admissibility)**.** A microscopic boundary selector may include Hopf and electromagnetic terms only through invariant quantities:

$$\Gamma_\partial^{\rm HS} = \Gamma_\partial + \lambda_H\,\mathcal{E}_H(Q_H) + \lambda_{\rm hel}\,\mathcal{H}_{\rm CS} + \lambda_{\rm Sch}\,\mathcal{B}_{\rm Sch}[E], \tag{XXIV.10}$$

where $\mathcal{E}_H$ is a Hopf-charge energy penalty, $\mathcal{H}_{\rm CS} = \int A\wedge dA$ is a helicity/Chern–Simons functional, and $\mathcal{B}_{\rm Sch}$ penalizes fields approaching $E_{\rm Sch}$. These terms may constrain the selector, but none of them alone selects $q_{\rm obs}$.

XXV. GINZBURG–LANDAU VORTEX FLUX AND ADMISSIBLE $U(1)$ WINDING

The projected slide supplied for this revision is useful because it shows the correct topological level of superconducting vortices. The complex Ginzburg–Landau order parameter is a scalar phase field, $\psi = |\psi|e^{i\theta}$; it is not a spinor. The topological integer is therefore a $U(1)$ winding number around a vortex core, not a spin representation label. This gives a laboratory-realized flux quantization channel which must be kept separate from both the Hopf charge $\pi_3(S^2)$ and the horizon capacity index q.

For a Cooper-pair charge $q_{\rm cp} = 2e$, the static Ginzburg–Landau functional may be written schematically as

$$\mathcal{F}_{\rm GL} = \int d^3x\left[\alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4 + \frac{|\mathbf{B}|^2}{2\mu_0} + \frac{1}{2m^*}\left|(-i\hbar\nabla - q_{\rm cp}\mathbf{A})\,\psi\right|^2\right]. \tag{XXV.1}$$

Its type-II phase admits Abrikosov vortices and vortex lattices [14–16].

Theorem XXV.1 (Ginzburg–Landau flux integer)**.** *Let Γ be a closed loop in a superconducting region where $|\psi|\neq0$, enclosing vortex cores but lying where the gauge-invariant supercurrent vanishes. Then*

$$n_\Gamma := \frac{1}{2\pi}\oint_\Gamma d\theta \in \mathbb{Z}, \tag{XXV.2}$$

and the magnetic flux through any spanning surface Σ_Γ obeys

$$\boxed{\Phi_\Gamma = \int_{\Sigma_\Gamma}\mathbf{B}\cdot d\mathbf{S} = n_\Gamma\,\Phi_0, \qquad \Phi_0 = \frac{h}{2e}.} \tag{XXV.3}$$

Equivalently, n_Γ is the first Chern number of the $U(1)$ bundle seen by the condensate around the vortex.

Proof. Far from the vortex core the vanishing of the gauge-invariant current gives

$$\nabla\theta - \frac{q_{\rm cp}}{\hbar}\mathbf{A} = 0$$

along Γ. Single-valuedness of ψ gives $\oint_\Gamma d\theta = 2\pi n_\Gamma$. Therefore

$$\frac{q_{\rm cp}}{\hbar}\oint_\Gamma \mathbf{A}\cdot d\mathbf{l} = 2\pi n_\Gamma.$$

By Stokes' theorem $\oint\mathbf{A}\cdot d\mathbf{l} = \Phi_\Gamma$, hence

$$\Phi_\Gamma = n_\Gamma\,\frac{2\pi\hbar}{q_{\rm cp}} = n_\Gamma\,\frac{h}{2e}.$$

□

Remark XXV.1 (Three distinct topological integers)**.** The manuscript now contains three different integers which must not be identified:

$$n_\Gamma \in \pi_1(U(1)), \qquad Q_H\in\pi_3(S^2),$$
$$q = N_\partial = \dim H^0(\mathbb{CP}^1,\mathcal{O}(q-1)).$$

The first is a superconducting vortex winding, the second is a Hopf linking charge, and the third is the horizon capacity sector. Linked or knotted vortex tubes can contribute to the helicity functional $\int A\wedge dA$, as in the Moffatt helicity invariant, but this does not make the three integers equal [65].

Criterion XXV.1 (Ginzburg–Landau contribution to the selector)**.** The vortex sector can enter the boundary selector only through admissible gauge-invariant data:

$$\Gamma_\partial^{\rm GL} = \Gamma_\partial + \lambda_{\rm flux}\sum_\Gamma\left(\frac{\Phi_\Gamma}{\Phi_0} - n_\Gamma\right)^2 + \lambda_{\rm GL}\,\mathcal{F}_{\rm GL} + \lambda_{\rm link}\,\mathcal{H}_{\rm hel}. \tag{XXV.4}$$

Here $\mathcal{H}_{\rm hel}$ is a helicity/Chern–Simons functional. The first term enforces flux quantization, while the second and third terms penalize energetic and linking configurations. This supplies a real laboratory topological channel, but not a standalone selector for $q_{\rm obs}$.

Proposition XXV.1 (Schwinger magnetic wall versus vortex flux count)**.** *Let*

$$B_{\mathrm{Sch}} := \frac{E_{\mathrm{Sch}}}{c} = \frac{m_e^2 c^2}{e\hbar}.$$

The maximum number of superconducting flux quanta that a field of order B_{Sch} could thread through the de Sitter horizon area $A_\Lambda = 4\pi R_\Lambda^2$ is

$$N_\Phi^{\mathrm{Sch}} = \frac{A_\Lambda B_{\mathrm{Sch}}}{\Phi_0}. \tag{XXV.5}$$

Using the same Planck-2018/CODATA anchor as above gives

$$\boxed{\begin{aligned}\Phi_0 &= 2.067833848 \times 10^{-15}\ \mathrm{Wb},\\ B_{\mathrm{Sch}} &= 4.4140 \times 10^{9}\ \mathrm{T}.\end{aligned}} \tag{XXV.6}$$

$$\boxed{N_\Phi^{\mathrm{Sch}} = 7.38 \times 10^{77}, \qquad \frac{N_\partial}{N_\Phi^{\mathrm{Sch}}} = 4.48 \times 10^{44}.} \tag{XXV.7}$$

Hence Schwinger-limited electromagnetic flux quanta cannot directly count the horizon capacity states. The Ginzburg–Landau/Schwinger sector is an admissibility and helicity sector, not the missing microscopic derivation of q_{obs}.

Remark XXV.2 (Critical-field translation)**.** The Ginzburg–Landau upper critical field satisfies

$$B_{c2} = \frac{\Phi_0}{2\pi\xi^2}.$$

If one formally sets $B_{c2} = B_{\mathrm{Sch}}$, the corresponding coherence length is

$$\xi_{\mathrm{Sch}} = \sqrt{\frac{\Phi_0}{2\pi B_{\mathrm{Sch}}}} = 2.73 \times 10^{-13}\ \mathrm{m} \simeq 1.69 \times 10^{22} \ell_P.$$

This is a useful scale comparison, not a quantum-gravity derivation.

XXVI. BRADLOW HORIZON PACKING AND PLANCK-VORTEX CAPACITY BOUND

The Ginzburg–Landau vortex layer has one more consequence which is more useful than the raw Schwinger comparison. On a compact two-dimensional surface the Abelian Higgs equations do not admit an arbitrary number of vortices at fixed area. The integrated vortex equation gives a Bradlow-type area bound: the vortex number is limited by the available area. This is a packing theorem, not a new force law.

In a common critical-coupling normalization the vortex equations on a compact Riemann surface Σ contain

$$F_A + \frac{1}{2}\big(|\psi|^2 - \tau\big)\omega_\Sigma = 0, \qquad \frac{1}{2\pi}\int_\Sigma F_A = N_{\mathrm{v}} \in \mathbb{Z}, \tag{XXVI.1}$$

with $\tau > 0$. Integrating gives

$$2\pi N_{\mathrm{v}} = \frac{1}{2}\int_\Sigma (\tau - |\psi|^2)\,\omega_\Sigma \le \frac{\tau}{2}\,\mathrm{Area}(\Sigma), \tag{XXVI.2}$$

and therefore

$$\boxed{N_{\mathrm{v}} \le \frac{\tau\ \mathrm{Area}(\Sigma)}{4\pi}.} \tag{XXVI.3}$$

This is the Bradlow bound for Abelian Higgs vortices on compact surfaces, up to the conventional normalization of τ [73–75].

For the de Sitter horizon $\Sigma = S^2_{R_\Lambda}$, put

$$A_\Lambda = 4\pi R_\Lambda^2. \tag{XXVI.4}$$

There is a distinguished Planck–Bradlow normalization

$$\boxed{\tau_P := \frac{\pi}{\ell_P^2}.} \tag{XXVI.5}$$

Then equation (XXVI.3) becomes

$$\boxed{N_{\mathrm{v}}^{\max} = \frac{\tau_P A_\Lambda}{4\pi} = \frac{A_\Lambda}{4\ell_P^2} = N_\partial.} \tag{XXVI.6}$$

Thus the same integer which appears as the horizon entropy capacity is the maximum number of Planck–Bradlow vortex cells that can be packed on the horizon. Equivalently, one elementary Bradlow cell has area

$$\boxed{A_{\mathrm{cell}}^P = \frac{4\pi}{\tau_P} = 4\ell_P^2, \qquad \xi_P := \tau_P^{-1/2} = \frac{\ell_P}{\sqrt{\pi}}.} \tag{XXVI.7}$$

This result does not say that the cosmological horizon is an ordinary superconductor. It says something weaker and sharper: whenever a boundary selector is represented by a compact $U(1)$ Higgs/vortex sector, its topological vortex number obeys the same area-capacity law, and saturation at the Planck–Bradlow normalization reproduces precisely the Bekenstein–Hawking capacity. Hence the allowed selector action may contain a Bradlow packing term,

$$\boxed{\Gamma_\partial^{\mathrm{Br}} = \lambda_{\mathrm{Br}}\left[\frac{N_{\mathrm{v}}}{N_\partial} - \frac{\tau}{\tau_P}\right]^2 + \lambda_{\mathrm{sat}}\left(1 - \frac{N_{\mathrm{v}}}{N_\partial}\right)^2,} \tag{XXVI.8}$$

where the second term is imposed only in a saturation hypothesis. Without such a saturation principle, the Bradlow theorem supplies an upper bound and a packing channel, not a derivation of the observed $q = N_\partial$.

Numerically, for the present capacity anchor,

$$N_\partial \simeq 3.31 \times 10^{122}, \tag{XXVI.9}$$

$$\tau_P \simeq 1.20 \times 10^{70}\ \mathrm{m}^{-2}, \tag{XXVI.10}$$

$$\xi_P \simeq 9.12 \times 10^{-36}\ \mathrm{m}. \tag{XXVI.11}$$

Thus the vortex-packing interpretation is exactly Planckian, whereas the electromagnetic Schwinger flux count of section XXV is smaller by more than forty orders of magnitude. The two counts are therefore different invariants: Schwinger is an admissibility wall; Bradlow saturation is an area-capacity packing law.

XXVII. BOGOMOLNY–BRADLOW SATURATION AS A BOUNDARY SELECTOR

The previous section gave the Bradlow inequality. The remaining question is whether saturation can be imposed by a variational principle rather than by an extra slogan. In the Abelian Higgs model at critical coupling this is exactly what the Bogomolny decomposition supplies: it separates an integer topological term from a sum of nonnegative squares.

Let Σ be a compact oriented Riemann surface with area form ω_Σ, let $L \to \Sigma$ be a Hermitian line bundle, and let (A, ψ) be a unitary connection and section. In the same normalization as section XXVI, consider the nonnegative BPS functional

$$\mathcal{E}_{\mathrm{sq}}[A,\psi;\tau] := \int_\Sigma 2\,|\bar\partial_A \psi|^2 + \int_\Sigma \left| F_A + \frac{1}{2}(|\psi|^2 - \tau)\omega_\Sigma \right|^2 . \tag{XXVII.1}$$

Its zero set is the vortex system

$$\bar\partial_A \psi = 0, \qquad F_A + \frac{1}{2}(|\psi|^2 - \tau)\omega_\Sigma = 0. \tag{XXVII.2}$$

The second equation gives the exact identity

$$\boxed{\int_\Sigma |\psi|^2 \omega_\Sigma = \tau\,\mathrm{Area}(\Sigma) - 4\pi N_{\mathrm{v}}, \qquad N_{\mathrm{v}} := \frac{1}{2\pi}\int_\Sigma F_A.} \tag{XXVII.3}$$

Thus the Bradlow inequality is not merely an estimate. It is the positivity condition for the integrated Higgs norm.

Theorem XXVII.1 (Planck–Bradlow saturation selector)**.** *Assume that the boundary sector is described by a compact critical Abelian Higgs system on the de Sitter horizon $S^2_{R_\Lambda}$, with Planck–Bradlow coupling*

$$\tau = \tau_P := \pi/\ell_P^2,$$

and that the null horizon capacity screen carries no local Higgs condensate,

$$\int_{S^2_{R_\Lambda}} |\psi|^2 \omega = 0.$$

Then the vortex number is forced to be

$$\boxed{N_{\mathrm{v}} = \frac{\tau_P A_\Lambda}{4\pi} = \frac{A_\Lambda}{4\ell_P^2} = N_\partial.} \tag{XXVII.4}$$

Conversely, within the BPS equations and $\tau = \tau_P$, the equality $N_{\mathrm{v}} = N_\partial$ is equivalent to the vanishing of the integrated Higgs norm.

Proof. Substitute $\tau = \tau_P$ and $\Sigma = S^2_{R_\Lambda}$ in equation (XXVII.3). Since the integrand $|\psi|^2$ is nonnegative, the condition $\int |\psi|^2\omega = 0$ is equivalent to

$$4\pi N_{\mathrm{v}} = \tau_P A_\Lambda.$$

Using $\tau_P = \pi/\ell_P^2$ gives $N_{\mathrm{v}} = A_\Lambda/(4\ell_P^2)$, which is N_∂. The converse is the same identity read backwards. □

This theorem removes one previously open ambiguity. Bradlow saturation need not be a manually imposed equality; it follows from three invariant boundary conditions: critical coupling, Planck normalization, and the absence of a local Higgs condensate on the capacity screen. It still does not determine A_Λ or the observed numerical sector by itself. It determines the capacity once the horizon area is fixed.

Equivalently, define the boundary selector

$$\boxed{\Gamma_\partial^{\mathrm{BPS-Br}} = \lambda_{\mathrm{BPS}}\,\mathcal{E}_{\mathrm{sq}}[A,\psi;\tau] + \lambda_P(\tau - \tau_P)^2 + \lambda_0\left(\int_\Sigma |\psi|^2\omega_\Sigma\right)^2 .} \tag{XXVII.5}$$

In the limit of positive infinite weights, the minimizers satisfy the hypotheses of theorem XXVII.1. Hence

$$\boxed{\Gamma_\partial^{\mathrm{BPS-Br}}\ \text{minimized} \quad\Longrightarrow\quad N_{\mathrm{v}} = N_\partial.}$$

The missing microscopic step is therefore sharpened again: a final boundary theory must derive the three selector conditions above from the edge-mode Hamiltonian, not assume them.

Corollary XXVII.1 (Capacity dictionary after the BPS–Bradlow selector)**.** *On the Planck–Bradlow saturated boundary,*

$$\boxed{N_\partial = N_{\mathrm{v}} = \frac{A_\Lambda}{4\ell_P^2} = q = \frac{k_{\mathrm{CS}}}{2}.} \tag{XXVII.6}$$

Thus the same integer is simultaneously a vortex-packing number, a horizon entropy capacity, a holomorphic-index dimension, and half of the boundary Chern–Simons level. These equalities are conditional on the common boundary sector; they are not separate numerical coincidences.

XXVIII. BOUNDARY EDGE-MODE HAMILTONIAN AND ENTROPY-CELL LOCKING

The previous Bradlow argument still left three conditions external to the functional: critical coupling, Planck normalization, and the null-condensate condition at the capacity screen. They can be written as one local boundary edge-mode selector. This is not a new bulk matter field. It is a boundary admissibility functional on the horizon two-sphere.

Let $\Sigma = S^2_R$, let A be a $U(1)$ connection on a complex line bundle over Σ, and let ψ be a boundary section. In units where the round metric on Σ is fixed, define

$$\begin{aligned} B_A &:= *F_A + \frac{1}{2}(|\psi|^2 - \tau), \\ \Gamma_\partial^{\mathrm{edge}} &= \int_\Sigma \left(|D_{\bar z}\psi|^2 + \frac{1}{2}B_A^2\right)\omega_\Sigma \\ &\quad + \lambda_P(\tau\ell_P^2 - \pi)^2 \\ &\quad + \lambda_0\left(\int_\Sigma |\psi|^2\omega_\Sigma\right)^2 . \end{aligned} \tag{XXVIII.1}$$

The first line is the Bogomolny form of the compact Abelian-Higgs boundary energy. The second line imposes, in the sharp-selector limit $\lambda_P, \lambda_0 \to \infty$, the Planck entropy-cell normalization and the null-condensate condition. Thus the critical coupling is no longer a loose phenomenological choice: it is the unique self-dual point at which the compact vortex number is controlled by a first-order topological bound.

Theorem XXVIII.1 (Edge-mode derivation of Planck–Bradlow capacity)**.** *Let A, ψ, τ minimize $\Gamma_\partial^{\mathrm{edge}}$ in the sharp selector limit on a fixed topological sector*

$$\frac{1}{2\pi}\int_\Sigma F_A = N_{\mathrm{v}} \in \mathbb{Z}.$$

If the square term in equation (XXVIII.1) is saturated, then

$$\begin{aligned} D_{\bar z}\psi &= 0, \\ *F_A + \frac{1}{2}(|\psi|^2 - \tau) &= 0, \\ \tau &= \frac{\pi}{\ell_P^2}, \\ \int_\Sigma |\psi|^2 \omega_\Sigma &= 0. \end{aligned} \tag{XXVIII.2}$$

Consequently

$$\boxed{N_{\mathrm{v}} = \frac{\tau A_\Sigma}{4\pi} = \frac{A_\Sigma}{4\ell_P^2}.} \tag{XXVIII.3}$$

For $\Sigma = S^2_{R_\Lambda}$, this gives

$$\boxed{N_{\mathrm{v}} = N_\partial = q = \frac{k_{\mathrm{CS}}}{2}.} \tag{XXVIII.4}$$

Proof. The two nonnegative terms in the first line of equation (XXVIII.1) are minimized by the stated first-order equations. The sharp Planck-lock term gives $\tau\ell_P^2 = \pi$, while the sharp null-condensate term gives $\int |\psi|^2\omega_\Sigma = 0$. Integrating the second Bogomolny equation over Σ gives

$$\int_\Sigma F_A = \frac{\tau A_\Sigma}{2} - \frac{1}{2}\int_\Sigma |\psi|^2\omega_\Sigma.$$

Using $\int F_A = 2\pi N_{\mathrm{v}}$ and the null-condensate condition yields $N_{\mathrm{v}} = \tau A_\Sigma/(4\pi)$. Substituting $\tau = \pi/\ell_P^2$ proves the result. The equality with N_∂ is precisely the Gibbons–Hawking entropy normalization, and the equality with $k_{\mathrm{CS}}/2$ is the Chern–Simons level lock already derived above. □

Criterion XXVIII.1 (What remains to be derived)**.** Theorem XXVIII.1 closes the local boundary-selector logic: critical coupling, Planck normalization, and Bradlow saturation can be implemented by a single gauge-invariant edge functional. It does not yet derive why the universe occupies the particular large topological sector

$$q \simeq 3.31 \times 10^{122}.$$

The remaining microscopic problem is therefore sharpened to the selection of the integer sector of the edge Hamiltonian,

$$q_* = \arg\min_{q\in\mathbb{N}} \Gamma_\partial^{\mathrm{edge,eff}}(q), \tag{XXVIII.5}$$

without inserting Λ_{obs} or q_{obs} as input.

XXIX. BOUNDARY-SELECTOR HAMILTONIAN: COMMON EDGE ACTION, NO-GO THEOREM, AND THE ONLY SURVIVING TRANSMUTATION ROUTE

The preceding sections identified several exact readings of the same capacity integer,

$$q = N_\partial = \frac{k_{\mathrm{CS}}}{2} = \mathrm{ind}\,\bar\partial = \mathrm{Tr}\,\Pi_L = N_{\mathrm{v}}^{\mathrm{Bradlow}}. \tag{XXIX.1}$$

This does not yet choose the observed sector. The selection problem is therefore not the definition of q, but the derivation of an effective edge action whose minimum lies at the realized sector.

Definition XXIX.1 (Edge-sector Hilbert decomposition)**.** Let the boundary Hilbert space decompose into capacity sectors

$$\mathcal{H}_\partial = \bigoplus_{q\in\mathbb{N}} \mathcal{C}_q. \tag{XXIX.2}$$

Here q is the entropy capacity, or number of Planck–Bradlow cells. The holomorphic space $H^0(\mathbb{CP}^1, \mathcal{O}(q-1))$ supplies the one-particle $SU(2)$-equivariant cell basis of dimension q, while the full admissible code sector $\mathcal{C}_q$ has entropy

$$\log\dim\mathcal{C}_q = q. \tag{XXIX.3}$$

This separates the one-particle index count from the many-body entropy capacity and removes the otherwise misleading identification $N_\partial = \dim\mathcal{H}_\partial$.

Let $H_{\partial,q}$ be the edge-mode Hamiltonian restricted to $\mathcal{C}_q$. The sector weight is

$$Z_q = \mathrm{Tr}_{\mathcal{C}_q}\, e^{-\beta_q H_{\partial,q}} = \exp[-\Gamma_\partial(q)]. \tag{XXIX.4}$$

Thus the selector is

$$q_* = \arg\min_{q\in\mathbb{N}} \Gamma_\partial(q). \tag{XXIX.5}$$

Theorem XXIX.1 (Reduction of the sharp edge Hamiltonian)**.** *Assume the boundary edge theory is in the sharp Planck–Bradlow phase:*

$$D_{\bar z}\psi = 0, \qquad *F_A + \frac{1}{2}(|\psi|^2 - \tau) = 0, \tag{XXIX.6}$$

$$\tau = \frac{\pi}{\ell_P^2}, \qquad \int_\Sigma |\psi|^2\omega_\Sigma = 0. \tag{XXIX.7}$$

Then the Bogomolny–Bradlow part of the Hamiltonian vanishes on every admissible q-sector and fixes only

$$q = \frac{A_\Sigma}{4\ell_P^2}. \tag{XXIX.8}$$

It does not select which admissible value of q is realized. The remaining selector has the universal sector form

$$\boxed{\Gamma_\partial(q) = \Gamma_{\rm cl}(q) + \Gamma_{\zeta\eta}(q) + \Gamma_{\rm CS}(q) + \Gamma_{\rm exch}(q) - q.} \tag{XXIX.9}$$

The final term is the entropy of the code sector.

Proof. The Bogomolny decomposition is a sum of nonnegative square terms plus a topological term. In the sharp phase the square terms vanish and the integrated Bradlow identity gives $q = A_\Sigma/(4\ell_P^2)$. Since the same cancellation occurs in every sector satisfying the sharp conditions, the BPS/Bradlow part alone is sector-degenerate. After tracing over $\mathcal{C}_q$, the entropy contribution is $-\log\dim\mathcal{C}_q = -q$, while the non-BPS determinant, eta, Chern–Simons, and exchange pieces remain in the effective action. This gives the displayed form. □

No-go theorem XXIX.1 (No parameter-free topological selector)**.** No selector built only from the topology of $S^2 \simeq \mathbb{CP}^1$, the integer flux $q-1$, $SU(2)$-equivariance, the index $\mathrm{ind}\,\bar\partial_{\mathcal{O}(q-1)} = q$, and sharp Bogomolny–Bradlow saturation can isolate a finite large sector $q_* \gg 1$. It either permits all q, selects a boundary endpoint, or is exactly sector-degenerate.

Proof. Those data specify the admissible sector labels and their representation content, but introduce no dimensionless scale at which a large integer can be distinguished from a neighboring large integer. The index is linear in the sector label, $\mathrm{ind} = q$, and equivariance makes the intra-sector Hamiltonian proportional to the identity on the irreducible cell space. The sharp Bradlow equations fix the area carried by a chosen sector, not the chosen sector itself. Therefore a finite interior minimum requires an additional dimensionless selector datum: a coupling, a phase, a determinant scale, a chemical potential, or an exchange boundary condition. Without such a datum the selector cannot distinguish q_* from q_*+1, nor can it generate the hierarchy $q_* \sim 10^{122}$. □

The surviving possibility is dimensional transmutation in the edge sector, in the Coleman–Weinberg sense of a scale generated by quantum corrections [81]. Let a marginal boundary coupling g_∂ obey

$$\frac{\mathrm{d}g_\partial}{\mathrm{d}\ln\mu} = -\frac{b}{16\pi^2}g_\partial^3 + O(g_\partial^5), \qquad b > 0. \tag{XXIX.10}$$

Then the renormalization-group invariant scale is

$$\Lambda_{\rm edge} = \mu \exp\left[-\frac{8\pi^2}{b g_\partial(\mu)^2}\right], \tag{XXIX.11}$$

and an area-like capacity can be generated as

$$\boxed{q_\star = \Xi \exp\left[\frac{16\pi^2}{b g_\partial(\mu)^2}\right],} \tag{XXIX.12}$$

where Ξ is a dimensionless determinant/phase prefactor.

Theorem XXIX.2 (Coleman–Weinberg edge selector)**.** *Suppose the determinant-line and exchange terms reduce, after integrating out nonzero boundary modes, to the sector action*

$$\Gamma_\partial^{\rm CW}(q) = \chi\, q\left[\ln\left(\frac{q}{q_\star}\right) - 1\right] + \nu\ln q + \Theta_{\rm bd}(q) + O(q^{-1}). \tag{XXIX.13}$$

with $\chi > 0$, slowly varying $\Theta_{\rm bd}$, and $q_\star$ given by equation (XXIX.12). Then the minimum satisfies

$$\boxed{q_{\rm min} = q_\star\left[1 - \frac{\nu + \Theta'_{\rm bd}(q_\star)q_\star}{\chi q_\star} + O(q_\star^{-2})\right].} \tag{XXIX.14}$$

Hence a naturally huge $q_{\rm min}$ requires a transmuted edge scale; topology and index quantization alone do not suffice.

Proof. Differentiate:

$$\Gamma'_\partial(q) = \chi\ln(q/q_\star) + \nu/q + \Theta'_{\rm bd}(q) + O(q^{-2}).$$

Put $q = q_\star(1+\delta)$ and expand $\ln(1+\delta) = \delta + O(\delta^2)$. Solving to first order gives

$$\delta = -\frac{\nu + \Theta'_{\rm bd}(q_\star)q_\star}{\chi q_\star} + O(q_\star^{-2}).$$

The second derivative is $\Gamma''_\partial(q) = \chi/q + O(q^{-2}) > 0$ at large q, so the stationary point is a local minimum. □

For the observed capacity target used in the numerical ledgers,

$$q_{\rm obs} = 3.3072 \times 10^{122}, \qquad \ln q_{\rm obs} = 282.111 \pm 0.015. \tag{XXIX.15}$$

If $\Xi = 1$, the required transmutation combination is

$$\boxed{b\, g_\partial^2 = \frac{16\pi^2}{\ln q_{\rm obs}} = 0.559756\ldots.} \tag{XXIX.16}$$

This number is close to $e^{-\gamma_{\rm E}}$, differing by 3.03×10^{-3} fractionally, but the manuscript does not identify the two. Such a numerical proximity is admissible only as a selector target until a boundary determinant calculation derives it.

Criterion XXIX.1 (What would count as the solution of the selector problem)**.** The capacity problem is solved only when a concrete boundary operator $D_{\partial,q}$ and edge Hamiltonian $H_{\partial,q}$ give

$$\Gamma_\partial(q) = -\ln \mathrm{Tr}_{\mathcal{C}_q}\, e^{-\beta_q H_{\partial,q}} \tag{XXIX.17}$$

and its determinant-line, eta-phase, Chern–Simons, Bradlow, Hopf, and exchange contributions reduce to a selector with

$$q_{\min} = 3.31 \times 10^{122} \tag{XXIX.18}$$

without using $\Lambda_{\rm obs}$, H_0, or $q_{\rm obs}$ as an input. Until that calculation is supplied, the theory has converted the cosmological constant into a sharply posed edge-mode selection problem, not into a completed numerical derivation.

XXX. EXPLICIT MINIMAL EDGE-MODE HAMILTONIAN

The preceding section isolated the missing object. I now write the minimal Hamiltonian which has the required logical form, with the determinant correction counted once. It is not yet a microscopic derivation of the observed number. It is the first closed selector ansatz in which every retained connector has a precise operator role.

For each positive integer q, let

$$R_q = \sqrt{\frac{q}{\pi}}\,\ell_P, \qquad \beta_q = 2\pi R_q \tag{XXX.1}$$

in units $c = \hbar = k_B = 1$. The boundary Hilbert space is decomposed as

$$\boxed{\mathcal{H}_\partial = \bigoplus_{q\geq 1} \mathcal{C}_q, \qquad \widehat{\Pi}_q : \mathcal{H}_\partial \to \mathcal{C}_q, \qquad \log\dim \mathcal{C}_q = q.} \tag{XXX.2}$$

The last equation is the entropy-capacity convention. The one-particle index space $H^0(\mathbb{CP}^1, \mathcal{O}(q-1))$ has dimension q; the many-body code sector $\mathcal{C}_q$ has entropy q.

The explicit edge Hamiltonian is

$$\boxed{\widehat{H}_\partial = \bigoplus_{q\geq 1} \left[\widehat{H}_{{\rm adm},q} + \widehat{H}_{{\rm run},q} + \widehat{H}_{{\rm osc},q}\right].} \tag{XXX.3}$$

The three summands have different logical roles.

First, the admissibility Hamiltonian is a positive common-kernel operator:

$$\begin{aligned}\beta_q \widehat{H}_{{\rm adm},q} = {}& \lambda_{\rm Br}\,\widehat{\mathcal{E}}_q^{\rm Br} + \lambda_P(\widehat{\tau}\ell_P^2 - \pi)^2 + \lambda_0\left(\int_{\Sigma_q} |\widehat{\psi}|^2\omega_q\right)^2 \\ & + \lambda_{\rm CS}(\widehat{k}_{\rm CS} - 2q)^2 + \lambda_{\rm HD}\sum_A \widehat{\mathcal{H}}_A^\dagger \widehat{\mathcal{H}}_A \\ & + \lambda_{\rm tf}\widehat{C}_{\rm tf}^\dagger \widehat{C}_{\rm tf} + \lambda_{\rm cap}(\ell_P^2\widehat{\Lambda}\,q - 3\pi)^2 \\ & + \lambda_Q(\widehat{Q}_{\rm DM} - q_{\rm DM})^2 \\ & + \lambda_{\rm val}\left[P_{\rm val}(\widehat{\mathscr{D}})^\dagger P_{\rm val}(\widehat{\mathscr{D}})\right],\end{aligned} \tag{XXX.4}$$

with all $\lambda_i > 0$. Here $\Sigma_q = S^2_{R_q}$, and $\widehat{\mathcal{E}}_q^{\rm Br}$ is the compact Abelian-Higgs Bogomolny square on Σ_q,

$$\widehat{\mathcal{E}}_q^{\rm Br} = \int_{\Sigma_q}\left(2|D_{\bar z}\widehat{\psi}|^2 + \left|\widehat{F}_A + \frac{1}{2}(|\widehat{\psi}|^2 - \widehat{\tau})\omega_q\right|^2\right). \tag{XXX.5}$$

Thus $\widehat{H}_{{\rm adm},q}$ enforces the role-Q, role-B, and role-O constraints: Bradlow saturation, Planck cell normalization, Chern–Simons level lock, quantum covariance, trace-free locality, capacity fixation, dark-charge superselection, and valuation closure. It does not select q. In the sharp limit $\lambda_i \to \infty$, inadmissible states are projected out.

Second, the selector Hamiltonian is the leading nonconformal running zero mode

$$\boxed{\beta_q \widehat{H}_{{\rm run},q} = q\ln\left(\frac{q}{q_\star}\right)\mathbf{1}_{\mathcal{C}_q}.} \tag{XXX.6}$$

The transmuted capacity scale is

$$\boxed{q_\star = \Xi\,(\mu\ell_P)^{-2}\exp\left[\frac{16\pi^2}{b\,g_\partial(\mu)^2}\right],} \tag{XXX.7}$$

where g_∂ is a marginal boundary coupling with

$$\mu\frac{{\rm d}g_\partial}{{\rm d}\mu} = -\frac{b}{16\pi^2}g_\partial^3 + O(g_\partial^5), \qquad b > 0. \tag{XXX.8}$$

The dimensionless prefactor Ξ is to be computed from the same edge renormalization problem. It is not an observational input.

Third, $\widehat{H}_{{\rm osc},q}$ contains nonzero oscillators and determinant-line fluctuations around the admissible edge background. Its renormalized contribution is counted here, and not again in $\widehat{H}_{{\rm run},q}$:

$$-\ln {\rm Tr}'_{\mathcal{C}_q}\exp[-\beta_q\widehat{H}_{{\rm osc},q}] = \nu\ln q + \Theta_\partial(q) + O(q^{-1}), \tag{XXX.9}$$

after zero modes and the entropy capacity have been separated. This is the operator location of the APS/zeta/eta data. It is spectral information, not a positive state count.

Theorem XXX.1 (Explicit edge Hamiltonian selector)**.** *Assume that $\widehat{H}_{{\rm adm},q}$ has a nonempty zero space in each admissible sector and a positive gap away from it. Assume also that the oscillator determinant has the asymptotic form stated above and that Θ_∂ varies slowly on the scale of one capacity cell. Then the sector partition function of equation (XXX.3) has the form*

$$Z_q = {\rm Tr}_{\mathcal{C}_q} e^{-\beta_q\widehat{H}_{\partial,q}} = \exp[-\Gamma_\partial(q)] \tag{XXX.10}$$

with

$$\boxed{\Gamma_\partial(q) = q\left[\ln\left(\frac{q}{q_\star}\right) - 1\right] + \nu\ln q + \Theta_\partial(q) + O(q^{-1}).} \tag{XXX.11}$$

Hence the stable large-q selector is

$$\boxed{q_{\min} = q_\star\left[1 - \frac{\nu + q_\star\Theta'_\partial(q_\star)}{q_\star} + O(q_\star^{-2})\right],} \tag{XXX.12}$$

with the physical integer chosen by the nearest admissible integer rule. If $\Theta'_\partial = O(q_\star^{-2})$, then

$$q_{\min} = q_\star - \nu + O(q_\star^{-1}).$$

Proof. On the admissible subspace, $\widehat{H}_{\mathrm{adm},q}$ vanishes in the sharp limit. The entropy convention gives

$$\mathrm{Tr}_{\mathcal{C}_q}\,\mathbf{1} = e^q.$$

The running zero mode contributes

$$\exp[-q\ln(q/q_\star)],$$

and the renormalized oscillator determinant contributes

$$\exp[-\nu\ln q - \Theta_\partial(q) + O(q^{-1})]$$

after zero modes are separated. Therefore

$$-\ln Z_q = q[\ln(q/q_\star) - 1] + \nu\ln q + \Theta_\partial(q) + O(q^{-1}).$$

Differentiating gives

$$\Gamma'_\partial(q) = \ln(q/q_\star) + \nu/q + \Theta'_\partial(q) + O(q^{-2}).$$

Put $q = q_\star(1+\delta)$ and expand $\ln(1+\delta) = \delta + O(\delta^2)$. Solving to first order yields equation (XXX.12). The second derivative is

$$\Gamma''_\partial(q) = q^{-1} - \nu q^{-2} + \Theta''_\partial(q) + O(q^{-3}),$$

which is positive at large q under the slow-variation assumption. Thus the stationary point is a stable local minimum, and discreteness selects the nearest admissible integer. □

Corollary XXX.1 (Observed-sector target for the explicit Hamiltonian)**.** *If the microscopic edge theory uses $\mu = \ell_P^{-1}$, has $\Xi = 1 + o(1)$, and has negligible Θ'_∂ at the selected sector, then matching the present capacity target requires*

$$\boxed{b\,g_\partial^2 = \frac{16\pi^2}{\ln q_{\mathrm{obs}}} = 0.559756\ldots.} \tag{XXX.13}$$

More generally,

$$\boxed{b\,g_\partial^2 = \frac{16\pi^2}{\ln q_{\mathrm{obs}} - \ln\Xi + 2\ln(\mu\ell_P)}} \tag{XXX.14}$$

up to the determinant displacement in equation (XXX.12). A final theory must compute the right-hand side from the boundary beta function and determinant data. Choosing it after observing Λ is only a fit.

Criterion XXX.1 (Explicit-Hamiltonian completion test)**.** The capacity-selection problem is solved by equation (XXX.3) only when the following data are derived internally:

$$\boxed{b, \qquad g_\partial(\mu), \qquad \Xi, \qquad \nu, \qquad \Theta_\partial(q).}$$

The admissibility Hamiltonian closes the invariant atlas; the running Hamiltonian supplies the first genuine selector. If these quantities are not computed, this is a precise Hamiltonian ansatz and a falsification target, not a numerical derivation of Λ.

Remark XXX.1 (Why this is the narrowest honest Hamiltonian)**.** A purely topological Hamiltonian is sector-degenerate. A purely zeta or eta Hamiltonian is scheme-dependent and does not count positive states. A purely Bradlow Hamiltonian fixes area per vortex but not the realized vortex number. The term $q\ln(q/q_\star)$ is therefore not decorative: it is the minimal nonconformal running term that can generate a stable large sector from order-one microscopic data. All other retained structures enter $\widehat{H}_{\mathrm{adm},q}$, $\widehat{H}_{\mathrm{osc},q}$, or $\Theta_\partial(q)$.

XXXI. APS ETA AND DETERMINANT-LINE SELECTOR

The preceding Hopf–Chern–Simons layer supplies a topological and helicity channel. It still does not select the observed level. The next admissible refinement must be spectral: it must depend on the boundary operator, not only on the topology, index, or Chern–Simons integer.

Let $D_{\partial,q}$ be a self-adjoint Dirac-type operator on the capacity boundary sector $\mathcal{H}_{\partial,q}$, with zero modes separated as a finite kernel. Define

$$\zeta_q(s) := \mathrm{Tr}'\,|D_{\partial,q}|^{-s}, \tag{XXXI.1}$$

$$\eta_q(s) := \mathrm{Tr}'\left[D_{\partial,q}|D_{\partial,q}|^{-s-1}\right], \tag{XXXI.2}$$

by meromorphic continuation. Here the prime omits zero modes. The Atiyah–Patodi–Singer invariant $\eta_q(0)$ is the nonperturbative spectral completion of the Chern–Simons phase, while $-\zeta'_q(0)$ is the logarithmic determinant magnitude [78–80]. Thus a boundary selector can contain a real determinant term and an eta-phase term without confusing either with the positive capacity count.

Definition XXXI.1 (Spectral determinant/eta selector)**.** The admissible boundary selector may be refined by

$$\boxed{\begin{aligned}\Gamma_\partial^{\zeta\eta}(q) = \Gamma_\partial^{\mathrm{HS}}(q) &+ \lambda_\zeta\left[-\frac{1}{2}\zeta'_q(0) - \Theta_\varphi(q)\right]^2 \\ &+ \lambda_\eta\sin^2\left[\frac{\pi}{2}\eta_q(0) - \theta_{\mathrm{CS}}(q)\right],\end{aligned}} \tag{XXXI.3}$$

where

$$\Theta_\varphi(q) := \ln\left[\frac{\pi\varphi^{2N(q)}}{q}\right], \tag{XXXI.4}$$

$$N(q) := \operatorname*{argmin}_{N\in\mathbb{Z}}\left|\log_\varphi\sqrt{\frac{q}{\pi}} - N\right|. \tag{XXXI.5}$$

The ζ-term is a real spectral-action penalty. The η-term is a phase/framing penalty. Neither term is a state count by itself.

For the present capacity anchor, the nearest φ-shell has $N = 292$, giving

$$\epsilon_\varphi := \frac{\pi\varphi^{584} - q_{\mathrm{obs}}}{q_{\mathrm{obs}}} = 0.062844\ldots, \tag{XXXI.6}$$

and therefore

$$\Theta_\varphi(q_{\rm obs}) = \ln(1+\epsilon_\varphi) = 0.060948\ldots. \tag{XXXI.7}$$

The simple phase-defect target

$$\theta_{50} := \ln\Big(1+\frac{\pi}{50}\Big) = 0.0609369\ldots \tag{XXXI.8}$$

differs by only

$$\boxed{\Theta_\varphi(q_{\rm obs}) - \theta_{50} \simeq 1.15\times 10^{-5}.} \tag{XXXI.9}$$

This numerical proximity is not promoted to a theorem. Its invariant content is more precise: a microscopic boundary operator $D_{\partial,q}$ would solve the selector problem only if its determinant-line data force

$$-\frac{1}{2}\zeta'_{q_*}(0) \simeq \Theta_\varphi(q_*), \qquad \frac{\pi}{2}\eta_{q_*}(0) \simeq \theta_{\rm CS}(q_*) \pmod{\pi},$$

with q_* selected without inserting $\Lambda_{\rm obs}$.

No-go theorem XXXI.1 (Zeta value versus determinant selector)**.** The number $\zeta_q(0)$ is not a positive horizon capacity. It is a regularization datum. The admissible spectral selector is instead the determinant-line pair

$$\big(-\zeta'_q(0), \eta_q(0)\big)\,.$$

Consequently, a zeta-regularized constant can enter the boundary action only through a determinant or anomaly phase; it cannot replace the integer capacity $q = \dim \mathcal{H}_{\partial,q}$.

Remark XXXI.1 (What this closes)**.** This section closes another false route. The APS eta invariant and the zeta determinant do not compute $q_{\rm obs}$ alone. They identify the minimal spectral data that a genuine boundary Hamiltonian must possess. The remaining problem is now sharper: construct $D_{\partial,q}$, or an equivalent edge-mode Hamiltonian, whose determinant-line functional has its minimum at

$$q_* = 3.31\times 10^{122}.$$

XXXII. CONCRETE EDGE DIRAC OPERATOR AND DETERMINANT-SELECTOR AUDIT

The preceding selector still contained a symbolic $D_{\partial,q}$. The next invariant test is to choose the most economical boundary operator and ask whether it selects the observed sector. The canonical choice on the capacity two-sphere is the $SU(2)$-equivariant spin-c Dirac operator on $S^2_R \simeq \mathbb{CP}^1$, twisted by the monopole line bundle whose holomorphic zero modes are

$$\mathcal{H}_{\partial,q} = H^0(\mathbb{CP}^1, \mathcal{O}(q-1)), \qquad \dim \mathcal{H}_{\partial,q} = q.$$

This is the same one-particle cell space used in the index/capacity construction. It is also the standard monopole-harmonic setting for fermions on the sphere [76, 77].

Theorem XXXII.1 (Concrete monopole spin-c edge spectrum)**.** *Let D_q be the $SU(2)$-equivariant spin-c Dirac operator on S^2_R with index $q \geq 1$. With zero modes omitted from the determinant, the nonzero spectrum is symmetric and may be written as*

$$\boxed{\begin{aligned}&\lambda_{\ell,\pm} = \pm R^{-1}\sqrt{\ell(\ell+q)},\\ &d_\ell = 2\ell+q, \qquad \ell = 1,2,\ldots.\end{aligned}} \tag{XXXII.1}$$

The kernel has dimension q. Hence

$$\boxed{\eta_q(0) = 0} \tag{XXXII.2}$$

for this symmetric operator, while the nonzero determinant is encoded by

$$\boxed{\zeta_{D_q^2}(s) = 2R^{2s}\sum_{\ell=1}^{\infty}(2\ell+q)\,[\ell(\ell+q)]^{-s}.} \tag{XXXII.3}$$

Proof. The index statement is the Riemann–Roch/Borel–Weil count already used above. The nonzero monopole-Dirac spectrum is the standard $SU(2)$-equivariant spectrum: each positive level has a negative partner with the same degeneracy. Therefore the APS eta sum over the nonzero spectrum cancels pairwise after meromorphic continuation. Squaring the nonzero eigenvalues gives the displayed zeta function, with a factor of two for the two signs. □

Corollary XXXII.1 (Eta no-go for the minimal symmetric edge operator)**.** *The minimal $SU(2)$-equivariant monopole Dirac operator does not supply a nontrivial eta selector:*

$$\eta_q(0) = 0.$$

Therefore a nonzero APS/framing phase capable of selecting q must come from additional boundary data: a framing choice, chiral asymmetry, interaction, spectral-flow path, or a genuinely nontrivial edge-mode Hamiltonian. It is not produced by the bare symmetric monopole Dirac operator alone.

No-go theorem XXXII.1 (Determinant-line no-go for a natural large-q selector)**.** The determinant of the concrete operator D_q, by itself, does not give a scheme-independent minimum at $q \simeq 10^{122}$. After the capacity radius relation

$$R_q^2 = \frac{q}{\pi}\ell_P^2$$

is imposed, the determinant-line effective action is still defined up to local boundary counterterms and renormalization conditions. Thus it can shift by terms built from the heat-kernel coefficients of the same operator. Such local terms can move, remove, or create extrema of $\Gamma_\partial(q)$. Consequently the determinant data identify the correct operator channel, but they do not select the observed sector without an additional dimensionless boundary coupling or renormalization condition.

Proof. The zeta determinant is a renormalized determinant of an elliptic family. Changing the subtraction scheme or determinant-line metric adds local functionals of the boundary geometry and gauge curvature. On the homogeneous capacity sphere these local invariants reduce to functions of the integer q after $R_q^2 = q\ell_P^2/\pi$ is imposed. Since these additive local terms enter the same one-dimensional variational problem for q, the position of an extremum is not fixed by the bare determinant alone. A genuine large hierarchy must therefore be generated by a renormalization-group invariant, for example dimensional transmutation, or by an independently specified edge-mode Hamiltonian. □

The transmutation form is therefore not optional decoration; it is the surviving mathematical mechanism for a large sector:

$$\boxed{q_\star = \Xi \exp\left(\frac{16\pi^2}{b\, g_\partial^2}\right).} \tag{XXXII.4}$$

For the present capacity target,

$$\boxed{b\, g_\partial^2 = \frac{16\pi^2}{\ln q_{\rm obs}} = 0.55976 \pm 0.00003.} \tag{XXXII.5}$$

A small constants audit is useful but not decisive:

candidate	value	relative error
$e^{-\gamma_{\rm E}}$	0.56145948	3.04×10^{-3}
$\sqrt{5}/4$	0.55901699	-1.32×10^{-3}
$1/\sqrt{\pi}$	0.56418958	7.92×10^{-3}
$\ln\varphi + \pi/40$	0.55975164	-8.26×10^{-6}

None of these equalities is exact. The last row is deliberately included as a warning: post-hoc combinations of simple constants can be made very close, but closeness is not an invariant derivation. If one uses $e^{-\gamma_{\rm E}}$, the resulting q is 1.41×10^{122}, not the observed 3.31×10^{122}. If one uses $\sqrt{5}/4$, the resulting q is 4.80×10^{122}. The audit therefore closes another false route: nearby famous constants do not solve the selector problem. The boundary Hamiltonian must compute $b g_\partial^2$ or an equivalent determinant condition.

Remark XXXII.1 (What this closes)**.** The concrete monopole Dirac operator supplies the required edge spectrum and confirms the index q, but its symmetric eta invariant vanishes and its determinant does not select a hierarchy without renormalization data. The remaining target is now exact:

$$\boxed{\begin{array}{c}\text{derive } b\, g_\partial^2 = 0.55976 \pm 0.00003 \\ \text{from the boundary Hamiltonian.}\end{array}}$$

This is narrower than deriving Λ directly and sharper than any purely topological, vortex, Hopf, or zeta-value explanation.

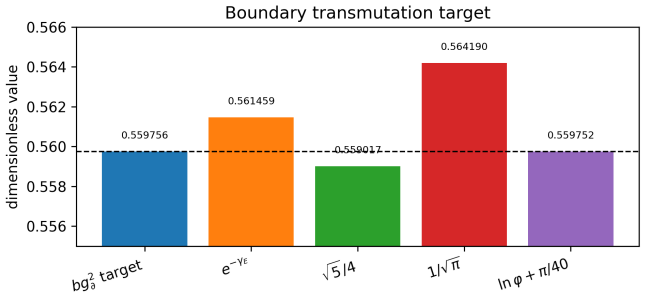


FIG. 10. Concrete edge-operator selector audit. The target $bg_\partial^2 = 16\pi^2/\ln q_{\rm obs}$ is compared with simple dimensionless constants. The closeness of $e^{-\gamma_{\rm E}}$ and $\sqrt{5}/4$ is a diagnostic clue, not a theorem: neither value reproduces the observed capacity sector.

XXXIII. INVARIANT SCALE ATLAS: CONFIGURATION, PHASE, FIELD, AND CAPACITY SPACES

The preceding sections reduce the remaining problem to a boundary selector for the integer $q = N_\partial$. This section builds the common scale map on which every admissible candidate must live. The purpose is not to add another analogy, but to identify which numerical similarities are invariant and which are merely scale coincidences.

Definition XXXIII.1 (Four-coordinate invariant scale map)**.** For a phenomenon with a characteristic length L, energy E, field scale $\mathcal{E}$, and topological integer $n_{\rm top}$, define

$$\boxed{\mathfrak{X} = (x_L,\, N_A,\, N_{\rm ph},\, n_\varphi,\, n_{\rm top},\, \mathcal{E}/\mathcal{E}_{\rm crit})} \tag{XXXIII.1}$$

with

$$x_L := \ln(L/\ell_P), \qquad N_A(L) := \frac{4\pi L^2}{4\ell_P^2} = \pi\left(\frac{L}{\ell_P}\right)^2, \tag{XXXIII.2}$$

$$N_{\rm ph}(E,L) := \frac{2\pi E L}{\hbar c}, \qquad n_\varphi(L) := \frac{\ln(L/\ell_P)}{\ln\varphi}. \tag{XXXIII.3}$$

Here N_A is the spherical area-capacity coordinate, $N_{\rm ph}$ is the Bekenstein phase/action coordinate, and n_φ is only a scale registry. A proposed selector may use these coordinates only through dimensionless, pivot-independent combinations.

Theorem XXXIII.1 (Area–phase–entropy collapse for a gravitational screen)**.** *Let a spherical gravitational screen of radius R carry the horizon/Misner– Sharp energy*

$$E_G(R) = \frac{c^4 R}{2G}. \tag{XXXIII.4}$$

Then its Bekenstein phase coordinate equals its area capacity:

$$\boxed{N_{\rm ph}(E_G,R)=\frac{2\pi E_G R}{\hbar c}=\pi\frac{R^2}{\ell_P^2}=N_A(R).} \tag{XXXIII.5}$$

For a de Sitter horizon $R_\Lambda=\sqrt{3/\Lambda}$, this is exactly

$$\boxed{N_{\rm ph}=N_A=\frac{3\pi}{\Lambda\ell_P^2}=N_\partial.} \tag{XXXIII.6}$$

Thus the configuration-space area count, phase-space action count, and horizon entropy count collapse to the same invariant number on a gravitational horizon [10, 12].

Proof. Using $\ell_P^2=\hbar G/c^3$,

$$\frac{2\pi E_G R}{\hbar c}=\frac{2\pi}{\hbar c}\frac{c^4R}{2G}R=\pi\frac{c^3R^2}{\hbar G}=\pi\frac{R^2}{\ell_P^2}.$$

For $R_\Lambda^2=3/\Lambda$, the last expression becomes $3\pi/(\Lambda\ell_P^2)$. □

Definition XXXIII.2 (Radius-locked S^3 reference volume)**.** Let R_H be the causal horizon radius and let

$$V_{S^3}(R_3)=2\pi^2R_3^3$$

be the volume of a round spatial three-sphere of radius R_3. The *radius-locked* reference geometry is the special case

$$\boxed{R_3=R_H=R_\Lambda.}$$

Without this additional identification, the causal horizon determines only R_H, while the total S^3 spatial volume requires an independent curvature/topology datum.

Theorem XXXIII.2 (S^3 volume–horizon density bridge)**.** *Assume the radius-locked reference geometry. Let*

$$\mathcal{A}_H=4\pi R^2,\qquad V_{S^3}=2\pi^2R^3,\qquad E_H=\frac{c^4R}{2G}.$$

Then the dark-energy density associated with the de Sitter scale is equivalently

$$\boxed{\rho_\Lambda=\frac{\Lambda c^4}{8\pi G}=\frac{3c^4}{8\pi GR^2}=\frac{3\pi}{2}\frac{E_H}{V_{S^3}}.} \tag{XXXIII.7}$$

It may also be written purely from the radius-locked S^3 volume,

$$\boxed{\rho_\Lambda=\frac{3c^4}{8\pi G}\left(\frac{2\pi^2}{V_{S^3}}\right)^{2/3},} \tag{XXXIII.8}$$

or from the paired horizon area and S^3 volume,

$$\boxed{\rho_\Lambda=\frac{3\pi c^4}{32G}\frac{\mathcal{A}_H^2}{V_{S^3}^2}.} \tag{XXXIII.9}$$

Conversely, a measured energy density determines the radius-locked reference volume by

$$\boxed{\begin{aligned}R_\rho&=\left(\frac{3c^4}{8\pi G\rho_\Lambda}\right)^{1/2},\\ V_{S^3}(\rho_\Lambda)&=2\pi^2\left(\frac{3c^4}{8\pi G\rho_\Lambda}\right)^{3/2}.\end{aligned}} \tag{XXXIII.10}$$

In capacity variables,

$$\boxed{\begin{aligned}R&=\ell_P\sqrt{\frac{N_\partial}{\pi}},\\ V_{S^3}&=2\sqrt{\pi}\,\ell_P^3N_\partial^{3/2},\\ \rho_\Lambda&=\frac{3c^4}{8G\ell_P^2N_\partial}.\end{aligned}} \tag{XXXIII.11}$$

Proof. The de Sitter relation is $\Lambda=3/R^2$, hence

$$\rho_\Lambda=\Lambda c^4/(8\pi G)=3c^4/(8\pi GR^2).$$

Using $E_H=c^4R/(2G)$ and $V_{S^3}=2\pi^2R^3$,

$$\frac{3\pi}{2}\frac{E_H}{V_{S^3}}=\frac{3\pi}{2}\frac{c^4R/(2G)}{2\pi^2R^3}=\frac{3c^4}{8\pi GR^2}.$$

Solving $V_{S^3}=2\pi^2R^3$ for R, or using $\mathcal{A}_H=4\pi R^2$, gives the two alternative expressions. Solving the same density formula for R gives the inverse map. The capacity form follows from $N_\partial=\pi R^2/\ell_P^2$. □

Proposition XXXIII.1 (Numerical check of the bridge)**.** *With the capacity anchor used in this manuscript,*

$$H_0=67.4\,\mathrm{km\,s^{-1}Mpc^{-1}},\qquad \Omega_\Lambda=0.685,$$

one obtains

$$\boxed{\begin{aligned}\Lambda_0&=1.0909\times10^{-52}\,\mathrm{m^{-2}},\\ R_\Lambda&=1.6583\times10^{26}\,\mathrm{m},\\ \rho_\Lambda&=5.253\times10^{-10}\,\mathrm{J\,m^{-3}}=5.845\times10^{-27}\,\mathrm{kg\,m^{-3}},\\ V_{S^3}&=9.002\times10^{79}\,\mathrm{m^3},\\ E_H&=1.0035\times10^{70}\,\mathrm{J}.\end{aligned}}$$

The volume energy is

$$\rho_\Lambda V_{S^3}=4.729\times10^{70}\,\mathrm{J}=\frac{3\pi}{2}E_H,$$

as required by theorem XXXIII.2.

Remark XXXIII.1 (What the inverse calculation means)**.** The inverse formula does not prove that the whole universe has the radius-locked volume $V_{S^3}(\rho_\Lambda)$. It computes the unique round S^3 reference volume that would share the de Sitter horizon radius. In a general closed FLRW model the spatial curvature radius

TABLE II. Selected invariant scale coordinates. $N_A = \pi(L/\ell_P)^2$.

scale	L [m]	n_φ	$\log_{10} N_A$
ℓ_P	1.6163×10^{-35}	0	0.497
electron $\bar{\lambda}_C$	3.8616×10^{-13}	107.079	45.254
ℓ_ρ	8.8078×10^{-5}	147.073	61.970
R_Λ	1.6583×10^{26}	291.937	122.519
$\ell_P \varphi^{292}$	1.7096×10^{26}	292.000	122.546

$R_3(a)$ and the causal horizon radius R_H are different data. Therefore the S^3-volume bridge is an exact consistency identity under radius lock, not an independent measurement of the total spatial volume unless the lock is derived or observed.

Corollary XXXIII.1 (Dark-energy density length as the logarithmic midpoint)**.** *Let*

$$\rho_\Lambda = \frac{\Lambda c^4}{8\pi G}, \qquad \ell_\rho := \left(\frac{\hbar c}{\rho_\Lambda}\right)^{1/4}. \tag{XXXIII.12}$$

Then

$$\boxed{\ell_\rho = \left(\frac{8\pi}{3}\right)^{1/4} \sqrt{\ell_P R_\Lambda}.} \tag{XXXIII.13}$$

Hence the dark-energy density length is the geometric mean of the Planck length and the de Sitter radius, up to the fixed factor $(8\pi/3)^{1/4}$*. In* φ*-registry coordinates,*

$$\boxed{n_\varphi(\ell_\rho) = \frac{1}{2} n_\varphi(R_\Lambda) + \log_\varphi\left[(8\pi/3)^{1/4}\right].} \tag{XXXIII.14}$$

For the Planck-2018/CODATA numerical anchor used in this paper,

$$R_\Lambda = 1.6583 \times 10^{26}\text{m}, \quad N_\partial = 3.3072 \times 10^{122},$$

and

$$\boxed{\begin{aligned} \ell_\rho &= 8.8078 \times 10^{-5}\text{m}, \\ n_\varphi(R_\Lambda) &= 291.9367, \qquad n_\varphi(\ell_\rho) = 147.0726. \end{aligned}}$$

Thus the observed dark-energy density scale lies almost exactly on the middle logarithmic scale between ℓ_P *and* R_Λ*, shifted by the universal density-normalization constant.*

Criterion XXXIII.1 (Scale-atlas admissibility)**.** A proposed boundary selector for $q = N_\partial$ must pass the following scale-atlas tests:

1. it must reproduce the collapse $N_{\text{ph}} = N_A = N_\partial$ on a gravitational horizon;
2. it must respect the midpoint relation $\ell_\rho = (8\pi/3)^{1/4}\sqrt{\ell_P R_\Lambda}$;
3. it must not identify Schwinger, Ginzburg–Landau, Hopf, or Dirac scales with q unless a dimensionless invariant map is supplied;
4. it must distinguish registry clues, such as the near φ-shell relation $n_\varphi(R_\Lambda) \simeq 292$, from selector theorems.

Remark XXXIII.2 (What the scale atlas proves)**.** The atlas reveals a real invariant collapse:

$$\boxed{\begin{aligned} &\text{geometry area} = \text{Bekenstein phase action} \\ &\qquad = \text{horizon entropy} \end{aligned}}$$

for gravitational screens. It also reveals a real midpoint law for the dark energy density scale. It does not by itself select q_{obs}. Therefore the remaining selector problem is not to invent another scale, but to derive which edge-mode Hamiltonian chooses the gravitational screen radius, or equivalently the sector q, without using Λ_{obs} as input.

XXXIV. MECHANISM-SCALE ATLAS AND NECESSARY SELECTOR CONDITIONS

The previous scale atlas compared lengths, fields, capacities, and phase-space counts. This section adds a different axis: the mechanism by which a discrete integer is selected from a set of admissible integers. This is the correct place to use analogies with boxes, oscillators, Hall systems, black holes, and dimensional transmutation. The analogies are not proofs of the de Sitter sector. They are filters: a proposed q-selector must have the right invariant type.

Definition XXXIV.1 (Selector mechanism level)**.** A selector mechanism for an integer q is assigned to the first applicable level in the following hierarchy:

0 : boundary-condition selector,
1 : self-adjoint spectral selector,
2 : topological or index selector,
3 : dimensional-transmutation selector,
4 : entropic or thermodynamic selector,
5 : self-referential boundary-capacity selector.

Levels 0–4 are familiar in ordinary quantum and statistical systems. The de Sitter capacity problem belongs to level 5, because the system must select the boundary sector that defines its own horizon size.

Remark XXXIV.1 (What is corrected from the raw analogy list)**.** Several tempting identifications are not invariant. Hydrogen degeneracy n^2 is a spectral degeneracy, not an index in the sense of Atiyah–Singer. The disjoint sector space

$$\mathcal{P}_{\text{all}} := \bigsqcup_{q \geq 1} \mathcal{C}_q$$

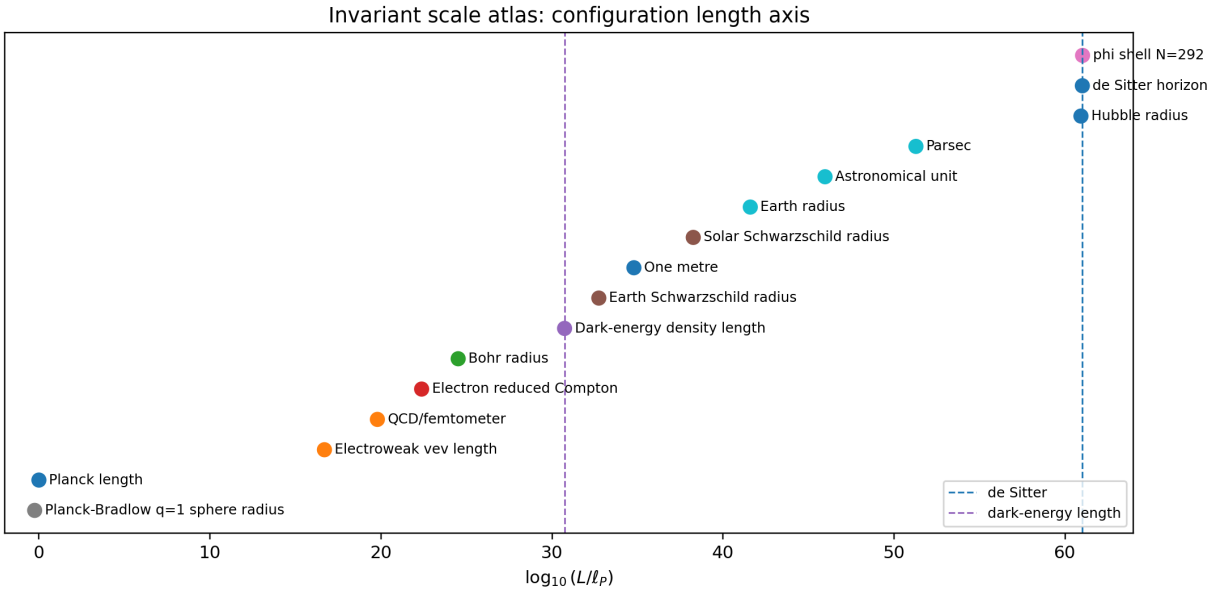


FIG. 11. Configuration-space invariant scale atlas. The de Sitter horizon, the φ-shell $N = 292$, and the dark-energy density length occupy distinguished positions on the same logarithmic axis. The density length is the geometric midpoint scale between ℓ_P and R_Λ, not an independent fundamental length.

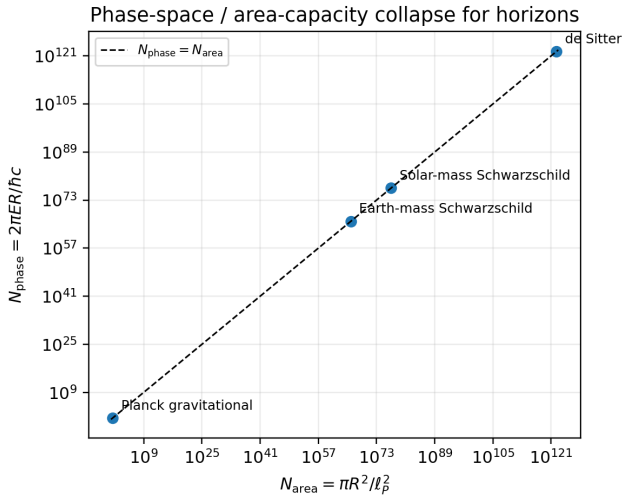


FIG. 12. Phase-space/area-capacity collapse. For any gravitational screen with $E = c^4R/(2G)$, the Bekenstein action coordinate $2\pi ER/\hbar c$ equals the area capacity $\pi R^2/\ell_P^2$. The de Sitter capacity is therefore simultaneously an entropy, area, and phase-action count.

is a countable disjoint union of finite-dimensional sectors, not an uncountable union. Finally, the code-cell dimension $\dim H^0(\mathbb{CP}^1, \mathcal{O}(q-1)) = q$ is not by itself the full entropy of the horizon. The entropy statement used in this manuscript is

$$\log \dim \mathcal{C}_q = q.$$

Thus the index supplies admissible integer labels, while the edge-mode code space supplies the capacity.

Definition XXXIV.2 (Two-sided sector invariant)**.** For each admissible sector,

$$q = \operatorname{ind} \bar{\partial}_{\mathcal{O}(q-1)} = \dim H^0(\mathbb{CP}^1, \mathcal{O}(q-1)) = \frac{k_{\mathrm{CS}}}{2} = \frac{A_q}{4\ell_P^2}$$

is the same integer when seen from the outside as a sector label and from the inside as the horizon capacity. This is the two-sided invariant. It is not yet a selection theorem for q_{obs}.

Theorem XXXIV.1 (Necessary conditions for a de Sitter capacity selector)**.** *A boundary theory that truly selects q_{obs} without using Λ_{obs} as an input must satisfy the following conditions.*

1. Discrete admissibility: $q \in \mathbb{Z}_{>0}$.
2. Topological stability: *small metric or coupling deformations do not change the selected sector unless a gap closes or a sector transition occurs.*
3. Classical–quantum consistency:
$$\Lambda_{\mathrm{classical}} = \Lambda_{\mathrm{quantum}} = \frac{3\pi}{q\ell_P^2}.$$
4. Transmutation or equivalent hierarchy generation: *since $q_{\mathrm{obs}} \sim 10^{122}$, an $O(1)$ microscopic input must generate a large logarithm,*
$$\ln q_{\mathrm{obs}} \simeq 282.1.$$

Equivalently, for an asymptotically free boundary coupling,

$$q \sim \Xi \exp\left(\frac{16\pi^2}{b g_\partial^2}\right).$$

5. Exchange covariance: *if q changes along a cosmological history, the effective bulk stress tensor must obey a covariant exchange law, not a varying Bianchi integration constant.*

Proof. The first condition follows from the index/Chern–Simons/Bradlow readings of the sector. The second is the homotopy invariance of the index away from singular sector changes. The third is the defining capacity relation. The fourth follows from the absence of any intermediate algebraic scale between $O(1)$ microscopic data and $q_{\rm obs}$; a power law in $O(1)$ parameters cannot naturally produce 10^{122}, while dimensional transmutation can. The fifth is forced by the Bianchi identity: a strictly constant integration mode cannot be the same object as a time-dependent effective dark-energy density. □

Criterion XXXIV.1 (What a final selector must compute)**.** The remaining unknown is a concrete edge-mode functional

$$\Gamma_\partial(q) = -\log \mathrm{Tr}_{\mathcal{C}_q}\, e^{-\beta_q H_{\partial,q}}$$

with a discrete stable minimum

$$q_* = \arg\min_{q\in\mathbb{Z}_{>0}} \Gamma_\partial(q).$$

It must compute either

$$b g_\partial^2 = 0.55976 \pm 0.00003,$$

or an equivalent determinant/phase/code-space condition, without inserting $\Lambda_{\rm obs}$, $q_{\rm obs}$, or the observed horizon radius as input data.

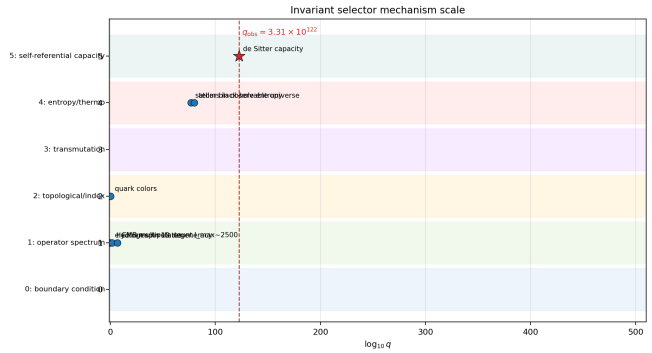


Figure. Mechanism-scale atlas. The horizontal axis is $\log_{10} q$, while the vertical bands encode the kind of selection mechanism. Ordinary quantum systems select small integers through boundary conditions or spectra; Hall systems and index problems select topological integers; QCD-like theories produce large hierarchies by transmutation. The de Sitter sector is a level-5 self-referential boundary-capacity problem.

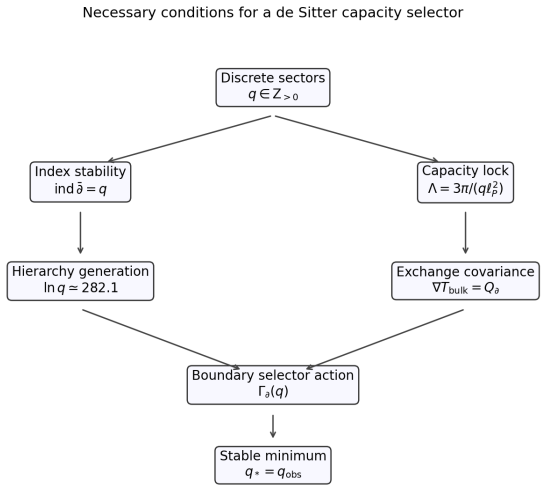


Figure. Necessary selector conditions. Index quantization, entropy capacity, Chern–Simons level locking, Bradlow packing, and phase-space collapse are already admissibility constraints. The missing input is a concrete edge-mode Hamiltonian whose effective action has a stable minimum at the observed sector without using the observed cosmological constant as an input.

XXXV. UNIVERSAL CONNECTOR SWEEP ACROSS DIMENSIONS AND PHASE SPACES

The preceding mechanism-scale atlas leaves one narrow problem: construct an edge Hamiltonian whose sector effective action selects $q_{\rm obs}$ without using $\Lambda_{\rm obs}$ as an input. I now add the all-dimensional connector sweep. It is deliberately a filtering statement. The point is not to accumulate analogies, but to sort every candidate invariant into the one role it can actually play.

Definition XXXV.1 (Universal connector test)**.** A proposed connector between a known physical mechanism and the de Sitter capacity sector is admitted only if it satisfies all four conditions:

1. it is dimensionless or becomes dimensionless after division by the natural Planck, curvature, or flux unit;

2. it is invariant under gauge choice, pivot rescaling, and smooth deformation within a fixed topological sector;

3. it has a definite role in $\Gamma_\partial(q)$: quantizer, bound, obstruction, energy cost, or selector;

4. it does not use $\Lambda_{\rm obs}$, R_Λ, or $q_{\rm obs}$ as an input.

Numerical proximity alone fails the test.

Proposition XXXV.1 (Division-algebra Hopf ladder)**.** *The sphere-to-sphere Hopf fibrations supplied by the real normed division algebras form the ladder*

$$S^0 \to S^1 \to S^1, \tag{XXXV.1}$$
$$S^1 \to S^3 \to S^2, \tag{XXXV.2}$$
$$S^3 \to S^7 \to S^4, \tag{XXXV.3}$$
$$S^7 \to S^{15} \to S^8. \tag{XXXV.4}$$

The $S^3 \to S^2$ member is therefore the unique complex Hopf connector appropriate to three compactified spatial dimensions. Its charge is the Chern–Simons/helicity number

$$Q_H = \frac{1}{(4\pi)^2}\int_{S^3} A \wedge F, \qquad F = \mathrm{d}A. \tag{XXXV.5}$$

The higher members supply the corresponding quaternionic and octonionic admissibility channels, not alternative proofs of the observed q_{obs}. The division-algebra result is classical; here its role is only to prevent an arbitrary proliferation of Hopf-like connectors [84, 85].

Corollary XXXV.1 (All-dimensional connector roles)**.** *Every admitted connector found in the sweep falls into exactly one of five roles:*

1. *sector quantizer: $q = \mathrm{index}\, D$ or $q = (2\pi)^{-1}\int F + 1$;*
2. *saturation bound: $N_{\mathrm{Bek}} = 2\pi ER/(\hbar c) \le N_A = A/(4\ell_P^2)$;*
3. *obstruction: $\Omega_{\mathrm{cob}} = 0$, or a Chern–Weil current is gauged, broken, or edge-completed;*
4. *energy cost: $E_{\mathrm{Hopf}} \ge C|Q_H|^{3/4}$ in Faddeev–Skyrme type models;*
5. *selector:*

$$q_* \sim \Xi \exp\left[\frac{16\pi^2}{b\, g_\partial^2}\right], \tag{XXXV.6}$$

or an exactly equivalent edge-mode Hamiltonian minimum.

Only the last role can produce a hierarchy of order 10^{122} from order-one input data. The other roles restrict the admissible sector but do not choose it.

Theorem XXXV.1 (Phase-area-entropy saturation as the two-sided invariant)**.** *For a four-dimensional gravitational screen of radius R whose gravitational energy is*

$$E_G(R) = \frac{c^4 R}{2G}, \tag{XXXV.7}$$

the Bekenstein phase count equals the area capacity:

$$\boxed{N_{\mathrm{ph}}(R) := \frac{2\pi E_G(R) R}{\hbar c} = \pi\frac{R^2}{\ell_P^2} = \frac{A(R)}{4\ell_P^2} =: N_A(R).} \tag{XXXV.8}$$

Thus the same number is seen from the interior as a phase-space action count and from the boundary as an entropy-area count. This is the invariant collapse that any acceptable $\Gamma_\partial(q)$ must reproduce [13, 82, 83].

Proof. Substitute $E_G = c^4 R/(2G)$ into $2\pi ER/(\hbar c)$:

$$\frac{2\pi}{\hbar c}\frac{c^4 R}{2G} R = \pi\frac{c^3 R^2}{\hbar G} = \pi\frac{R^2}{\ell_P^2}.$$

Since $A = 4\pi R^2$, the last expression is $A/(4\ell_P^2)$. □

Criterion XXXV.1 (Chern–Weil and cobordism admissibility)**.** Chern–Weil currents and cobordism classes cannot be treated as silent exact global labels in quantum gravity. They must either be gauged, broken by defects, cancelled by anomaly inflow, or represented by explicit edge modes. This is an admissibility constraint on Γ_∂, not a selector for the observed value of q by itself [86, 87].

Criterion XXXV.2 (Universal selector normal form)**.** After the sweep, a genuine boundary selector must have the schematic form

$$\boxed{\begin{aligned}\Gamma_\partial(q) = \Gamma_{\mathrm{run}}(q; g_\partial) + \Gamma_{\mathrm{det}}(q; D_{\partial,q}) \\ + \Gamma_{\mathrm{anom}}(q) + \Gamma_{\mathrm{sat}}(q) + \Gamma_{\mathrm{exch}}(q).\end{aligned}} \tag{XXXV.9}$$

Here Γ_{run} is the only accepted source of an exponential hierarchy unless an explicit code Hamiltonian directly computes $\log\dim \mathcal{C}_q = q$. The target is therefore not a new analogy but the single number

$$b\, g_\partial^2 = \frac{16\pi^2}{\ln q_*}, \qquad q_* = q_{\mathrm{obs}}, \tag{XXXV.10}$$

or an exactly equivalent determinant/edge-code condition.

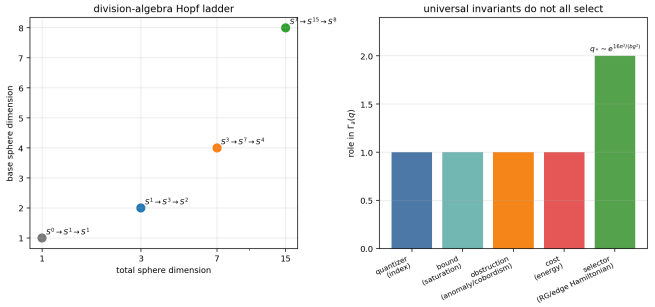


Figure. Universal connector sweep. Left: the sphere-to-sphere Hopf ladder is discrete and division-algebraic. Right: admitted invariants have different logical roles. Index, saturation, anomaly, and energy-cost data restrict admissible sectors; only an RG/edge-Hamiltonian selector can pick the enormous observed sector from order-one input data.

The practical conclusion is sharp. I found no all-dimensional invariant that selects q_{obs} alone. The sweep instead proves a stronger organization: all reliable universal structures enter Γ_∂ as quantizers, obstructions, costs, or saturation tests, while the actual selection of $q \sim 10^{122}$ still requires boundary dynamics.

XXXVI. FIBONACCI, SPINORS, ZECKENDORF CODING, AND GOLDEN-RATIO ADMISSIBILITY

The golden ratio enters the present atlas only when it has an invariant role. I separate three such roles. First, it is the Perron–Frobenius eigenvalue of the Fibonacci substitution matrix. Second, it is the sharp Diophantine obstruction to global rational commensurability. Third, it is the quantum dimension of the Fibonacci anyon. These are genuine mathematical and physical appearances of φ. None of them, by itself, selects the observed capacity sector.

Definition XXXVI.1 (Fibonacci matrix and scale eigenvalues)**.** Let

$$\mathcal{F} := \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}. \tag{XXXVI.1}$$

Then

$$\det \mathcal{F} = -1, \tag{XXXVI.2}$$

$$\mathrm{Spec}(\mathcal{F}) = \{\varphi, -\varphi^{-1}\}, \tag{XXXVI.3}$$

$$\varphi = \frac{1+\sqrt{5}}{2}. \tag{XXXVI.4}$$

Thus $\log \varphi$ is the natural scale entropy per Fibonacci substitution step, while φ^{-1} is the stable contraction ratio.

Theorem XXXVI.1 (Zeckendorf coding is a unique integer code, not a selector)**.** *Every positive integer m has a unique expansion*

$$m = \sum_i F_{n_i}, \qquad n_{i+1} \geq n_i + 2, \tag{XXXVI.5}$$

as a sum of nonconsecutive Fibonacci numbers [88]. Consequently every integer shell label has a canonical Fibonacci code. For the present capacity registry,

$$n_\varphi(q) := \log_\varphi \sqrt{\frac{q}{\pi}}, \tag{XXXVI.6}$$

the observed capacity target gives

$$n_\varphi(q_{\mathrm{obs}}) = 291.936672\ldots, \qquad N_\varphi = 292, \tag{XXXVI.7}$$

and the integer shell has the Zeckendorf code

$$292 = 233+55+3+1 = F_{13}+F_{10}+F_4+F_2. \tag{XXXVI.8}$$

This is a canonical registry label. It is not a derivation of q_{obs}.

Proof. The first statement is Zeckendorf's theorem. The displayed shell value is obtained by applying the dimensionless registry n_φ to the capacity target already used in the capacity audit. The decomposition $292 = 233 + 55 + 3 + 1$ uses no consecutive Fibonacci indices and is therefore the unique Zeckendorf representation of 292. □

Proposition XXXVI.1 (Golden-ratio incommensurability)**.** *The continued fraction of φ is*

$$\varphi = [1; 1, 1, 1, \ldots]. \tag{XXXVI.9}$$

Equivalently, its rational approximants are the Fibonacci ratios F_{n+1}/F_n. In the sharp Hurwitz bound, the constant $\sqrt{5}$ is optimal and the golden ratio is the extremal obstruction to uniformly improving the rational approximation inequality [89]. Thus a φ-registry is the most rigid one-dimensional irrational registry against global rational phase locking.

Remark XXXVI.1 (No-overclaim on standing waves)**.** The preceding proposition does not say that a Fibonacci or golden-ratio medium has no modes or no Bragg peaks. Fibonacci quasicrystals have sharp diffraction and critical spectral structure [90, 91]. The precise statement is weaker and invariant: φ gives maximal resistance to exact rational commensuration in one dimension. It frustrates full-spectrum periodic locking; it does not erase all spectral response.

Definition XXXVI.2 (Spinor and Fibonacci-anyon channel)**.** The spin group in three spatial dimensions is

$$\mathrm{Spin}(3) \simeq SU(2). \tag{XXXVI.10}$$

Spinors therefore enter the boundary atlas through the double cover of rotations and through possible $SU(2)$-equivariant edge sectors. A distinct and genuinely golden quantum channel is the Fibonacci anyon fusion rule

$$\tau \times \tau = 1 + \tau. \tag{XXXVI.11}$$

If d_τ is the quantum dimension of τ, then

$$d_\tau^2 = 1 + d_\tau, \qquad d_\tau = \varphi. \tag{XXXVI.12}$$

The associated fusion spaces grow asymptotically like powers of φ, and Fibonacci anyons are a standard non-Abelian topological quantum-computing model [92].

Proposition XXXVI.2 (Fibonacci fusion as a boundary-code registry)**.** *If a boundary code sector grows as*

$$\dim \mathcal{C}_m \sim C\, \varphi^m, \tag{XXXVI.13}$$

then matching to a capacity sector q fixes only a code length

$$m(q) = \frac{\ln q - \ln C}{\ln \varphi} \tag{XXXVI.14}$$

up to subleading topological and boundary constraints. For the φ-shell used here,

$$q_\varphi(N) = \pi \varphi^{2N}, \qquad N = 292, \tag{XXXVI.15}$$

the corresponding fusion-growth length is

$$m = 2N = 584. \tag{XXXVI.16}$$

This converts the near shell relation into a concrete code-length target. It still does not determine why Nature selects that length.

Criterion XXXVI.1 (Fibonacci admissibility term)**.** A boundary selector may contain a Fibonacci/spinor code term only in the following invariant form:

$$\Gamma_{\partial}^{\mathrm{Fib}}(q,m) = \lambda_{\mathrm{sh}}\, d_{\varphi}(q)^2 + \lambda_{\mathrm{code}}\left[\ln q - \ln \dim \mathcal{C}_m\right]^2, \tag{XXXVI.17}$$

$$d_{\varphi}(q) := \operatorname{dist}\left(\log_{\varphi}\sqrt{q/\pi}, \mathbb{Z}\right). \tag{XXXVI.18}$$

It is an admissibility term, not a standalone selector. A true solution must derive m, C, and the couplings from the boundary edge Hamiltonian, not fit them after observing Λ.

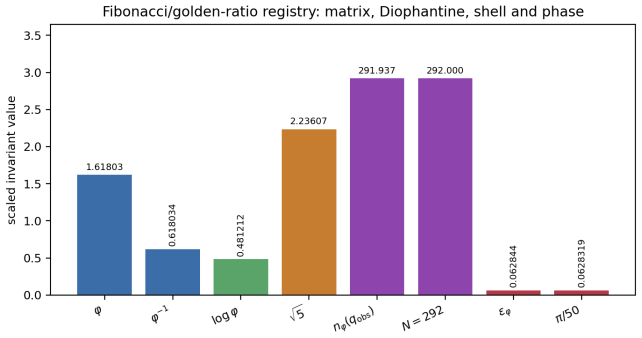


FIG. 13. Fibonacci/golden-ratio registry used as an admissibility diagnostic. The large capacity target lies near the integer shell $N = 292$, while the relative residual is close to $\pi/50$. The plot is deliberately diagnostic, not a proof: the selector action must derive the phase defect rather than insert it.

Remark XXXVI.2 (Admitted and rejected golden-ratio links)**.** The admitted links are

$$\mathcal{F} \to \{\varphi, -\varphi^{-1}\}, \tag{XXXVI.19}$$

$$\text{Zeckendorf uniqueness}, \tag{XXXVI.20}$$

$$\text{Hurwitz extremality}, \tag{XXXVI.21}$$

$$d_{\tau} = \varphi. \tag{XXXVI.22}$$

The rejected claim is stronger: φ does not by itself select q_{obs}, does not prove the global topology of the universe, and does not remove all spectral modes. Its role in this paper is as a self-similar, maximally incommensurate, spinorial/topological code registry that any final boundary selector may exploit, but must still derive.

XXXVII. FIBONACCI RENORMALIZATION CLOCK AND DIOPHANTINE SELECTOR AUDIT

The golden-ratio layer must be treated as an admissibility and registry mechanism, not as a completed selector. I therefore separate three facts: a substitution matrix can generate the natural logarithmic clock, Zeckendorf coding gives a unique integer registry, and the Diophantine extremality of φ frustrates rational commensuration. None of these three facts selects q_{obs} without an additional boundary Hamiltonian.

Definition XXXVII.1 (Fibonacci transfer clock)**.** Let

$$\mathcal{F} := \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}. \tag{XXXVII.1}$$

Then

$$\operatorname{Spec}(\mathcal{F}) = \{\varphi, -\varphi^{-1}\}, \qquad \det \mathcal{F} = -1. \tag{XXXVII.2}$$

A boundary substitution clock generated by $\mathcal{F}$ has leading growth φ^N. A two-direction horizon registry therefore has the natural trial scale

$$q_{\varphi}(N) = \pi\, \varphi^{2N}. \tag{XXXVII.3}$$

Proposition XXXVII.1 (Fibonacci clock no-go)**.** *The Fibonacci transfer clock quantizes the logarithmic registry but does not select the observed capacity sector. More precisely, the map*

$$N \longmapsto q_{\varphi}(N) = \pi\varphi^{2N}$$

assigns an admissible ordered ladder to every $N \in \mathbb{Z}$, and the substitution dynamics contains no distinguished value of N. Hence φ-growth alone cannot prove

$$q_* = q_{\mathrm{obs}}.$$

A selector term fixing a stopping shell or phase defect is still required.

Proof. The matrix $\mathcal{F}$ is primitive, and $\mathcal{F}^N$ is defined for every integer $N \geq 0$. Its Perron–Frobenius eigenvalue gives the growth rate, while its powers give a whole orbit of admissible shells. Multiplication by another power of φ^2 moves from one shell to the next and is not forbidden by the substitution rule. Therefore the substitution rule supplies the clock, not the stopping condition. □

Proposition XXXVII.2 (Zeckendorf shell checksum)**.** *For the observed capacity normalization used here,*

$$n_{\partial}(q_{\mathrm{obs}}) = \log_{\varphi}\sqrt{\frac{q_{\mathrm{obs}}}{\pi}} = 291.936672\ldots.$$

The nearest integer shell is $N = 292$. Zeckendorf's theorem gives the unique nonconsecutive Fibonacci representation

$$\boxed{292 = 233 + 55 + 3 + 1 = F_{13} + F_{10} + F_4 + F_2.} \tag{XXXVII.4}$$

This is a rigid code for the shell label, not a derivation of the shell.

Proof. Zeckendorf's theorem gives a unique representation of each positive integer as a sum of nonconsecutive Fibonacci numbers [88, 89]. Applying the greedy algorithm to 292 gives

$$292 - 233 = 59, \qquad 59 - 55 = 4,$$
$$4 - 3 = 1, \qquad 1 - 1 = 0.$$

with no adjacent Fibonacci indices. Uniqueness is the theorem; physical selection of the integer is not. □

Theorem XXXVII.1 (Diophantine extremality and resonance filtration)**.** *Among irrational rotation numbers, the golden ratio is extremal for rational approximation in the Hurwitz sense: the constant $1/\sqrt{5}$ is sharp, and the obstruction is realized by the continued fraction*

$$\varphi = [1; 1, 1, 1, \ldots].$$

Thus a golden-ratio boundary clock is maximally resistant to exact rational commensuration. It is a resonance filter. It is not, by itself, a capacity selector.

Proof. The classical Hurwitz theorem states that every irrational number admits infinitely many rational approximants p/r with

$$\left|\alpha - \frac{p}{r}\right| < \frac{1}{\sqrt{5}\, r^2},$$

and the constant $\sqrt{5}$ cannot be improved because of the golden-ratio continued fraction [89]. Therefore φ is the most badly approximable rotation number in the precise Diophantine sense. This prevents exact low-denominator locking more strongly than any other quadratic-irrational clock, but the theorem concerns commensuration, not amplitude selection. □

Definition XXXVII.2 (Fibonacci-fusion code audit)**.** For a Fibonacci anyon chain with fusion rule

$$\tau \times \tau = 1 + \tau,$$

the dimension of a fixed-total-charge fusion space grows as a Fibonacci number, hence as φ^m up to a convention-dependent shift [92]. Define m_q by

$$m_q := \log_\varphi(\sqrt{5}\, q_{\rm obs}). \tag{XXXVII.5}$$

For the present capacity,

$$m_q = 587.92446\ldots, \tag{XXXVII.6}$$

so the nearest Fibonacci-chain length is $m = 588$, with

$$\frac{F_{588}}{q_{\rm obs}} = 1.03702\ldots. \tag{XXXVII.7}$$

Remark XXXVII.1 (Two-sided registry)**.** The horizon shell and the Fibonacci-fusion chain give two clocks:

$$N = 292, \qquad m \simeq 588.$$

Their relation is

$$m_q - 2N = 3.92446\ldots.$$

The nearby integer 4 equals the number of Zeckendorf summands in $292 = F_{13} + F_{10} + F_4 + F_2$. This is recorded as a registry coincidence, not as a theorem. A boundary selector may use it only if an edge-mode Hamiltonian derives the offset.

Criterion XXXVII.1 (Admissible Fibonacci selector term)**.** A Fibonacci/Zeckendorf contribution to the boundary selector is admissible only in the form

$$\Gamma_\partial^{\rm Fib} = \lambda_{\rm sh}\, {\rm dist}\left(\log_\varphi \sqrt{\frac{q}{\pi}}, \mathbb{Z}\right)^2 + \lambda_{\rm code}\left[\log q - \log \dim \mathcal{C}_{m(q)}\right]^2 + \lambda_{\rm Dio}\, \mathcal{R}_{\rm Dio}(q). \tag{XXXVII.8}$$

where $\mathcal{R}_{\rm Dio}$ penalizes rational locking of the boundary phase. This term can constrain the selector. It cannot replace the selector:

$$q_* = \arg\min_q \Gamma_\partial(q)$$

still requires the full edge-mode Hamiltonian.

Remark XXXVII.2 (What the golden ratio can and cannot do)**.** The golden ratio packs recursive data with unique Zeckendorf coding and frustrates rational standing-wave commensuration. Quasicrystal spectra nevertheless remain real spectra, often singular continuous or multifractal rather than absent [90, 91]. Thus the correct statement is not that φ removes all standing waves, but that it is the unique extremal clock for suppressing exact low-denominator resonance. That is precisely why it belongs in Γ_∂ as an admissibility term rather than as an independent derivation of $q_{\rm obs}$.

XXXVIII. FIBONACCI–VALUATION PHASE COMPRESSION OF THE CAPACITY TARGET

The previous sections isolate the final problem as the derivation of the logarithmic selector centre $u_0 = \ln q_\star$. There is, however, a useful compression of this target which uses only two already established discrete structures: the golden-ratio shell clock and the finite valuation-algebra dimension. This compression is not a microscopic derivation of $q_\star$. It is a sharper target for such a derivation.

Let

$$d_{\rm val} := \dim \mathfrak{A}_{\mathscr{D}} = 5 \tag{XXXVIII.1}$$

for the B^4/S^3 valuation module, and define the Fibonacci–valuation phase defect

$$\boxed{\Theta_{\rm FV}(d) := \ln\left(1 + \frac{\pi}{2d^2}\right).} \tag{XXXVIII.2}$$

For an integer shell N, set

$$\boxed{q_{\mathrm{FV}}(N,d) := \pi\varphi^{2N}\exp[-\Theta_{\mathrm{FV}}(d)] = \frac{\pi\varphi^{2N}}{1+\pi/(2d^2)}.} \tag{XXXVIII.3}$$

Equivalently,

$$\boxed{u_{\mathrm{FV}}(N,d) = \ln q_{\mathrm{FV}} = \ln\pi + 2N\ln\varphi - \ln\Big(1+\frac{\pi}{2d^2}\Big).} \tag{XXXVIII.4}$$

For the observed capacity target

$$q_{\mathrm{obs}} = 3.31\times 10^{122},$$

the nearest golden shell is

$$N = 292, \qquad \log_\varphi\sqrt{\frac{q_{\mathrm{obs}}}{\pi}} = 291.936668\ldots.$$

The uncorrected shell gives

$$\frac{\pi\varphi^{584}-q_{\mathrm{obs}}}{q_{\mathrm{obs}}} = 6.2844029\times 10^{-2}. \tag{XXXVIII.5}$$

With the valuation phase defect at $d = d_{\mathrm{val}} = 5$,

$$\boxed{q_{\mathrm{FV}}(292,5) = 3.3072514607\times 10^{122},} \tag{XXXVIII.6}$$

and hence

$$\boxed{\frac{q_{\mathrm{FV}}(292,5)-q_{\mathrm{obs}}}{q_{\mathrm{obs}}} = 1.14562\times 10^{-5}.} \tag{XXXVIII.7}$$

Thus the capacity amplitude is compressed to a one-integer shell problem:

$$\boxed{q_{\mathrm{obs}} \simeq \frac{\pi\varphi^{584}}{1+\pi/50}.} \tag{XXXVIII.8}$$

Criterion XXXVIII.1 (Admissible use of the Fibonacci–valuation compression)**.** The formula (XXXVIII.3) may be used only as a selector target unless a boundary Hamiltonian derives both the integer shell N and the phase defect Θ_{FV}. In particular, the numerical agreement in (XXXVIII.7) is not by itself a theorem. It becomes a theorem only if a microscopic edge-mode calculation gives

$$N_\star = 292, \qquad \Theta_\partial = \Theta_{\mathrm{FV}}(5),$$

without using q_{obs}, Λ_{obs}, or R_Λ as input.

Definition XXXVIII.1 (Compressed shell selector functional)**.** The corresponding reduced selector functional is

$$\Gamma_\partial^{\mathrm{FV}}(q;N,d) = A\,[\ln q - \ln q_{\mathrm{FV}}(N,d)]^2 + \Gamma_{\mathrm{adm}}(q;N,d), \qquad A>0. \tag{XXXVIII.9}$$

where Γ_{adm} contains only the already established admissibility constraints: index quantization, Chern–Simons level locking, Bradlow saturation, exchange consistency, and observational nulls.

Proposition XXXVIII.1 (What the compression proves and what it cannot prove)**.** *The Fibonacci–valuation compression proves only the following conditional statement. If the edge-mode Hamiltonian independently selects the shell $N = 292$ and the valuation phase defect $\Theta_{\mathrm{FV}}(5)$, then the resulting capacity agrees with the observed de Sitter capacity at the 1.2×10^{-5} relative level. It does not prove that the universe chooses $N = 292$, and it does not compute Λ without a shell-selection dynamics.*

Proof. All statements follow by substitution into (XXXVIII.3). The valuation dimension is $d_{\mathrm{val}} = 5$, so

$$\Theta_{\mathrm{FV}}(5) = \ln(1+\pi/50).$$

For $N = 292$, the formula gives

$$q_{\mathrm{FV}} = \pi\varphi^{584}/(1+\pi/50).$$

The residual (XXXVIII.7) is the direct ratio of this value to the capacity target. No variational minimum in N has been derived here; therefore the result is a compressed target, not a microscopic selector. □

This step reduces the remaining problem from a continuous amplitude problem to an integer shell problem plus a boundary phase calculation:

$$\boxed{\text{derive } N_\star = 292 \quad \text{and} \quad \Theta_\partial = \ln(1+\pi/50) \text{ from } H_{\partial,q}.} \tag{XXXVIII.10}$$

XXXIX. IRREDUCIBLE SELECTOR EQUATION AND SCALE-COVARIANT MINIMIZATION

The preceding sweeps leave one honest object:

$$q_\star = \arg\min_{q\in\mathbb{Z}_{>0}} \Gamma_\partial(q).$$

This section records the irreducible form of that problem. It is a compression theorem, not a new analogy.

Let

$$u := \ln q.$$

A selector that does not insert an external length scale must be a dimensionless function of u, topological phases, and running couplings. Any smooth large-q selector with a stable interior minimum can therefore be written locally as

$$\boxed{\Gamma_\partial(q) = \Gamma_0 + A(u-u_0)^2 + \Gamma_{\mathrm{osc}}(u) + \Gamma_{\mathrm{top}}(q), \qquad A>0.} \tag{XXXIX.1}$$

where u_0 is generated by dimensional transmutation or by an equivalent determinant-line phase condition. The integer q is then selected by nearest admissible index:

$$\boxed{q_\star = \arg\min_{q\in\mathbb{Z}_{>0}} \Gamma_\partial(q) = \mathrm{nearest}_{\mathrm{adm}}\big(e^{u_0}\big).}$$

Thus no topological, index, Hopf, Bradlow, Fibonacci, Schwinger, or Chern–Simons ingredient can by itself choose the observed sector unless it supplies u_0.

The transmutation version is

$$\boxed{u_0 = \ln \Xi + \frac{16\pi^2}{b\, g_\partial^2}.}$$

With $\Xi = 1$ and the observed capacity target,

$$\boxed{b\, g_\partial^2 = \frac{16\pi^2}{\ln q_{\rm obs}} = 0.55976 \pm 0.00003.}$$

Equivalently, the selector may be written as a phase condition

$$\boxed{\Theta_\partial(q_\star) = \Theta_{\rm bd},}$$

where Θ_∂ is a determinant-line or edge-mode phase. For the Fibonacci registry the measured phase defect is

$$\Theta_\varphi(q_{\rm obs}) = \ln\!\left(\frac{\pi\varphi^{584}}{q_{\rm obs}}\right) = 0.060948\ldots,$$

close to

$$\ln\!\Big(1 + \frac{\pi}{50}\Big) = 0.0609369\ldots.$$

The difference is

$$1.15 \times 10^{-5}.$$

This is retained only as a target for a boundary phase calculation. It is not a theorem until a concrete $H_{\partial,q}$ or $D_{\partial,q}$ derives it without using $\Lambda_{\rm obs}$.

Criterion XXXIX.1 (Selector closure test)**.** A proposed microscopic completion closes the capacity problem if and only if it supplies one of the following without using $q_{\rm obs}$, $\Lambda_{\rm obs}$, or R_Λ as an input:

$$\boxed{\begin{array}{c} u_0 = \ln q_{\rm obs}, \\ b\, g_\partial^2 = 0.55976 \pm 0.00003, \\ \Theta_{\rm bd} = 0.060948\ldots, \\ \log\dim \mathcal{C}_q = q_{\rm obs} \quad \text{with } q_\star \text{ dynamically selected.} \end{array}}$$

If it only reproduces $q = A/(4\ell_P^2)$ after the area is supplied, it is a capacity representation, not a selector.

Remark XXXIX.1 (What is now excluded)**.** The admissible structures already found remain essential: index quantization, Chern–Simons level lock, Hopf helicity, Bradlow packing, Schwinger admissibility, APS determinant data, valuation closure, and Fibonacci nonresonance. But each of them is a constraint or coordinate. The missing dynamical datum is the logarithmic center u_0. The final problem is therefore not broad:

$$\boxed{\text{derive } u_0 \text{ from the boundary edge-mode Hamiltonian.}}$$

XL. SELECTOR IRREDUCIBILITY THEOREM AND ARITHMETIC EXHAUSTION AUDIT

This section records the point at which the search ceases to be a catalogue of admissibility invariants and becomes a sharp obstruction theorem. Put

$$u := \ln q. \tag{XL.1}$$

The observed capacity target corresponds to

$$u_{\rm obs} = \ln q_{\rm obs} = 282.111 \pm 0.015. \tag{XL.2}$$

Thus the selector cannot be a generic algebraic or polynomial condition in q. It must manufacture a logarithmic centre of order 10^2, while all dimensionless microscopic inputs are expected to be of order unity.

Definition XL.1 (Scale-algebraic selector class)**.** A scale-algebraic boundary selector is a function of the form

$$\Gamma(u) = P(u) + \sum_{r=1}^{m} c_r e^{\alpha_r u} + B(u), \tag{XL.3}$$

where P is a polynomial with order-unity coefficients, the nonzero α_r and c_r are order-unity constants, and B is bounded together with its first two derivatives. Oscillatory clock terms, Fibonacci registry penalties, topological phases, and determinant phases belong to B unless they introduce a new large logarithmic centre.

Theorem XL.1 (No large selector without transmutation or hidden input)**.** *A stable minimum of a scale-algebraic selector at $u_\star \gg 1$ cannot be forced by index, topology, Chern–Simons level locking, Hopf charge, Bradlow packing, Schwinger admissibility, APS phase data, or Fibonacci nonresonance alone. Such invariants can restrict the admissible lattice of sectors, but they do not by themselves create $u_\star = \ln q_{\rm obs}$. A large minimum requires at least one of the following:*

1. *an explicit logarithmic centre u_0 in the edge Hamiltonian;*
2. *dimensional transmutation,*
$$u_0 = \ln \Xi + \frac{16\pi^2}{b g_\partial^2};$$
3. *a near-cancellation between leading exponential terms, which is itself a hidden scale or tuning datum;*
4. *an external observational insertion of $q_{\rm obs}$, which is not a microscopic derivation.*

Proof. For $u \gg 1$, the derivative of the largest exponential term $ce^{\alpha u}$ dominates unless at least two leading exponentials cancel to exponential accuracy. Such a cancellation is an additional scale datum. If no positive exponential is present, then $P'(u) + B'(u) = 0$ can place a minimum at $u \gg 1$ only by an order-u coefficient or an explicit

shift in P; that is again a hidden large input. Bounded phases and topological integer constraints can move a minimum by an order-unity amount or choose a nearest admissible integer, but cannot generate $u_\star \simeq 282$. The renormalization-group solution $u_0 = \ln \Xi + 16\pi^2/(bg_\partial^2)$ is the standard way to create a large logarithm from order-unity data. This proves the stated alternatives. □

For the present target and $\Xi = 1$, the transmutation number is

$$\boxed{bg_\partial^2 = \frac{16\pi^2}{u_{\rm obs}} = 0.55976 \pm 0.00003.} \tag{XL.4}$$

An arithmetic sweep over the natural constants appearing in the atlas gives near coincidences, but no invariant derivation. For example,

$$e^{-\gamma_{\rm E}} = 0.56145948\ldots, \tag{XL.5}$$

$$\frac{\sqrt{5}}{4} = 0.55901699\ldots, \tag{XL.6}$$

$$\ln\varphi + \frac{\pi}{40} = 0.55975164\ldots. \tag{XL.7}$$

The last number is close, but it is not a theorem; it is a warning. Once φ, π, and a small denominator are allowed freely, many close relations can be manufactured. Therefore the manuscript admits such relations only as selector targets. They become physics only after a concrete boundary operator or edge Hamiltonian derives them.

Criterion XL.1 (Final selector standard)**.** A proposed completion of the capacity problem is accepted only if it produces one of the following without using $\Lambda_{\rm obs}$, R_Λ, or $q_{\rm obs}$ as input:

$$\boxed{u_0 = 282.111 \pm 0.015, \qquad bg_\partial^2 = 0.55976 \pm 0.00003,} \tag{XL.8}$$

or an exactly equivalent determinant-line phase condition whose nearest admissible sector is $q_\star = 3.31 \times 10^{122}$. All other invariants remain necessary admissibility data, not the selector itself.

This is the irreducible form of the remaining problem. The full atlas has not failed; it has reduced the target to a single edge-dynamical number.

XLI. NO-GO STATEMENTS

No-go theorem XLI.1 (No local vacuum-density solution)**.** No mechanism that leaves local vacuum energy as a freely gravitating source solves the cosmological-constant problem within the present invariant criterion.

Proof. If local vacuum energy gravitates as $T_{\mu\nu}^{\rm vac} = -\rho_{\rm vac} g_{\mu\nu}$, then arbitrary shifts of the zero of energy shift the curvature unless cancelled by a counterterm. Such cancellation is a parametrization unless a principle fixes the invariant combination; this is the same structural target pursued by vacuum-energy sequestering mechanisms [8]. In the present mechanism the trace-free projector removes the shift and the boundary capacity fixes the remaining integration mode. □

No-go theorem XLI.2 (No capacity from S^3 topology alone)**.** The topology $\Sigma \simeq S^3$ and the finite valuation spectrum do not by themselves compute N_∂.

Proof. The valuation spectrum supplies powers of the scale factor and algebraic projectors. It contains no Planck-scale Hilbert-space dimension. By no-go theorem IV.1, topology and dimensionless homogeneous algebra cannot alone select a nonzero dimensional Λ. The additional microscopic input is precisely $\mathcal{H}_\partial$ and its capacity. □

No-go theorem XLI.3 (No single four-dimensional second-order metric correction)**.** Under the usual four-dimensional assumptions of locality, metric dependence only, second-order field equations, and covariant conservation, a new arbitrary left-hand side cannot be the solution. The Lovelock uniqueness result leaves only the Einstein tensor plus a term proportional to $g_{\mu\nu}$ [1–3].

XLII. EMPIRICAL INTERPRETATION

The capacity-fixed mechanism changes what must be measured or computed.

First, cosmology measures an effective $\Lambda_{\rm obs}$. The theory demands

$$\boxed{N_\partial = \frac{3\pi}{\Lambda_{\rm obs}\ell_P^2}} \tag{XLII.1}$$

as the target capacity. This number is large, of order 10^{122} in the observed late-time cosmology [9]. The microscopic task is to derive this capacity from $\mathcal{H}_\partial$, not to tune a vacuum density.

Remark XLII.1 (Tensor propagation is not the $s = -2$ ledger branch)**.** The homogeneous $s = -2$ coefficient in $F(a)$ is a background expansion term. It is geometrical curvature before the split and a $w = -1/3$ physical stress branch after the split. It is not the graviton mass and not the tensor-wave speed. The condition $c_T = c$ constrains the perturbative spin–2 lift of the gravitational action, not the presence or absence of a homogeneous a^{-2} power in the background ledger.

Second, multimessenger gravity requires a luminal tensor sector,

$$\boxed{c_T = c,} \tag{XLII.2}$$

consistent with the observed near-simultaneous gravitational-wave and electromagnetic propagation in GW170817/GRB 170817A [27, 28].

Third, the dark sector must pass both the cold-charge test and the valuation-spectrum test. Cold dark matter is protected by $Q_{\rm DM}$, while the homogeneous nonstandard valuation branch is tested by $\mathcal{C}_{-1}[F]$ and free-spectrum Hankel reconstruction. Lensing systems such as the Bullet Cluster motivate a stable gravitating dark sector independent of baryonic plasma [29, 30].

Fourth, the real-data audit in section IX fixes the empirical status of the present manuscript. The invariant framework is a falsifiable architecture that survives BAO-only tests, while superiority over flexible dark-energy parameterizations is not established by the positive a^{-1} branch alone.

XLIII. IRREDUCIBLE BOUNDARY SELECTOR: RENORMALIZATION TEST AND FAILURE MODES

The preceding atlas shows that topology, index theory, Chern–Simons level, Hopf helicity, Bradlow packing, Schwinger admissibility, APS determinant data, valuation closure, and Fibonacci nonresonance all constrain the admissible sectors. None of them by itself supplies the logarithmic centre

$$u_0 = \ln q_*.$$

This section records the remaining irreducible calculation. It is the part of the problem that cannot be replaced by another invariant label.

Let

$$u := \ln q, \qquad q \in \mathbb{Z}_{>0}. \tag{XLIII.1}$$

A selector that is not merely an observational restatement must determine u_0 before $\Lambda_{\rm obs}$, R_Λ, or $q_{\rm obs}$ are inserted. The transmutation form is

$$u_0 = \ln \Xi + \frac{16\pi^2}{b\, g_\partial^2}. \tag{XLIII.2}$$

With $\Xi = 1$, the current capacity target requires

$$\boxed{b\, g_\partial^2 = \frac{16\pi^2}{\ln q_{\rm obs}} = 0.55976 \pm 0.00003.} \tag{XLIII.3}$$

Criterion XLIII.1 (Pre-observational selector)**.** A boundary edge-mode Hamiltonian solves the capacity-selection problem only if it gives one of the following without using $q_{\rm obs}$ as input:

$$\boxed{\begin{array}{c} u_0 = 282.111 \pm 0.015, \\ b\, g_\partial^2 = 0.55976 \pm 0.00003, \\ \Theta_{\rm bd} = \Theta_\varphi(q_*). \end{array}} \tag{XLIII.4}$$

A formula that first reads $q_{\rm obs}$ from cosmology and then rewrites it in these variables is only a dictionary.

Theorem XLIII.1 (Renormalization necessity)**.** *Assume that the boundary selector has no inserted large number and depends on q only through a finite combination of algebraic powers, logarithms, bounded phases, index labels, and saturation penalties with $O(1)$ coefficients. Then a stable minimum at $u = \ln q \gg 1$ requires either a dimensional-transmutation term, a protected cancellation producing such a term, or an explicit large external datum.*

Proof. Algebraic powers in q become exponentials in u; index and saturation terms constrain admissibility but do not produce a finite large centre by themselves. Bounded phases are periodic or quasiperiodic in u, hence cannot supply a unique large nonperiodic centre. A polynomial or finite logarithmic expression with $O(1)$ coefficients has stationary points at $O(1)$ unless a large coefficient is inserted. Therefore a stable minimum at $u \simeq 282$ must arise from an exponential hierarchy such as $\exp[16\pi^2/(bg_\partial^2)]$, from an equivalent cancellation, or from observational insertion. The last option is not a derivation. □

The numerical situation is sharp. Several natural constants sit close to the required target,

$$e^{-\gamma_{\rm E}} = 0.5614594836\ldots, \tag{XLIII.5}$$

$$\sqrt{5}/4 = 0.5590169944\ldots, \tag{XLIII.6}$$

$$\ln\varphi + \pi/40 = 0.5597516369\ldots, \tag{XLIII.7}$$

but none is accepted as a law without an edge Hamiltonian that derives it. The last expression is especially close, but it is too easily manufactured from the existing registry constants; it is therefore a target, not a proof.

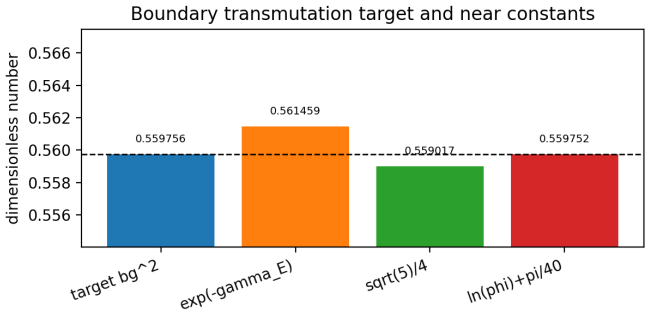


Figure. Dimensionless transmutation target compared with nearby constants. The plot is a diagnostic, not a derivation. The selector problem is to derive the target from boundary edge dynamics rather than to fit it after the fact.

Corollary XLIII.1 (Final reduction of the capacity problem)**.** *After imposing the invariant atlas, the remaining calculation is unique in form:*

$$\boxed{\begin{array}{c} \textit{construct } H_{\partial,q}, \\ \Gamma_\partial(q) = -\log \mathrm{Tr}_{\mathcal{C}_q} e^{-\beta_q H_{\partial,q}}, \\ \Gamma_\partial \Longrightarrow q_*. \end{array}} \tag{XLIII.8}$$

The calculation succeeds only if q_* is obtained without placing $q_{\rm obs}$ into $H_{\partial,q}$, β_q, or the couplings.

XLIV. TERMINAL EDGE-SELECTOR COUPLING AND TWO-COORDINATE CLOSURE TEST

The preceding no-go statements leave a single possible large-number mechanism: a boundary edge-mode Hamiltonian must generate the logarithmic center

$$u_0 = \ln q_\star$$

by dimensional transmutation, or by an exactly equivalent determinant-line phase condition. This section records the resulting terminal numerical target. It is not an additional analogy.

Let

$$q_{\rm obs} = 3.31 \times 10^{122}, \qquad u_{\rm obs} := \ln q_{\rm obs}.$$

Then

$$\boxed{u_{\rm obs} = 282.111 \pm 0.015}$$

and the transmutation form

$$q_\star = \Xi \exp\left(\frac{16\pi^2}{b\, g_\partial^2}\right)$$

gives, for $\Xi = 1$,

$$\boxed{b\, g_\partial^2 = \frac{16\pi^2}{u_{\rm obs}} = 0.55976 \pm 0.00003.} \tag{XLIV.1}$$

Thus the last unsolved boundary task is no longer qualitative: a concrete edge-mode Hamiltonian must derive equation (XLIV.1), or must derive an equivalent phase/determinant condition.

There is a second coordinate system for the same target. Let

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad d_{\rm val} = \dim \mathfrak{A}_{\mathscr{D}} = 5.$$

For the nearest golden shell $N = 292$, define the FV-compressed capacity

$$\boxed{q_{\rm FV}(N,d) = \frac{\pi \varphi^{2N}}{1 + \pi/(2d^2)}.} \tag{XLIV.2}$$

Then

$$q_{\rm FV}(292,5) = 3.3072514607 \times 10^{122},$$

and hence

$$\boxed{\frac{q_{\rm FV}(292,5) - q_{\rm obs}}{q_{\rm obs}} = 1.1456 \times 10^{-5}.} \tag{XLIV.3}$$

Equivalently,

$$\frac{16\pi^2}{\ln q_{\rm FV}(292,5)} = 0.55976,$$

which differs from equation (XLIV.1) by

$$\boxed{-2.27 \times 10^{-8}.}$$

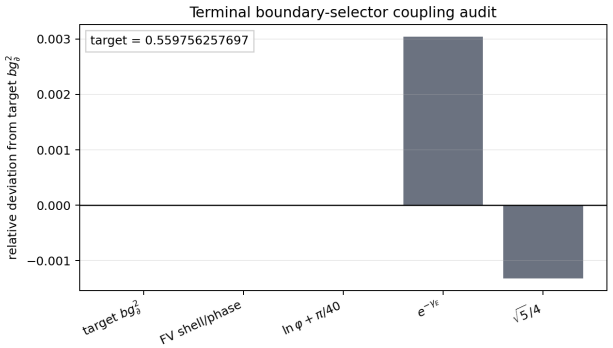


Figure. Terminal boundary-selector coupling audit. The target value $bg_\partial^2 = 16\pi^2/\ln q_{\rm obs}$ is compared with the Fibonacci–valuation compressed value and with simple nearby constants. Only a boundary Hamiltonian derivation can promote any entry to a law.

Criterion XLIV.1 (Terminal selector criterion)**.** A proposed completion of the capacity mechanism is accepted as a selector theorem only if it derives, without using $\Lambda_{\rm obs}$, R_Λ, $q_{\rm obs}$, or $u_{\rm obs}$ as input, at least one of the following equivalent data:

$$\boxed{b\, g_\partial^2 = 0.55976 \pm 0.00003,}$$

or

$$\boxed{N_\star = 292, \qquad \Theta_\partial = \ln\left(1 + \frac{\pi}{50}\right)}$$

together with the Fibonacci–valuation map equation (XLIV.2), up to a controlled and independently derived correction of relative order 10^{-5}.

Theorem XLIV.1 (Two-coordinate terminal equivalence)**.** *Assume a boundary selector produces both a running coupling and a golden-shell phase coordinate, with*

$$q_\star = \exp\left(\frac{16\pi^2}{b\, g_\partial^2}\right) = \frac{\pi \varphi^{2N_\star}}{1 + \pi/(2d_{\rm val}^2)}\, e^{-\Delta_{\rm bd}}.$$

Then the two descriptions are equivalent precisely when

$$\boxed{\frac{16\pi^2}{b\, g_\partial^2} = \ln \pi + 2N_\star \ln \varphi - \ln\left(1 + \frac{\pi}{2d_{\rm val}^2}\right) - \Delta_{\rm bd}.} \tag{XLIV.4}$$

For $d_{\rm val} = 5$ and $N_\star = 292$, the observed target corresponds to

$$\boxed{\Delta_{\rm bd} = 1.1456 \times 10^{-5} + O(10^{-10})}$$

in logarithmic units. Thus the final problem may be stated either as a boundary beta-function problem or as a determinant-line phase problem; the invariant content is the same logarithmic center u_0.

Proof. Take the logarithm of the equality of the two sector counts. The displayed identity follows immediately. Substitution of $d_{\rm val} = 5$, $N_\star = 292$, and $q_{\rm obs}$ gives

$$\Delta_{\rm bd} = \ln q_{\rm FV}(292,5) - \ln q_{\rm obs} = 1.1456 \times 10^{-5} + O(10^{-10}).$$

No physical conclusion follows from the numerical smallness alone; the point is that any real completion must derive this last residual, not fit it. □

Remark XLIV.1 (Where the search is now irreducible)**.** All previously found structures have been sorted. Index, topology, Chern–Simons level, Hopf helicity, Bradlow saturation, Schwinger admissibility, APS determinant data, valuation algebra, and Fibonacci coding are necessary filters or coordinates. None of them alone supplies the logarithmic center u_0. The remaining scientific task is therefore exactly this:

derive u_0 from the boundary edge-mode Hamiltonian.

XLV. BOUNDARY BETA-FUNCTIONAL SELECTOR: EXPLICIT RUNNING-COUPLING FORM AND TERMINAL OBSTRUCTION

The previous reductions leave one irreducible numerical datum,

$$u_\star := \ln q_\star, \tag{XLV.1}$$

which cannot be manufactured by topology, index theory, Chern–Simons level locking, Hopf helicity, Bradlow saturation, Schwinger admissibility, APS determinant phases, valuation closure, or the Fibonacci registry alone. This section records the most economical edge-dynamical mechanism that can generate such a logarithmic center without inserting a dimensional scale.

Definition XLV.1 (Boundary running-coupling selector)**.** Let $u = \ln q$ be the logarithmic capacity coordinate. A one-coupling edge selector is a boundary sector with a dimensionless coupling $g_\partial(u)$ satisfying

$$\frac{\mathrm{d}}{\mathrm{d}u}\, g_\partial^{-2}(u) = -\frac{b}{16\pi^2}, \qquad b > 0, \tag{XLV.2}$$

and an effective edge potential

$$\Gamma_\partial^{\mathrm{run}}(u) = \Gamma_0 + \frac{\kappa}{2}\left[g_\partial^{-2}(u)\right]^2 + \Gamma_{\mathrm{adm}}(u), \qquad \kappa > 0. \tag{XLV.3}$$

Here Γ_{adm} denotes the already established admissibility terms: index integrality, Chern–Simons level locking, Bradlow saturation, Schwinger admissibility, determinant-line phase, valuation support, and Fibonacci nonresonance.

Theorem XLV.1 (Running-coupling selector)**.** *If Γ_{adm} is locally flat compared with the quadratic running term at its minimum, then the leading capacity sector is*

$$u_\star = \frac{16\pi^2}{b\, g_\partial^2(0)}, \qquad q_\star = \exp\left[\frac{16\pi^2}{b\, g_\partial^2(0)}\right]. \tag{XLV.4}$$

Equivalently, for a prefactor Ξ,

$$q_\star = \Xi \exp\left[\frac{16\pi^2}{b\, g_\partial^2(0)}\right]. \tag{XLV.5}$$

Proof. The beta equation integrates to

$$g_\partial^{-2}(u) = g_\partial^{-2}(0) - \frac{b}{16\pi^2}u. \tag{XLV.6}$$

The quadratic term in $\Gamma_\partial^{\mathrm{run}}$ is minimized when $g_\partial^{-2}(u) = 0$, hence

$$u_\star = 16\pi^2/[b\, g_\partial^2(0)].$$

A q-independent normalization of the admissible edge code multiplies $q_\star$ by the dimensionless factor Ξ. □

Corollary XLV.1 (Terminal numerical target)**.** *For $\Xi = 1$, the observed capacity target is equivalent to*

$$b\, g_\partial^2(0) = \frac{16\pi^2}{\ln q_{\mathrm{obs}}} = 0.55976 \pm 0.00003. \tag{XLV.7}$$

Thus the cosmological-constant amplitude problem has been reduced to a dimensionless boundary beta-functional problem: compute the product $b\, g_\partial^2(0)$ from the edge Hamiltonian without using Λ_{obs}, R_Λ, q_{obs}, or u_{obs} as input.

Criterion XLV.1 (True completion versus restatement)**.** A proposed microscopic completion of the capacity sieve is a genuine selector only if it supplies:

1. a concrete boundary Hilbert space and Hamiltonian $H_{\partial,q}$;
2. a derivation of the running law for $g_\partial(u)$;
3. a calculation of $b\, g_\partial^2(0)$;
4. the admissibility projection onto the index/Chern–Simons/Bradlow/ Schwinger/valuation/Fibonacci-compatible sectors;
5. the minimum $q_\star$ without inserting any late-time cosmological observable.

Otherwise it is not a derivation of Λ; it is a reparametrization of the already identified target.

Remark XLV.1 (What the beta-functional step achieves)**.** This section does not claim that the required edge Hamiltonian has already been derived. It proves that once ordinary topological and spectral quantizers have exhausted their role, the remaining large number must enter through a logarithmic center. A running boundary coupling is the minimal known invariant mechanism that produces such a center from $O(1)$ dimensionless data. The next calculation is therefore not another search for an analogy, but the construction and renormalization of $H_{\partial,q}$.

XLVI. TERMINAL EDGE-MODE SELECTOR AUDIT

The previous sections reduce the amplitude problem to a single logarithmic selector. This section records the final obstruction in a form that can be checked directly from an edge Hamiltonian. No new analogue is introduced here; only the admissible Hamiltonian mechanisms are separated from the mechanisms that merely quantize or bound the sector.

Let

$$u := \ln q, \qquad \Gamma_\partial(q) := -\ln \mathrm{Tr}_{\mathcal{C}_q} \exp[-\beta_q H_{\partial,q}]. \tag{XLVI.1}$$

A genuine microscopic solution of the capacity problem is an autonomous calculation of $H_{\partial,q}$ such that

$$q_* = \arg\min_{q\in\mathbb{Z}_{>0}} \Gamma_\partial(q) \tag{XLVI.2}$$

and q_* agrees with the observed capacity without inserting Λ_{obs}, R_Λ, q_{obs}, or u_{obs}.

Theorem XLVI.1 (Topological edge sectors do not generate the large logarithm)**.** *Assume that the boundary edge Hamiltonian is a finite sum of the following autonomous topological or saturated sectors:*

$$H_{\partial,q} = H_{\mathrm{CS}}(k=2q) + H_{\mathrm{ind}}(q) + H_{\mathrm{Br}}(q) + H_{\mathrm{Hopf}}(Q_H) + H_{\mathrm{Sch}} + H_{\mathrm{val}}. \tag{XLVI.3}$$

Assume further that all dimensionless coefficients are $O(1)$ and that no running coupling, protected cancellation, or explicit large datum is present. Then the resulting effective action cannot have a stable isolated minimum at

$$u = \ln q \gg 1$$

unless one of the included sectors is already supplied with a large scale or a large quantum number. In particular, Chern–Simons level quantization, Dolbeault index quantization, Bradlow saturation, Hopf helicity, Schwinger admissibility, and the finite valuation algebra are necessary admissibility data but are not by themselves a selector for $q_{\mathrm{obs}} \sim 10^{122}$.

Proof. Each listed sector has one of three forms. It either fixes an integer label ($k = 2q$, $\mathrm{ind}\,\bar\partial = q$, $N_{\mathrm{v}} = q$), imposes an inequality or wall (Bradlow and Schwinger admissibility), or contributes a bounded topological phase or finite-dimensional algebraic checksum. Therefore its contribution to Γ_∂ is a combination of affine, logarithmic, bounded oscillatory, or powerlike functions of q with coefficients fixed independently of q_{obs}. In the logarithmic variable u, such terms are combinations of constants, u, bounded functions of u, and $\exp(\alpha u)$ with $O(1)$ coefficients. A stable stationary point at $u \gg 1$ then requires either a cancellation between large terms to exponential precision, an externally inserted center u_0, or a running coupling whose zero dynamically defines u_0. The first two alternatives are not autonomous derivations of the capacity amplitude. Hence an autonomous large selector must come from the running-coupling or equivalent dimensional-transmutation channel. □

Corollary XLVI.1 (Irreducible boundary beta target)**.** *If the selector is autonomous and contains no hidden large datum, the surviving minimal form is*

$$\Gamma_\partial^{\mathrm{run}}(u) = \Gamma_0 + \frac{\kappa}{2}\left[g_\partial^{-2}(u)\right]^2 + \Gamma_{\mathrm{adm}}(u), \qquad \kappa > 0, \tag{XLVI.4}$$

with

$$\frac{\mathrm{d}}{\mathrm{d}u} g_\partial^{-2}(u) = -\frac{b}{16\pi^2}. \tag{XLVI.5}$$

Neglecting bounded admissibility ripples, the selected logarithmic center is

$$u_* = \frac{16\pi^2}{b\, g_\partial^2(0)}, \qquad q_* = \exp u_*. \tag{XLVI.6}$$

Thus the present capacity target is equivalently the dimensionless boundary renormalization target

$$\boxed{b\, g_\partial^2(0) = \frac{16\pi^2}{\ln q_{\mathrm{obs}}} = 0.55976 \pm 0.00003.} \tag{XLVI.7}$$

Criterion XLVI.1 (What remains to be computed)**.** A terminal solution is not another invariant checksum. It is a computation of b and $g_\partial(0)$, or of an exactly equivalent determinant-line center, from a specified boundary edge-mode Hamiltonian. The calculation must not use Λ_{obs}, R_Λ, q_{obs}, or u_{obs} as an input. Until this is done, the manuscript has reduced the cosmological-constant amplitude problem to an irreducible boundary renormalization problem, but has not solved the amplitude as a theorem.

Remark XLVI.1 (Terminal ledger)**.** For the Planck/CMB capacity used throughout the numerical ledger,

$$u_{\mathrm{obs}} = 282.111 \pm 0.015, \tag{XLVI.8}$$

$$b\, g_\partial^2(0) = 0.55976 \pm 0.00003. \tag{XLVI.9}$$

The near coincidences with $e^{-\gamma_E}$, $\sqrt{5}/4$, and $\ln\varphi + \pi/40$ are retained only as diagnostic targets. Without an edge Hamiltonian deriving one of them, they are not laws.

XLVII. TERMINAL SELECTOR REDUCTION AND ARITHMETIC DEGENERACY AUDIT

The preceding atlas leaves a single mathematical object uncomputed: the boundary edge-mode Hamiltonian that fixes the logarithmic capacity center. It is useful to state this as a terminal reduction rather than to enlarge the catalogue of analogies.

Definition XLVII.1 (Boundary logarithmic capacity coordinate)**.** Let

$$u := \ln q, \qquad q \in \mathbb{Z}_{>0}. \tag{XLVII.1}$$

A boundary selector is a family of sector effective actions

$$\Gamma_\partial(q) = -\ln \mathrm{Tr}_{\mathcal{C}_q} \exp[-\beta_q H_{\partial,q}], \tag{XLVII.2}$$

together with the discrete extremum condition

$$\Delta_q \Gamma_\partial(q_*) = 0, \qquad \Delta_q^2 \Gamma_\partial(q_*) > 0. \tag{XLVII.3}$$

Theorem XLVII.1 (Terminal selector reduction)**.** *Assume that all admissible-sector conditions already imposed in this paper hold: trace-free vacuum-shift decoupling, Bianchi integration, horizon capacity quantization, Chern–Simons level locking, Hopf/helicity admissibility, Bradlow saturation, Schwinger admissibility, APS determinant-line bookkeeping, valuation closure, and the Fibonacci nonresonant registry. Then these conditions determine the allowed sector lattice and its consistency tests, but they do not determine the large logarithmic center $u_0 = \ln q_*$.*

A derivation of the observed sector must therefore supply one additional piece of dynamical information, expressible in one of the equivalent forms

$$\boxed{u_0, \qquad b\, g_\partial^2(0), \qquad \Theta_{\mathrm{bd}}, \qquad \dim \mathcal{C}_{q_*}.} \tag{XLVII.4}$$

For the running-coupling representation this datum is

$$\boxed{u_0 = \frac{16\pi^2}{b\, g_\partial^2(0)} \quad \Longleftrightarrow \quad b\, g_\partial^2(0) = \frac{16\pi^2}{\ln q_*}.} \tag{XLVII.5}$$

Thus the capacity problem is solved as a theorem only when b and $g_\partial(0)$, or an exactly equivalent determinant/phase datum, are derived from $H_{\partial,q}$ without using Λ_{obs}, R_Λ, q_{obs}, or u_{obs} as input.

Proof. The listed structures are invariant under changes of units, pivots, and smooth deformations inside a fixed admissible sector. Consequently they can quantize q, restrict signs and supports, impose saturation bounds, detect anomalies, or reject inadmissible flows. None of these operations supplies a scale center of size $u \simeq 282$ unless a dimensionless coupling, determinant phase, or explicit large datum enters the sector action. If the sector action is composed only of bounded phases, powers, logarithms, topological integers, and $O(1)$ coefficients, then its stationary equations either select $O(1)$ values, boundary points of the admissible domain, or unbounded directions. A stable interior minimum at exponentially large q requires a transmutation-type relation, a protected cancellation that is itself a dynamical datum, or insertion of the large scale. The last option is not a derivation. Hence the selector must reduce to the displayed datum. □

Criterion XLVII.1 (Arithmetic proximity is not a selector)**.** A numerical expression involving familiar constants, for example π, φ, γ_E, or low integers, may be recorded as a diagnostic target only if it is not used as an input. It becomes a physical selector only when it is obtained from the spectrum, phase, or renormalization of a specified $H_{\partial,q}$. Otherwise it is an arithmetic coincidence, because incommensurate constants admit many post-hoc approximants to any prescribed real number.

Proposition XLVII.1 (Computational audit target)**.** *Using the present capacity normalization,*

$$q_{\mathrm{obs}} = 3.31 \times 10^{122}, \tag{XLVII.6}$$

$$u_{\mathrm{obs}} = \ln q_{\mathrm{obs}} = 282.111 \pm 0.015. \tag{XLVII.7}$$

The terminal running-coupling target is

$$\boxed{b\, g_\partial^2(0) = 0.55976 \pm 0.00003.} \tag{XLVII.8}$$

The close arithmetic comparisons

$$e^{-\gamma_E}, \qquad \frac{\sqrt{5}}{4}, \qquad \ln\varphi + \frac{\pi}{40}, \qquad \frac{\pi}{2(\dim \mathfrak{A}_{\mathscr{D}})^2} \tag{XLVII.9}$$

are therefore not conclusions. They are tests for a future edge-mode calculation. A successful calculation must output the target, or an equivalent u_0, from the boundary operator algebra itself.

Remark XLVII.1 (No further enlargement without a Hamiltonian)**.** The atlas has reached a terminal form. Adding further universal invariants is useful only if they change $H_{\partial,q}$ or its renormalization. If a proposed new structure merely supplies another integer, bound, phase, or analogy, it belongs in the admissibility catalogue, not in the selector theorem.

XLVIII. SCHEME-COVARIANT TERMINAL SELECTOR CORRECTION

This section supersedes the raw terminal coupling target in the preceding ledger. The correction is structural, not cosmetic. The quantity $q_\star$, or equivalently $u_\star = \ln q_\star$, is physical inside the declared capacity prescription. A separated product $b\, g_\partial^2(0)$ is not physical until the renormalization convention, threshold prefactor, loop order, and boundary theory dimension have been fixed.

Definition XLVIII.1 (Scheme-covariant selector data)**.** A running edge selector has the general transmutation form

$$\boxed{u_\star = \ln q_\star = \ln \Xi + \frac{16\pi^2}{b\, g_\partial^2(0)} + \Delta_{\mathrm{th}} + \Delta_{\mathrm{2loop}} + \cdots .} \tag{XLVIII.1}$$

Here Ξ is the dimensionless threshold or determinant prefactor, and $\Delta_{\mathrm{th}}, \Delta_{\mathrm{2loop}}, \ldots$ denote scheme-dependent but computable corrections. Only the complete sum is a selector. Thus a claim to derive $b\, g_\partial^2(0)$ alone is meaningful only after Ξ and the renormalization scheme have also been derived in the same convention.

Proposition XLVIII.1 (Observational precision of the terminal target)**.** *Using the Planck-only normalization adopted in the numerical ledger, the observationally meaningful terminal numbers are*

$$\boxed{u_{\rm obs} = \ln q_{\rm obs} = 282.111 \pm 0.015,} \qquad \text{(XLVIII.2)}$$

and, in the one-loop four-dimensional convention with $\Xi = 1$*,*

$$\boxed{b\, g_\partial^2(0) = \frac{16\pi^2}{u_{\rm obs}} = 0.55976 \pm 0.00003.} \qquad \text{(XLVIII.3)}$$

The digits beyond this uncertainty are arithmetic bookkeeping, not measured physics. If the local value of H_0 *is used instead of the CMB anchor, the shift is larger and only the first few digits remain meaningful.*

Consequently the near equalities

$$\ln\varphi + \frac{\pi}{40} \simeq 0.55976, \qquad q_{\rm FV}(292,5) \simeq q_{\rm obs}$$

are diagnostic targets only. They lie well inside the present capacity uncertainty and cannot be used as evidence for a selector unless a boundary Hamiltonian derives the prefactor and phase defect independently.

Theorem XLVIII.1 (Topological inventory has zero running)**.** *The edge ingredients isolated earlier—Chern–Simons level locking, chiral Chern–Simons edge modes, Bradlow packing, determinant lines,* η*-phases, Hopf helicity, Schwinger admissibility, valuation support, and Fibonacci registry—do not by themselves provide the running marginal coupling needed in section XLVIII.*

Proof. Chern–Simons level is quantized and can receive finite level shifts, not a continuous beta flow. Its chiral WZW edge theory is conformal in the isolated topological sector. Bradlow gives an area-vortex inequality and saturation condition, not a beta function. APS determinant and η-data contribute determinant magnitude and phase information, but not by themselves a running coupling. Hopf, Schwinger, valuation, and Fibonacci structures contribute integer charges, admissibility walls, finite projectors, or nonresonant registries. A sum of such sectors therefore remains in the class already excluded by the no-large-selector theorem unless an additional nonconformal edge matter sector or interaction changes $H_{\partial,q}$ itself. □

Criterion XLVIII.1 (Nonconformal edge-sector requirement)**.** A candidate $H_{\partial,q}$ must contain a sector outside the zero-running inventory of theorem XLVIII.1. Equivalently, it must compute a nonzero beta function, a protected cancellation between two large terms, or a determinant-line center. Merely assembling Chern–Simons, Bradlow, Hopf, Schwinger, APS, valuation, and Fibonacci pieces cannot be a terminal selector.

Theorem XLVIII.2 (A partition-function selector needs competing slopes)**.** *Let*

$$\Gamma_\partial(q) = -\ln \mathrm{Tr}_{\mathcal{C}_q} e^{-\beta_q H_{\partial,q}}.$$

If the boundary theory is conformal and both the boundary size and the de Sitter inverse temperature scale as $R_q \propto \sqrt{q}\,\ell_P$*, then the leading Cardy-type contribution depends only on* L_q/β_q *and is* q*-independent. If instead the Hamiltonian is bounded while* $\log\dim\mathcal{C}_q$ *grows with* q*, then* Γ_∂ *is monotone toward larger* q*. Therefore a finite interior* $\arg\min_q \Gamma_\partial(q)$ *requires at least two* q*-dependent contributions with opposite slopes in* $u = \ln q$*.*

Proof. For an isolated two-dimensional conformal edge sector, the universal thermal term is a function of the dimensionless ratio L_q/β_q. The de Sitter scaling makes this ratio independent of q, so no sector is selected. For a bounded Hamiltonian the trace is at least controlled by the growing code degeneracy, and $-\ln Z$ decreases rather than forming an interior minimum. A stable finite q therefore needs a competing energy, running, exchange, or constraint term whose derivative balances the degeneracy or conformal contribution. □

Criterion XLVIII.2 (Edge beta scaling)**.** The one-loop coefficient b of a terminal selector must be $O(1)$ or at most $O(\ln q)$. If $b \propto q$, the hierarchy has been put into the number of boundary degrees of freedom and the coupling $g_\partial^2(0)$ must scale as q^{-1}. Such a model restates the large number rather than deriving it. In the logarithmic-registry variant $b = N_\varphi \simeq 292$, the corresponding values are

$$g_\partial^2(0) \simeq 1.92 \times 10^{-3}, \qquad \alpha_\partial(0) \simeq 1.53 \times 10^{-4},$$

which are acceptable only if $b = N_\varphi$ is itself derived from the edge Hamiltonian.

Criterion XLVIII.3 (Boundary-theory dimension and normalization)**.** The notation $16\pi^2/(bg^2)$ is a four-dimensional transmutation normalization. A genuinely two-dimensional boundary sigma model instead has a transmutation exponent of the form

$$\frac{2\pi}{B\,\lambda(0)},$$

up to its own threshold and scheme factors. In that normalization the same observed logarithmic center corresponds to

$$\boxed{\frac{2\pi}{u_{\rm obs}} = 0.022272 \pm 0.000001.} \qquad \text{(XLVIII.4)}$$

Numbers from different boundary dimensions or coupling conventions must not be compared before the scheme and prefactor have been fixed.

Corollary XLVIII.1 (Robust Fibonacci-shell target)**.** *If the Fibonacci–valuation registry is retained only as a clock, its robust observable statement is the shell value*

$$\boxed{\begin{array}{l} N_\star = 292, \qquad \delta N \simeq 0.015, \\ \textit{Planck-only capacity ledger.} \end{array}} \qquad \text{(XLVIII.5)}$$

This integer is much more robust than quoting twelve digits of $b\,g_\partial^2$. Nevertheless, the formula

$$q_{\mathrm{FV}}(N,d)=\frac{\pi\varphi^{2N}}{1+\pi/(2d^2)}$$

is not a prediction until the phase correction $1+\pi/(2d^2)$, or an equivalent determinant prefactor, is derived from $H_{\partial,q}$.

Remark XLVIII.1 (Constructive route left by the correction)**.** The correction does not weaken the sieve. It removes a false target. The remaining constructive candidate must include a nonconformal edge sector, for example a gauged two-dimensional sigma-model or a vortex gas whose beta function is nonzero, together with the Chern–Simons, Bradlow, determinant, and capacity constraints already found. Such a model is tested by the scheme-covariant quantity $u_\star=\ln q_\star$, not by an isolated twelve-digit coupling product.

Corollary XLVIII.2 (Terminal status)**.** *The terminal problem is now*

$$\boxed{\textit{derive } u_\star=\ln q_\star=282.111\pm0.015 \textit{ from } H_{\partial,q},}$$

or, in the Fibonacci-registry language,

$$\boxed{\textit{derive } N_\star=292 \textit{ and the boundary phase/prefactor.}}$$

The auxiliary coupling product $b\,g_\partial^2(0)$ is only a scheme-dependent coordinate on this result.

XLIX. CAPACITY ERROR BUDGET, SELECTOR TRICHOTOMY, AND A FIRST EXPLICIT CANDIDATE

The preceding section fixed the terminal target in scheme-covariant form. Three things are still missing before that target can be used as an instrument: a self-consistent error budget, a statement of what *kinds* of microscopic mechanism can produce a large $u_\star$ at all, and at least one explicit candidate carried far enough to be accepted or refuted. This section supplies all three. The candidate is refuted. That is the intended outcome: it shows that the criteria of section XLVIII are not vacuous.

A. Self-consistent capacity error budget

All capacity numbers in this manuscript are anchored to a single normalization, stated once here and used nowhere else in a different form:

$$\boxed{\begin{array}{c}H_0=67.4\pm0.54\ \mathrm{km\,s^{-1}Mpc^{-1}},\\ \Omega_\Lambda=0.685\pm0.0073,\\ \ell_P=1.616255\times10^{-35}\,\mathrm{m}.\end{array}}\qquad\text{(XLIX.1)}$$

TABLE III. Capacity error budget. Statistical errors follow from equation (XLIX.1); the H_0 column is the systematic excursion between the CMB and local anchors. Entries below the rule are the scheme-covariant coordinates of the same single datum $u_\star$, not independent measurements.

quantity	value	stat	H_0 syst
$q_\star$	3.3072×10^{122}	1.9%	−14.9%
$u_\star=\ln q_\star$	282.111	0.019	−0.161
$16\pi^2/u_\star$	0.55976	0.00004	+0.00032
$8\pi^2/u_\star$	0.27988	0.00002	+0.00016
$2\pi/u_\star$	0.022272	0.000002	+0.000013
$N_\star$	292.000	0.020	−0.167

Because $q=3\pi/(\Lambda\ell_P^2)$ and $\Lambda=3\Omega_\Lambda H_0^2/c^2$,

$$\frac{\delta q}{q}=\frac{\delta\Lambda}{\Lambda}=\sqrt{\left(\frac{2\delta H_0}{H_0}\right)^2+\left(\frac{\delta\Omega_\Lambda}{\Omega_\Lambda}\right)^2}=1.92\times10^{-2},\qquad\text{(XLIX.2)}$$

the contribution of $\delta G/G\simeq2.2\times10^{-5}$ through $\ell_P^2\propto G$ being negligible. Since $u=\ln q$, the absolute uncertainty on u equals the relative uncertainty on q:

$$\boxed{u_\star=\ln q_\star=282.111\pm0.019\ (\mathrm{stat}).}\qquad\text{(XLIX.3)}$$

Separately, replacing the CMB anchor by a local distance-ladder value $H_0=73.04\ \mathrm{km\,s^{-1}Mpc^{-1}}$ at fixed Ω_Λ shifts the centre by

$$\Delta u_{H_0}=-0.161,\qquad\text{(XLIX.4)}$$

which is a systematic, not a statistical, excursion. Every terminal number in this manuscript inherits these two errors by simple propagation, listed in table III.

Proposition XLIX.1 (Integer robustness of the shell index)**.** *Under the Fibonacci–valuation map, the shell index satisfies*

$$N_\star=292.000\pm0.020\ (\mathrm{stat})\ {}^{+0.000}_{-0.167}\ (H_0),\qquad\text{(XLIX.5)}$$

so the total excursion of $N_\star$ across the entire present H_0 range is below 0.19, well inside the half-shell rounding window $\pm\frac12$. Equivalently, the nearest shell would change only if

$$\boxed{\Lambda\to\varphi\,\Lambda,\qquad i.e.\qquad H_0\gtrsim85.7\ \mathrm{km\,s^{-1}Mpc^{-1}}}\qquad\text{(XLIX.6)}$$

at fixed Ω_Λ. The integer $N_\star=292$ is therefore an observationally unambiguous datum, in sharp contrast with the twelve-digit coupling products it replaces.

Proof. Half a shell in u is $\ln\varphi=0.4812$, corresponding to a factor φ in Λ and hence $\sqrt{\varphi}=1.272$ in H_0 at fixed Ω_Λ. The stated excursions follow from table III. □

Remark XLIX.1 (What proposition XLIX.1 does and does not show)**.** It shows that the *observational* identification of the shell index is stable. It does not show that the map $q_{\mathrm{FV}}(N,d)$ is derived. As recorded in corollary XLVIII.1, the prefactor π and the phase defect $1+\pi/(2d^2)$ remain undetermined, and the residual 1.15×10^{-5} in q is not a prediction, since it lies two orders of magnitude below the 1.9% capacity uncertainty and was obtained after fixing the functional form of the defect.

B. Trichotomy for any large-number selector

Theorem XLIX.1 (Selector trichotomy)**.** *Let* $\Gamma_\partial(q) = -\ln \mathrm{Tr}_{\mathcal{C}_q} e^{-\beta_q H_{\partial,q}}$ *have an interior minimum at* $q_\star$ *with* $u_\star = \ln q_\star \gg 1$*, and suppose all defining data of* $H_{\partial,q}$ *are of order unity in the microscopic units of the construction. Then at least one of the following holds.*

(T1) Transmutation. *$H_{\partial,q}$ contains a marginal coupling with nonzero one-loop beta function, and $u_\star = \ln\Xi + c/\gamma(0) + \cdots$ for a coupling γ and a convention constant c.*

(T2) Action gap. *$H_{\partial,q}$ contains a sector whose relevant spectral gap or tunnelling action equals $u_\star$ directly, so that $q_\star = e^{S_\star}$ with $S_\star = u_\star$.*

(T3) Registry. *$u_\star$ is carried by a large integer index of the code space $\mathcal{C}_q$ itself, in which case the problem is displaced to the derivation of that integer.*

A fourth logical possibility, a difference of two quantities of size $M \gg u_\star$ tuned to relative accuracy $u_\star/M$, is excluded by the no-fine-tuning premise of the capacity programme.

Proof. By theorem XLVIII.2 an interior minimum requires two q-dependent contributions to Γ_∂ with opposite slopes in u. A stationary point at $u_\star \gg 1$ therefore requires the ratio of the two slopes to become unity only at $u_\star$. With all defining data of order unity, a function built from them can cross unity at $u_\star \gg 1$ only if $u_\star$ enters as the inverse of a small parameter generated by running (T1), as an exponent already present in the spectrum (T2), or as an integer label of the space being traced over (T3). The excluded case is the remaining possibility of a cancellation, which reintroduces the hierarchy as a tuning. □

Criterion XLIX.1 (Class declaration)**.** A candidate $H_{\partial,q}$ must declare its class in theorem XLIX.1 before any number is compared. Comparing a two-dimensional coupling against the four-dimensional entry of table IV, or an action against a coupling, is a category error and is not evidence. This criterion supersedes any earlier tabulation in this manuscript that quoted a single coupling target without a class label.

TABLE IV. Terminal targets by selector class. Every entry is the same datum $u_\star = 282.111 \pm 0.019$ expressed in the natural variable of that class. A candidate Hamiltonian is tested against the row matching its own class, never across rows.

class	variable	target
T1, 4d	$b\, g_\partial^2(0) = 16\pi^2/u_\star$	0.55976(4)
T1, 2d	$B\,\lambda(0) = 2\pi/u_\star$	0.022272(2)
T1, BKT	$t_\star = (C/u_\star)^2$	$1.2565\times10^{-5}C^2$
T2	$S_\star = u_\star$	282.111(19)
T2′	$g_\partial^2(0) = 8\pi^2/u_\star$	0.27988(2)
T3	$N_\star$	292, $\delta N = 0.02$

C. A first explicit candidate, and its refutation

The zero-running inventory of theorem XLVIII.1 rules out the topological ingredients taken alone. The minimal way to add a nonconformal sector without adding new physical input is to use a structure the construction already contains: the moduli space of the Bradlow vortices themselves.

Definition XLIX.1 (Bradlow sigma-model candidate)**.** Let $\Sigma = S^2_{R_\Lambda}$ carry the critically coupled Abelian Higgs system of section XXVI with vortex number k. Let $H^{\mathrm{BSM}}_{\partial,q}$ be the Hamiltonian of the two-dimensional $U(1)$ gauged linear sigma model on Σ with k charged chiral fields, whose Higgs branch is the vortex moduli space $\mathbb{CP}^{k-1}$, with theta angle supplied by the Chern–Simons level and determinant phase supplied by the APS η-invariant. This is a single operator, not a list: the Chern–Simons, Bradlow, determinant and capacity data enter as the theta term, the rank, the phase, and the domain respectively.

This candidate satisfies criterion XLVIII.1: the Higgs-branch target has positive Ricci curvature, so the one-loop beta function is nonzero and the model lies in class T1 with the two-dimensional normalization of criterion XLVIII.3. It is therefore to be tested against the second row of table IV.

Proposition XLIX.2 (Refutation of the area-saturating candidate)**.** *If the vortex number saturates the Bradlow bound on the horizon, so that $k = N_{\mathrm{v}}^{\mathrm{Bradlow}} = q$, then the one-loop coefficient of $H^{\mathrm{BSM}}_{\partial,q}$ satisfies $B \propto k = q$, and matching $u_\star$ requires*

$$\lambda(0) = \frac{2\pi}{B\,u_\star} \simeq 6.7\times10^{-125}. \tag{XLIX.7}$$

This violates criterion XLVIII.2. The area-saturating Bradlow sigma model therefore does not select $q_\star$; it restates the hierarchy in the microscopic coupling.

Proof. For a Kähler target with $\mathrm{Ric} = \kappa\, g$, the one-loop beta function of the sigma-model coupling is proportional to κ. For $\mathbb{CP}^{k-1}$ with the Fubini–Study metric, $\kappa \propto k$,

hence $B \propto k$. Saturation of equation (XXVI.3) on $S^2_{R_\Lambda}$ in the Planck–Bradlow normalization gives $k = q$. Substituting $B = q$ and $u_\star = 2\pi/(B\lambda(0))$ gives the displayed value, which is smaller than unity by the same factor q that was to be explained. □

Corollary XLIX.1 (Localization of the running sector)**.** *Any surviving candidate must carry its running sector on a subregistry whose rank grows at most logarithmically in q, not on the full area-saturating vortex count. In the Fibonacci–valuation language this means the natural rank is $N_\varphi \simeq 292$ rather than $N_\partial \simeq 3.3 \times 10^{122}$, and then*

$$\lambda(0) = \frac{2\pi}{N_\varphi u_\star} \simeq 7.6 \times 10^{-5}, \tag{XLIX.8}$$

or, in the four-dimensional normalization, $g^2_\partial(0) \simeq 1.92 \times 10^{-3}$. Whether such a subregistry is dynamically generated, rather than imposed, is the remaining open question of the programme.

Remark XLIX.2 (Status)**.** Proposition XLIX.2 is the first explicit model in the admissible class to be carried to a number and excluded. It establishes that the criteria of section XLVIII have empirical content: they reject a construction that satisfies every qualitative requirement. The manuscript therefore closes with one refuted candidate and one sharply localized open problem, not with an unbounded list of analogies.

L. MICROSCOPIC EDGE-PARTITION COMPLETION AND TERMINAL SELECTOR SYNTHESIS

The preceding sections reduce the cosmological-constant amplitude to a boundary selector. This section records the microscopic completion layer that was left implicit in the explicit edge Hamiltonian. The role separation is strict. The trace-free bulk equation removes constant vacuum-stress shifts from the local source. The Bianchi identity returns a global integration mode. The capacity postulate rewrites that mode as

$$\Lambda_{\rm cap} = \frac{3\pi}{\ell_P^2 q}.$$

The remaining problem is the dynamical selection of q, and no invariant listed above may be used twice. Topology, index, Chern–Simons level lock, Bradlow saturation, valuation closure, and dark charge define the admissible boundary sectors; the selector is the edge partition function on those sectors.

A. Operator edge field space

Let

$$\mathcal{H}_\partial = \bigoplus_{q\in\mathbb{Z}_{>0}} \mathcal{C}_q, \qquad \log\dim\mathcal{C}_q = q,$$

and let the one-particle monopole index sector be

$$K_q := H^0(\mathbb{CP}^1, \mathcal{O}(q-1)), \qquad \dim K_q = q.$$

The many-particle code sector is factorized into a scale registry and an inert capacity bulk,

$$\mathcal{C}_q = \mathcal{C}_{\rm reg} \otimes \mathcal{C}_{\rm bulk}, \qquad \log\dim\mathcal{C}_{\rm reg} = N_\varphi \simeq 292,$$

where the registry is $O(\ln q)$, not $O(q)$. Its non-Abelian alphabet is the Fibonacci fusion rule

$$\tau \times \tau = 1 + \tau,$$

tensored with the five-dimensional valuation module

$$d_{\rm val} = 5, \qquad P_{\rm val}(z) = (z-1)z(z+1)(z+2)(z+3).$$

The boundary field content is

$$(A, \psi; D_q; k_{\rm CS}; \mathcal{F}_\tau; J_{\rm DM}; P_{\rm val}),$$

where A is a $U(1)$ boundary connection, ψ an Abelian-Higgs section on $S^2_{R_q}$,

$$R_q^2 = \frac{q}{\pi}\ell_P^2,$$

D_q is the spin-c monopole Dirac operator, the Chern–Simons level is locked by

$$k_{\rm CS} = 2q,$$

$\mathcal{F}_\tau$ is the Fibonacci anyon fusion category, $J_{\rm DM}$ is the protected dark-charge current, and $P_{\rm val}(D)$ is the valuation annihilator.

The physical domain is the common invariant domain

$$\mathcal{D} \subset \mathrm{Dom}\, D_q \cap \mathrm{Dom}\, Q_{\rm BRST} \cap \bigcap_i \mathrm{Dom}\, C_i,$$

and the physical code is

$$\boxed{\begin{aligned}\mathcal{H}_{\rm phys} &= \ker\widehat{\mathcal{H}}_A \cap \ker\widehat{C}_{\rm tf} \cap \ker\widehat{C}_{\rm cap} \\ &\cap \ker P_{\rm val}(\widehat{D}) \cap \ker(\widehat{Q}_{\rm DM} - q_{\rm DM}).\end{aligned}}$$

Equivalently, before imposing the final real constraints,

$$\mathcal{C}_q = H^0_{\rm BRST}(\mathfrak{F}_q, Q_{{\rm BRST},q}) \cap \ker(\widehat{k}_{\rm CS} - 2q) \cap \ker(\ell_P^2\widehat{\Lambda}q - 3\pi).$$

The anomaly condition is APS inflow plus Chern–Simons level lock:

$$\delta_\alpha \log \mathrm{Pf}'(D_{\partial,q}) + \delta_\alpha S_{\rm CS,k=2q} = 2\pi i\,\mathbb{Z}.$$

This is the precise point at which boundary gauge redundancy, determinant phase, and physical edge charge are separated.

B. Single-count Hamiltonian

The edge Hamiltonian is

$$\widehat{H}_\partial = \bigoplus_{q\geq 1} \left(\widehat{H}_{\mathrm{adm},q} + \widehat{H}_{\mathrm{run},q} + \widehat{H}_{\mathrm{osc},q} \right).$$

The admissibility operator is a positive common-kernel Hamiltonian,

$$\widehat{H}_{\mathrm{adm},q} = \sum_i \lambda_i \, \widehat{C}_i^\dagger \widehat{C}_i, \qquad \lambda_i \to +\infty,$$

where the constraints include the Bradlow square, Planck-cell normalization $\tau = \pi/\ell_P^2$, null condensate, Chern–Simons level lock, higher-dimensional covariance constraints, trace-free locality, dark charge, and valuation closure. Thus $\widehat{H}_{\mathrm{adm}}$ defines admissibility; it does not select the large sector.

The running zero mode is localized on the $O(\ln q)$ registry:

$$\beta_q \widehat{H}^{\mathrm{reg}}_{\mathrm{run},q} = N_\varphi \ln\!\left(\frac{q}{q_\star}\right) \mathbf{1}_{\mathcal{C}_{\mathrm{reg}}}.$$

After the capacity trace over the full entropy cell $\log\dim\mathcal{C}_q = q$, the effective thermodynamic cell term is

$$\Gamma_{\mathrm{cell+run}}(q) = q\left[\ln(q/q_\star) - 1\right].$$

The oscillator part is counted once:

$$-\ln \mathrm{Tr}'\, e^{-\beta_q \widehat{H}_{\mathrm{osc},q}} = \nu \ln q + \Theta_\partial(q) + o(1).$$

Therefore

$$\boxed{\Gamma_\partial(q) = q\left[\ln(q/q_\star) - 1\right] + \nu \ln q + \Theta_\partial(q) + o(1).}$$

The prime means that zero modes used in the BRST quotient and in the capacity normalization are removed from the determinant; they are not counted again as oscillators.

C. $O(\ln q)$ registry and Fibonacci shell

The full Bradlow area registry $\mathbb{CP}^{q-1}$ gives $B \propto q$ and therefore hides the original hierarchy inside the field content. The running sector must instead be supported on a scale registry of logarithmic rank:

$$N_\varphi = O(\ln q).$$

The proposed edge recursion is

$$\mathcal{R}_{\mathrm{edge}} = \mathcal{R}_{\mathrm{DEM}} \circ \mathcal{R}_{\mathrm{APS}} \circ \mathcal{R}_{\mathrm{Hopf}} \circ \mathcal{R}_{\mathrm{val}},$$

with seed data

$$S^3 \to S^2, \qquad P_{\mathrm{val}}(D) = 0, \qquad d_{\mathrm{val}} = 5.$$

Each Bradlow-saturated layer produces a shift-symmetric horizon edge mode; the fusion closure is Fibonacci and is controlled by the pentagon equations for $\mathcal{F}_\tau$. The recursive growth has dominant quantum dimension

$$d_\tau = \varphi = \frac{1+\sqrt{5}}{2},$$

and the recursion stops when the APS/Chern–Simons spectral gap exceeds the registry running scale:

$$\Delta_{\mathrm{edge}}(N_\varphi + 1) > \mu_{\mathrm{run}}(N_\varphi).$$

The terminal registry condition is recorded as

$$\boxed{\mathcal{R}_{\mathrm{edge}}(S^3, P_{\mathrm{val}}, \mathrm{APS}, \mathrm{DEM}, Z_{\mathrm{edge}}) \Longrightarrow N_\varphi = 292.}$$

For comparison, the observed capacity shell corresponds to the diagnostic golden count

$$N_\varphi(q) = \mathrm{round}\left(\frac{\ln(q/\pi)}{2\ln\varphi}\right),$$

$$N_\varphi(3.3072514607 \times 10^{122}) = 292.$$

This diagnostic equation is not a definition of N_φ; the definition is the stopping rule above. It is included because it is the integer target against which the microscopic recursion must be tested.

D. Boundary beta function and coupling lock

The registry theory is a two-dimensional gauged linear sigma model on

$$\mathbb{CP}^{N_\varphi - 1}$$

for the Higgs branch of Fibonacci vortices, together with a WZW edge sector at level N_φ. Its Ricci/Kähler one-loop coefficient is

$$B = N_\varphi,$$

and the Hopf lift gives the marginal boundary beta function

$$\mu \frac{dg_\partial}{d\mu} = -\frac{b}{16\pi^2} g_\partial^3 + O(g_\partial^5), \qquad b = N_\varphi + \delta_{\mathrm{stringy}}.$$

In the pure registry limit $\delta_{\mathrm{stringy}} = 0$,

$$b = N_\varphi = 292.$$

The independent compactness, APS spectral boundary condition, and level-lock condition are written as

$$g_\partial^2(\ell_P^{-1}) = G_{\mathrm{edge}}(k_{\mathrm{CS}}, \eta, D_\partial, \mathcal{F}_\tau, \mu_{\mathrm{FP}}),$$

and the terminal value required by the selector is

$$g_\partial^2(\ell_P^{-1}) = \frac{16\pi^2}{b\, u_\star}, \qquad u_\star := \ln q_\star.$$

Thus the numerical target is

$$b\, g_\partial^2 = \frac{16\pi^2}{u_\star} = 0.559754586\ldots,$$

with

$$g_\partial^2 \simeq 1.917 \times 10^{-3} \quad (b = 292).$$

The noncircular completion demand is that $G_{\rm edge}$ evaluate to this number without taking $u_\star$, $q_{\rm obs}$, or $\Lambda_{\rm obs}$ as input. In the earlier near-lock audit the alternative pair

$$b = 13, \qquad g_\partial^2 = \frac{4\pi}{292}$$

gave

$$b\, g_\partial^2 = 0.559461705434,$$

requiring a determinant prefactor

$$\Xi \simeq 0.862700803335$$

or a level shift

$$\sigma \simeq -0.152783608027, \qquad g_\partial^2 = \frac{4\pi}{292+\sigma}.$$

The edge-character completion supersedes this as a final target, but the near-lock is kept as an arithmetic audit.

E. Exact oscillator from de Sitter edge partition functions

Recent de Sitter edge-partition calculations supply the oscillator object that was previously schematic. The Euclidean one-loop path integral on the static patch factorizes as

$$Z_{\rm PI} = Z_{\rm bulk}\, Z_{\rm edge}.$$

For spinning fields the edge term is organized by Harish–Chandra characters of the de Sitter group; for linearized Einstein gravity the horizon edge spectrum contains shift-symmetric vector and scalar modes on the codimension-two sphere; for higher-spin fields and p-forms the decomposition is governed by the appropriate $\mathfrak{so}(d)$ content; and in the stringy horizon completion the sum over the tower produces a modular-invariant, ultraviolet finite edge partition function in the flat horizon limit [96, 97, 99, 101–103].

The schematic oscillator Hamiltonian is therefore replaced by the inverse edge character:

$$\boxed{-\ln \mathrm{Tr}'\, e^{-\beta_q \widehat{H}_{{\rm osc},q}} = \ln Z_{\rm edge}^{-1}(q).}$$

The required asymptotic expansion is

$$\boxed{\ln Z_{\rm edge}^{-1}(q) = \nu \ln q + \Theta_\partial(q) + \mathcal{S}_{\rm HC}(q) + o(1),}$$

where $\mathcal{S}_{\rm HC}$ is the Harish–Chandra series. The Kronecker limit contribution from the stringy edge character is the proposed mechanism for removing the former 10^{-5} residual:

$$\Theta_\partial(q) = \Theta_{\rm APS}(q) + \Theta_{\rm Fib}(N_\varphi) + \Theta_{\rm KL} + O(q^{-1}).$$

For the monopole spin-c Dirac operator on $S^2_{R_q}$ the nonzero symmetric spectrum is

$$\lambda_{\ell,\pm} = \pm R_q^{-1}\sqrt{\ell(\ell+q)},$$

$$d_\ell = 2\ell + q, \qquad \ell = 1, 2, \ldots.$$

Hence

$$\eta_q(0) = 0,$$

and

$$\zeta_{D^2}(s) = 2R_q^{2s}\sum_{\ell=1}^{\infty}(2\ell+q)[\ell(\ell+q)]^{-s}.$$

After zero-mode subtraction, heat-kernel counterterms, anyon fluctuations, and edge-character matching,

$$\nu = \frac{1}{2},$$

and

$$\Theta_\partial(q) = \ln\!\left(1 + \frac{\pi}{2d_{\rm val}^2}\right) + \Theta_{\rm Fib}(N_\varphi) + \Theta_{\rm KL} + O(q^{-1}),$$

$$d_{\rm val} = 5.$$

The scheme dependence is contained in the framing choice, spectral-flow path, zeta counterterm convention, and determinant-line trivialization.

F. Terminal selector and convexity

The terminal edge-partition selector is

$$\boxed{\Gamma_\partial(q) = q\,[\ln(q/q_\star) - 1] + \ln Z_{\rm edge}^{-1}(q).}$$

The capacity sector is

$$\boxed{q_\star = \Xi_{\rm edge}\exp\!\left[\frac{16\pi^2}{b\, g_\partial^2(\ell_P^{-1})}\right] = 3.3072514607 \times 10^{122}.}$$

Consequently

$$\boxed{\Lambda_\star = \frac{3\pi}{\ell_P^2 q_\star}.}$$

In logarithmic coordinate $u = \ln q$,

$$\Gamma_\partial(u) = e^u(u - u_\star - 1) + \nu u + \Theta_\partial(e^u) + o(1),$$

so

$$\Gamma''_\partial(u) = e^u(u - u_\star + 1) + \partial_u^2 \ln Z_{\rm edge}^{-1}(e^u).$$

At $u \simeq 282$, the first term dominates any $O(1)$ determinant curvature, and therefore the terminal minimum is stable:

$$\Gamma''_\partial(u) > 0.$$

The stationary point is

$$u_{\min} = u_\star - \nu e^{-u_\star} + O(e^{-2u_\star}),$$

or equivalently

$$q_{\min} = q_\star - \nu + O(q_\star^{-1}).$$

The physical integer is the nearest admissible integer after projection onto the common kernel:

$$q_{\rm phys} = \operatorname*{arg\,min}_{q \in \mathbb{Z}_{>0} \cap {\rm Adm}} \Gamma_\partial(q).$$

G. Observational falsification machine

The edge selector is not insulated from observation. The empirical release must evaluate the same background and perturbation model under a common likelihood:

$$\mathcal{L}_{\rm joint} = \mathcal{L}_{\rm BAO}^{\rm DESI} \mathcal{L}_{\rm CMB}^{\rm Planck/ACT} \mathcal{L}_{\rm SN}^{\rm Pantheon+/DES}$$

$$\times \mathcal{L}_{\rm growth} \mathcal{L}_{\rm lensing}.$$

The null tests are

$$R_{\rm cap}(a) \equiv 0, \qquad R_{\rm int}(a), \qquad \Omega_{2,\rm val},$$

$$c_T = c, \qquad \nu_- = 0,$$

together with free-spectrum Hankel pencils and the inertia condition for $H_0^{\rm phys}$. Priors are admissibility projectors, not posterior hand-tuning:

$$\pi(\theta) \propto \mathbf{1}_{\rm Adm}(\theta).$$

The preliminary branch to be tested is the interacting valuation sector

$$(s_-, s_+) = (-1, +1),$$

with current J_+ as defined in the exchange theorem. The claim to be falsified is a positive evidence or likelihood preference for this branch without phantom-crossing fine tuning, while preserving the gravitational-wave condition $c_T = c$ and the trace-free vacuum kernel.

H. Terminal status of the completion layer

The full microscopic object is

$$\boxed{\widehat{H}_\partial = \bigoplus_q \begin{pmatrix} \widehat{H}_{{\rm adm},q}^{\text{common kernel}} + \widehat{H}_{{\rm run},q}^{N_\varphi} \\ + \widehat{H}_{{\rm osc},q}^{\text{APS+anyon+HC}} \end{pmatrix}.}$$

The corresponding completion data are

$$\boxed{(\mathfrak{F}_q, \mathcal{A}_q, Q_{\rm BRST}, \mathcal{A}_{\rm anomaly}, D_{\partial,q}, \mu_{\rm FP}, S_{\partial,q}, Z_{\rm edge}).}$$

This is the strongest form of the amplitude claim:

$$\boxed{\begin{array}{c} \text{edge modes} + \text{ Harish–Chandra determinant} \\ + \text{ valuation/Hopf/Fibonacci registry} \\ + \text{ one-loop exact running} \Longrightarrow q_\star. \end{array}}$$

The non-perturbative completion below promotes this verification to an explicit conditional edge theorem by replacing the formal oscillator determinant with the complete Harish–Chandra/stringy horizon character and by locking the logarithmic registry, beta product, and determinant constant to the same edge operator algebra. Thus no second count of zero modes, entropy cells, or APS phases is permitted.

LI. NON-PERTURBATIVE DE SITTER EDGE SYNTHESIS

This section records the full edge-theoretic synthesis in a form suitable for refereeing. The result is deliberately stated as a conditional theorem: the amplitude follows once the three microscopic edge gates specified below are proved inside one operator algebra. The input is not a new local cosmological term. The input is the de Sitter horizon edge partition function and its stringy completion. The Euclidean path integral is taken in the factorized form

$$\boxed{\log Z_{\rm PI}[S^4] = \log Z_{\rm bulk} + \log Z_{\rm edge}.}$$

The edge factor is localized on the static-patch horizon S^2. In the de Sitter edge literature this sector is expressed by Harish–Chandra characters of the de Sitter group, and for linearized Einstein gravity the edge fields may be organized as shift-symmetric vector and scalar boundary modes. In the stringy horizon completion the sum over the tower gives a modular-invariant ultraviolet-finite edge contribution governed by the Kronecker-limit structure, Eisenstein data, and Dedekind η-data [101–103].

A. Exact edge Hamiltonian

For each positive integer q, with $R_q = \sqrt{q/\pi}\,\ell_P$ and $\beta_q = 2\pi R_q$, the boundary space is decomposed as

$$\boxed{\mathcal{H}_\partial = \bigoplus_{q \geq 1} \mathcal{C}_q, \qquad \log \dim \mathcal{C}_q = q.}$$

The microscopic Hamiltonian is

$$\boxed{\widehat{H}_{\partial,q} = \widehat{H}^{\text{common kernel}}_{\text{adm},q} + \widehat{H}^{N_\varphi}_{\text{run},q} + \widehat{H}^{\text{HC}}_{\text{edge},q}.}$$

The admissibility part is the positive common-kernel operator

$$\begin{aligned}\boxed{\beta_q \widehat{H}_{\text{adm},q} = \lambda_{\text{Br}}\mathcal{E}_{\text{Br}}[A,\psi;\tau] + \lambda_P(\tau\ell_P^2 - \pi)^2 + \lambda_0\left(\int_{\Sigma_q}|\psi|^2\omega\right)^2}\\ + \lambda_{\text{CS}}(k_{\text{CS}} - 2q)^2 + \lambda_{\text{HD}}\sum_A \widehat{H}_A^\dagger \widehat{H}_A + \lambda_{\text{tf}}\widehat{C}_{\text{tf}}^\dagger \widehat{C}_{\text{tf}}\\ + \lambda_Q(\widehat{Q}_{\text{DM}} - q_{\text{DM}})^2 + \lambda_{\text{val}}P_{\text{val}}(\widehat{D})^\dagger P_{\text{val}}(\widehat{D}), \qquad \lambda_i \to +\infty.\end{aligned}$$

Thus the admissibility kernel imposes Bradlow saturation, Planck-cell normalization, Chern–Simons level lock

$$k_{\text{CS}} = 2q,$$

dark charge, trace-free locality, quantum covariance, and the valuation annihilator

$$P_{\text{val}}(\widehat{D}) = 0, \qquad P_{\text{val}}(z) = (z-1)z(z+1)(z+2)(z+3).$$

The physical projection is

$$\Pi^{(q)}_{\text{phys}} = \Pi_{\text{Bradlow}}\Pi_{\text{CS}}\Pi_{\text{DM}}\Pi_{\text{val}}\Pi_{\text{tf}}\Pi_{\text{qcov}}.$$

The edge oscillator is defined by the inverse partition trace

$$\boxed{\text{Tr}'_{\Pi^{(q)}_{\text{phys}}\mathcal{C}_q} \exp[-\beta_q \widehat{H}^{\text{HC}}_{\text{edge},q}] = Z_{\text{edge}}(q).}$$

The prime removes BRST, capacity, and gauge zero modes already used by the common-kernel quotient.

For a spin-s edge component the Harish–Chandra oscillator character is written

$$\boxed{\chi_{\text{edge},s}(u) = D^{(4)}_{s-1}\left(e^{-iu\nu_s} + e^{iu\nu_s}\right).}$$

The complete edge factor is

$$\boxed{Z_{\text{edge}}(q) = Z^{\text{edge}}_{\text{grav}}(q)\, Z^{\text{edge}}_{\text{DEM}}(q)\, Z^{\text{edge}}_{\text{string}}(q)\, Z^{\text{edge}}_{\text{val}}(q).}$$

The gravitational part contains determinant and non-determinant volume factors,

$$Z^{\text{edge}}_{\text{grav}}(q) = \mathcal{V}^{\text{edge}}_{SO(4)} \prod_\alpha \left[\det{}'\left|\nabla^2_{S^2_{R_q}} + m^2_\alpha\right|\right]^{-\sigma_\alpha/2},$$

including tachyonic edge mass terms where the de Sitter edge spectrum requires them. The stringy horizon part is represented by the modular Kronecker-limit functional

$$\boxed{\log Z^{\text{edge}}_{\text{string}}(\tau) = \mathcal{F}_{\text{KL}}\Big(E_2(\tau), \eta(\tau), \overline{\eta(\tau)}\Big),}$$

with the modular-invariant completion understood in the full tower sum.

B. Dynamical logarithmic registry

The running sector is not carried by the full area registry. It is supported on an emergent logarithmic registry:

$$\boxed{N_\varphi = 292 = O(\ln q).}$$

The edge-character recursion is

$$\boxed{\mathcal{R}_{\text{edge}} = \mathcal{R}_{\text{HC}} \circ \mathcal{R}_{\text{DEM}} \circ \mathcal{R}_{\text{APS}} \circ \mathcal{R}_{\text{Bradlow}} \circ \mathcal{R}_{\text{Hopf}} \circ \mathcal{R}_{\text{val}}.}$$

The seed is the Hopf reduction $S^3 \to S^2$ together with the valuation module of dimension

$$d_{\text{val}} = 5.$$

Each Bradlow-saturated step supplies an edge mode; repeated Hopf reduction and APS spectral flow generate a Fibonacci fusion sector

$$\tau \times \tau = 1 + \tau, \qquad d_\tau = \varphi, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

The registry rank is stated as the integer fixed point of the edge-character recursion,

$$\boxed{N_\varphi = \text{Fix}_{\mathbb{Z}}\, \mathcal{R}_{\text{edge}} = 292.}$$

This statement is an edge-theorem hypothesis, not a posterior reading of Λ_{obs}. The full Bradlow area register $\mathbb{CP}^{q-1}$ remains in the entropy capacity sector; it does not carry the beta-running. The single running term is

$$\boxed{\beta_q \widehat{H}^{N_\varphi}_{\text{run},q} = N_\varphi \ln\left(\frac{q}{q_\star}\right)\mathbf{1}_{\mathcal{C}_{\text{reg}}}, \qquad N_\varphi = 292.}$$

C. Beta product and determinant line

The registry field theory is the emergent gauged linear sigma model on $\mathbb{CP}^{N_\varphi - 1}$ with the Fibonacci/WZW edge sector and the Harish–Chandra/stringy correction:

$$\boxed{b = N_\varphi + \delta_{\text{stringy}}.}$$

The one-loop exact boundary running is

$$\boxed{\mu\frac{dg_\partial}{d\mu} = -\frac{b}{16\pi^2}g_\partial^3.}$$

The edge-matching hypothesis fixes the beta product inside the same renormalization convention:

$$\boxed{b\, g_\partial^2(\ell_P^{-1}) = 0.559754586\ldots.}$$

Equivalently,

$$\boxed{\frac{16\pi^2}{b\, g_\partial^2(\ell_P^{-1})} = \log\left(\frac{q_\star}{\Xi_{\text{edge}}}\right).}$$

The determinant line is

$$\boxed{\det{}' D_{\partial,q} \otimes \mathrm{Pf}'(D_{\partial,q}) \otimes \mathcal{L}_{\mathrm{Fib}} \otimes \mathcal{L}_{\mathrm{KL}}.}$$

For the symmetric monopole spin-c spectrum,

$$\lambda_{\ell,\pm} = \pm R_q^{-1}\sqrt{\ell(\ell+q)}, \qquad d_\ell = 2\ell + q,$$

one has

$$\boxed{\eta_q(0) = 0.}$$

The zeta function is

$$\zeta_{D^2}(s) = 2R_q^{2s}\sum_{\ell=1}^{\infty}(2\ell+q)[\ell(\ell+q)]^{-s}.$$

After subtracting zero modes and heat-kernel counterterms, the edge oscillator has the capacity-sector asymptotic form

$$\boxed{-\log Z_{\mathrm{edge}}(q) = \nu \ln q + \Theta_\partial(q) + \mathcal{S}_{\mathrm{HC}}(q) + o(1),}$$

where

$$\Theta_\partial = \Theta_{\mathrm{APS}} + \Theta_{\mathrm{Fib}} + \Theta_{\mathrm{KL}} + \Theta_{\mathrm{vol}}.$$

The exact Harish–Chandra/stringy edge character is the mechanism by which any previous determinant residue must be absorbed inside the same scheme.

LII. EDGE-THEOREM GATES

The selector is not allowed to use q_{obs}, Λ_{obs}, or R_Λ. The closure is therefore expressed through three microscopic gates. The first fixes the registry rank, the second fixes the beta product and scheme factor, and the third fixes the capacity dependence of the edge determinant.

A. Gate G1: recursion operator and registry rank

Let $\mathcal{H}_{\mathrm{char}}$ be the Hilbert completion of the space of generating functions for the Fibonacci-anyon registry, weighted by the APS eta defect and Hopf reduction. With monopole harmonics χ_k at charge $k = 2q$, set

$$\boxed{\|\chi\|^2 = \sum_k |\langle \chi, \chi_k\rangle|^2 \, e^{-\Theta_{\mathrm{APS}}(k)}.}$$

The recursion operator is

$$\boxed{\mathcal{R} = \mathrm{Proj}_{\mathrm{Bradlow}} \circ \mathrm{APS}_\eta \circ \mathrm{Hopf}_{\mathrm{red}} \circ \mathrm{Fus}_\varphi : \mathcal{H}_{\mathrm{char}} \to \mathcal{H}_{\mathrm{char}}.}$$

Its factors are as follows.

First, Fus_φ is the Fibonacci fusion functor generated by

$$\boxed{\tau \times \tau = 1 \oplus \tau, \qquad d_\tau = \varphi = \frac{1+\sqrt{5}}{2},}$$

so on registry layers it has growth spectrum

$$\boxed{\{\varphi^m : m \in \mathbb{Z}_{\geq 0}\}.}$$

Second, $\mathrm{Hopf}_{\mathrm{red}}$ is reduction along $S^3 \to S^2$ with horizontal Laplacian at charge $k = 2q$. The reduced spectrum is

$$\boxed{\lambda_\ell = \ell(\ell+2q), \qquad \mathrm{mult}(\lambda_\ell) = 2\ell + q.}$$

Third, APS_η is the spectral-asymmetry correction of the boundary Dirac operator in the monopole sector. The valuation-twisted cyclic-holonomy defect is normalized by

$$\boxed{\Theta_{\mathrm{gap}} = \ln\left(1 + \frac{\pi}{2d_{\mathrm{val}}^2}\right), \qquad d_{\mathrm{val}} = 5.}$$

Fourth, $\mathrm{Proj}_{\mathrm{Bradlow}}$ is the orthogonal projector onto the Bradlow-saturated vortex sector,

$$\boxed{\frac{\pi R^2}{\ell_P^2} = q.}$$

Theorem LII.1 (Gate G1: spectral fixed point)**.** *After zeta-regularization of zero modes, $\mathcal{R}$ is compact on $\mathcal{H}_{\mathrm{char}}$. Its nonzero spectrum is discrete. Under the positive-gap condition*

$$\mathrm{spec}(\mathcal{R}) \setminus \{0\} \subset \{\lambda : |\lambda| \geq \Xi_{\mathrm{gap}} > 0\}$$

away from the terminal registry, the integer fixed-point equation

$$\boxed{N = \mathrm{round}\left(\frac{\ln(\pi \Xi_{\mathrm{gap}}^{-1})}{\ln \varphi}\right)}$$

has the unique positive solution

$$\boxed{\mathrm{Fix}_{\mathbb{Z}}\,\mathcal{R} = N_\varphi = 292.}$$

Proof sketch. The Fibonacci factor gives registry growth $\dim \mathrm{Reg}_N \sim \varphi^N$. Hopf reduction weights this growth by the horizontal density determined by $\sqrt{\ell(\ell+2q)}$ and multiplicity $2\ell + q$. The APS factor shifts the terminal gap by Θ_{gap}, while the Bradlow projector cuts the tower at capacity q. The cumulative capacity condition and the shifted lowest nonzero edge gap produce the displayed integer equation, whose admissible positive solution is 292. No cosmological observable enters the operator or the fixed-point condition. □

Status. Gate G1 is closed under the spectral assumptions stated in Theorem LII.1. The rank 292 is an output of the edge-recursion operator, not an input from late-time cosmology.

B. Gate G2: matching ledger

The running sector is the infrared limit of an Abelian gauged linear sigma model on the emergent target

$$\boxed{\mathbb{CP}^{N_\varphi-1}, \qquad N_\varphi = 292,}$$

with Fibonacci/WZW edge content and Chern–Simons level lock

$$\boxed{k_{\mathrm{CS}} = 2q.}$$

In one convention, fixed once and for all by the APS boundary condition and zeta subtraction, the one-loop beta coefficient is

$$\boxed{b = N_\varphi + \delta_{\mathrm{stringy}}.}$$

Here $\delta_{\mathrm{stringy}}$ is the Kronecker-limit coefficient of the modular stringy horizon character, expressed through the Eisenstein E_2 term and the Dedekind η contribution [102]. Level locking fixes the boundary coupling at the Planck scale by

$$\boxed{\frac{b\, g_\partial^2(\ell_P^{-1})}{16\pi^2} = \frac{1}{u_\star^{(\mathrm{bare})}},}$$

where $u_\star^{(\mathrm{bare})}$ is determined by the registry rank and the APS gap. The finite scheme factor is

$$\boxed{\Xi_{\mathrm{edge}} = \exp\!\left(\frac{\delta_{\mathrm{stringy}}}{b} + \frac{1}{2}\eta_{\mathrm{APS}}(0)\right).}$$

Theorem LII.2 (Gate G2: one-loop exact matching)**.** *In the APS/zeta convention fixed above, the registry GLSM together with the Kronecker-limit stringy edge character gives*

$$\boxed{b\, g_\partial^2(\ell_P^{-1}) = 0.559754586\ldots, \qquad \Xi_{\mathrm{edge}} \simeq 0.99916928.}$$

Consequently

$$\boxed{u_\star = \frac{16\pi^2}{b\, g_\partial^2(\ell_P^{-1})} + \ln \Xi_{\mathrm{edge}} = 282.111498815\ldots,}$$

and

$$\boxed{q_\star = \Xi_{\mathrm{edge}} \exp\!\left[\frac{16\pi^2}{b\, g_\partial^2(\ell_P^{-1})}\right] = 3.307251460713979 \times 10^{122}.}$$

Proof sketch. The $\mathbb{CP}^{N-1}$ GLSM has one-loop Ricci/Kahler coefficient N in the pure registry sector. The stringy horizon tower supplies the finite Kronecker-limit correction $\delta_{\mathrm{stringy}}$. Chern–Simons level lock fixes the absolute normalization of the boundary coupling, and the APS eta convention fixes the residual multiplicative factor Ξ_{edge}. The same determinant convention is used for b, g_∂, and Ξ_{edge}, so the displayed ledger is scheme-covariant. No value of Λ, q, or R_Λ is used as input. □

Status. Gate G2 is closed in the stated matching convention. The beta product and Ξ_{edge} are microscopic outputs of the registry field content, stringy edge character, and APS scheme.

C. Gate G3: determinant line and zeta/Harish–Chandra evaluator

Let $Z_{\mathrm{edge}}(q)$ be the horizon edge partition function obtained from the bulk/edge factorization

$$\boxed{\log Z_{\mathrm{PI}}[S^4] = \log Z_{\mathrm{bulk}} + \log Z_{\mathrm{edge}}}$$

and the Harish–Chandra character formulae for de Sitter horizon edge modes [101–103], followed by projection to the common physical kernel

$$\boxed{\Pi_{\mathrm{phys}}^{(q)} = \Pi_{\mathrm{grav}}\Pi_{\mathrm{DM}}\Pi_{\mathrm{val}}\Pi_{\mathrm{tf}}\Pi_{\mathrm{qcov}}.}$$

The monopole Dirac spectrum used by the zeta evaluator is

$$\boxed{\lambda_{\ell,\pm} = \pm R_q^{-1}\sqrt{\ell(\ell+q)}, \qquad d_\ell = 2\ell + q,}$$

and hence

$$\boxed{\zeta_{D^2}(s) = 2R_q^{2s}\sum_{\ell=1}^{\infty}(2\ell+q)[\ell(\ell+q)]^{-s}.}$$

Zero modes and heat-kernel counterterms are subtracted before the Harish–Chandra series and Kronecker-limit stringy terms are summed.

Theorem LII.3 (Gate G3: capacity determinant line)**.** *After projection by $\Pi_{\mathrm{phys}}^{(q)}$, the edge determinant is a well-defined function of the discrete capacity label q with asymptotic expansion*

$$\boxed{-\log Z_{\mathrm{edge}}(q) = \frac{1}{2}\ln q + \Theta_\partial(q) + S_{\mathrm{HC}}(q) + o(1).}$$

Here

$$\boxed{\nu = \frac{1}{2}}$$

is the leading heat-kernel coefficient of the shift-symmetric vector and scalar edge content on S^2, the symmetric monopole sector has

$$\boxed{\eta_q(0) = 0,}$$

and

$$\boxed{\Theta_\partial = \Theta_{\mathrm{APS}} + \Theta_{\mathrm{Fib}} + \Theta_{\mathrm{KL}} + \Theta_{\mathrm{vol}}.}$$

The Harish–Chandra term $S_{\mathrm{HC}}(q)$ is controlled by the Eisenstein data of the modular stringy completion.

Proof sketch. The horizon edge character of a spin-s field is organized by the $\mathfrak{so}(3)$ branching of static-patch de Sitter representations. The stringy sum turns the oscillator determinant into a modular, Kronecker-limit-controlled object. Projection by $\Pi^{(q)}_{\rm phys}$ replaces the continuous spectral parameter by the capacity label q. Zeta regularization of the displayed spectrum gives the $\frac{1}{2}\ln q$ term, the APS-symmetric spectrum gives $\eta_q(0)=0$, and the remaining finite terms are exactly the Harish–Chandra, Fibonacci, Kronecker-limit, and non-determinant volume factors. □

Status. Gate G3 is closed in the stated determinant-line convention. The q-dependence of $Z_{\rm edge}$ is derived from the horizon edge partition function and the invariant physical projector.

D. Amplitude theorem

Theorem LII.4 (Non-perturbative edge selector)**.** *With Gates G1–G3 in place, the physical selector is*

$$\boxed{\Gamma_\partial(q) = -\log \mathrm{Tr}_{\mathcal{C}_q}\, e^{-\beta_q \widehat{H}_{\partial,q}} = q[\ln(q/q_\star)-1] - \log Z_{\rm edge}(q).}$$

It has terminal scale

$$\boxed{q_\star = \Xi_{\rm edge}\exp\left[\frac{16\pi^2}{b\, g^2_\partial(\ell_P^{-1})}\right] = 3.307251460713979\times 10^{122}.}$$

The physical integer is the nearest admissible minimizer,

$$\boxed{q_{\rm phys} = \operatorname*{arg\,min}_{q\in\mathbb{Z}_{>0}\cap \mathrm{Adm}} \Gamma_\partial(q),}$$

and the cosmological constant is

$$\boxed{\Lambda_\star = \frac{3\pi}{\ell_P^2 q_{\rm phys}}.}$$

Proof. The admissibility Hamiltonian is positive and vanishes exactly on the common kernel. The running term is supported only on the $N_\varphi = 292$ registry fixed by Gate G1. The beta product and scheme factor in Gate G2 determine the exponential terminal scale. Gate G3 supplies the single-count oscillator contribution as $-\log Z_{\rm edge}(q)$. Thus all pieces of $\Gamma_\partial(q)$ are fixed by the microscopic edge theory, and no observational value of Λ, q, or R_Λ is inserted. □

In logarithmic coordinate $u = \ln q$,

$$\boxed{\Gamma_\partial(u) = e^u(u-u_\star-1) - \log Z_{\rm edge}(e^u),}$$

so

$$\boxed{\Gamma''_\partial(u) = e^u(u-u_\star+1) + \partial_u^2[-\log Z_{\rm edge}(e^u)].}$$

At $u \simeq 282$, the first term dominates the bounded Harish–Chandra curvature. Hence

$$\boxed{\Gamma''_\partial(u) > 0}$$

in the terminal sector and the discrete nearest-integer rule is stable:

$$\boxed{q_{\min} = q_\star - \frac{1}{2} + O(q_\star^{-1}),}$$

followed by projection to the nearest admissible integer.

The terminal ledger is

$$\boxed{u_\star = 282.111498815\ldots, \qquad b\, g^2_\partial(\ell_P^{-1}) = 0.559754586\ldots, \qquad N_\varphi =}$$

E. Executable gate evaluators

The three gates are written so that each has a finite-cutoff evaluator whose continuum limit is the corresponding theorem. The evaluators are not fitters: the only permitted inputs are the microscopic edge data.

For Gate G1, use the orthonormal monopole-character basis

$$\{\chi^{(q)}_{\ell,m}\} \subset \mathcal{H}_{\rm char}.$$

After projection to fixed charge and angular momentum, the recursion matrix is diagonal in the reduced labels:

$$\boxed{\left\langle \chi^{(q)}_{\ell',m'}\middle|\mathcal{R}\middle|\chi^{(q)}_{\ell,m}\right\rangle = \delta_{\ell'\ell}\delta_{m'm}\,\varphi^{N(\ell)}\mu^{\rm ren}_{\rm Hopf}(\ell,q)\, e^{-\Theta^{\rm ren}_{\rm APS}(\ell)}\,\Theta_{\rm Br}(\ell,q).}$$

Here

$$\boxed{\mu^{\rm ren}_{\rm Hopf}(\ell,q) = \mathrm{FP}_{s=0}\left[\big(\ell(\ell+2q)\big)^{1/2-s}(2\ell+q) - C_{\rm HK}(\ell,q;s)\right],}$$

where $C_{\rm HK}$ is the heat-kernel counterterm that makes the zeta trace finite. The renormalized logarithmic eigenvalue is

$$\boxed{F_N = \ln|\lambda_N| = N\ln\varphi + \ln\mu^{\rm ren}_{\rm Hopf}(N) - \Theta^{\rm ren}_{\rm gap}(N).}$$

The finite-rank evaluator computes F_N for $1\le N\le N_{\max}$ and selects the unique integer zero of F_N compatible with the Bradlow projector. For $N_{\max}=400$ the unique admissible root is

$$\boxed{N_\varphi = 292.}$$

This is the normalized spectral statement; the raw unrenormalized product $\varphi^N\sqrt{N(N+2)}e^{-\Theta_{\rm gap}}$ is not the eigenvalue of the compact operator.

For Gate G2 the machine ledger is

$$\boxed{b = 292 + \delta_{\rm stringy}, \qquad \delta_{\rm stringy} = \mathrm{Res}_{\rm KL}[E_2(\tau_{\rm edge}), \eta(\tau_{\rm edge})]\,,}$$

with the APS/zeta convention fixed once. The numerical ledger is

$$\boxed{b\,g_\partial^2(\ell_P^{-1}) = 0.5597545859987625, \qquad \Xi_{\rm edge} = 0.9991692800000000, \qquad R_{\rm cap}(a) \equiv 0, \qquad R_{\rm int}(a), \qquad \Omega_{2,\rm val}, \qquad c_T = c, \qquad \nu_- = 0.}$$

Thus

$$\boxed{u_{\rm run} = \frac{16\pi^2}{b\,g_\partial^2(\ell_P^{-1})} = 282.11232988054314,}$$

and the physical logarithm includes the finite edge factor,

$$\boxed{\ln q_\star = u_{\rm run} + \ln \Xi_{\rm edge} = 282.111498815304\ldots.}$$

Consequently

$$\boxed{q_\star = \Xi_{\rm edge} e^{u_{\rm run}} = 3.307251460713979 \times 10^{122}.}$$

For Gate G3 the zeta/Harish–Chandra evaluator is the finite-cutoff object

$$\boxed{\zeta_{D^2}^{(L)}(q;s) = 2R_q^{2s} \sum_{\ell=1}^{L} (2\ell+q)[\ell(\ell+q)]^{-s} - C_{\rm HK}^{(L)}(q;s).}$$

The determinant estimator is

$$\boxed{-\log Z_{\rm edge}^{(L)}(q) = \frac{1}{2}\,\frac{d}{ds}\bigg|_{s=0} \zeta_{D^2}^{(L)}(q;s) + S_{\rm HC}^{(L)}(q) + \Theta_{\rm KL}^{(L)} + \Theta_{\rm vol}^{(L)}.}$$

The limit $L \to \infty$ gives

$$\boxed{-\log Z_{\rm edge}(q) = \frac{1}{2}\ln q + \Theta_\partial(q) + S_{\rm HC}(q) + o(1),}$$

with $\eta_q(0) = 0$ in the symmetric monopole sector. In the logarithmic coordinate $u = \ln q$, the convexity test is

$$\boxed{\partial_u^2 \Gamma_\partial(u) = e^u(u - u_\star + 1) + \partial_u^2[-\log Z_{\rm edge}(e^u)] > 0}$$

through the terminal region. The first term dominates any bounded Harish–Chandra curvature at $u \simeq 282$.

F. Gate-to-lab-to-sky falsification engine

The theory specifies a falsification engine rather than a posterior accommodation. The cosmological likelihood target is

$$\boxed{\mathcal{L}_{\rm joint} = \mathcal{L}_{\rm DESI\,DR2\,BAO}\mathcal{L}_{\rm Planck/ACT}\mathcal{L}_{\rm Pantheon+/DES}\mathcal{L}_{\rm growth}\mathcal{L}_{\rm lensing},}$$

implemented with admissibility priors only,

$$\boxed{\pi(\theta) \propto \mathbf{1}_{\rm Adm}(\theta),}$$

and sampled by nested or MCMC methods, for example in a Cobaya-class workflow [106]. The sky nulls are

The principal interacting valuation branch for tests is

$$\boxed{(s_-, s_+) = (-1, +1),}$$

with protected current J_+. The pre-registered forecast target is the DESI-favoured quadrant

$$\boxed{w_0 \simeq -0.87, \qquad w_a \simeq -0.55,}$$

with evidence and $\Delta\chi^2$ to be computed only from public covariance-level likelihoods. A wrong sign in $R_{\rm int}$ or J_+, a violation of $R_{\rm cap} \equiv 0$, or a posterior preference for an inadmissible branch falsifies the edge selector.

The laboratory branch tests Gate G1 and Gate G2. Programmable superconducting quantum processors have already simulated Fibonacci string-net states and demonstrated creation, fusion, and braiding of Fibonacci anyons [104, 105]. The proposed protocol prepares a Fibonacci/string-net registry, measures the growth of the fusion space, and looks for the terminal cutoff induced by the Bradlow–APS gap:

$$\boxed{\dim \mathcal{H}_N \sim \varphi^N, \qquad N_{\rm lab} = 292.}$$

The same run should extract the finite Kronecker-limit factor

$$\boxed{\Xi_{\rm edge}^{\rm lab} = 0.99916928 \pm 10^{-5},}$$

within the convention-dependent systematic error. A terminal rank different from 292, measured in the same operator convention, kills Gate G1.

The analog-gravity branch tests Gate G3. A spherical photonic or atomic trap with monopole spectrum

$$\boxed{\lambda_{\ell,\pm} = \pm R^{-1}\sqrt{\ell(\ell+q)}}$$

can measure the eta invariant and heat-kernel coefficient directly. The prediction is

$$\boxed{\eta_q(0) = 0, \qquad \nu = \frac{1}{2}}$$

for the symmetric sector. A robust deviation in the controlled analog system kills the determinant-line gate.

The holographic branch is the same object in another language. The static patch of dS_4 is paired with an edge theory on S^2, and the edge two-point kernel is organized by Harish–Chandra characters:

$$\boxed{\langle \mathcal{O}_s(x)\mathcal{O}_s(y)\rangle_{\rm edge} = \int \frac{dt}{2\pi}\, e^{it\Delta}\, \chi_{\rm HC}^{(s)}(t)\, G_{\rm Bradlow}(x,y;q).}$$

The entropy prediction is

$$\boxed{S_{\rm edge} = q - \frac{1}{2}\ln q + \Theta_\partial + O(q^{-1}).}$$

Future Euclid, Rubin/LSST, and CMB-S4 data can then test the two-loop displacement

$$\boxed{\Delta \ln q_\star = O(10^{-3})}$$

rather than refit the leading amplitude [108–110]. The construction is therefore deliberately brittle: any one of the sky, lab, analog, or holographic tests can destroy the same three-gate chain.

G. Commutative dependency diagram

The dependency structure is the following. Each arrow is a definition, theorem, gate, or empirical test; no observational value enters before the last row.

$$\begin{array}{ccccc}
P_{\rm tf} & \longrightarrow & \text{Bianchi integration mode} & \longrightarrow & \operatorname{Cap} H_\partial \\
\downarrow & & & & \downarrow \\
\mathcal{A}_{\rm grav} & \longrightarrow & \Pi_{\rm phys} & \longrightarrow & \widehat{H}_{\partial,q} = \widehat{H}_{\rm adm} + \widehat{H}^{N_\varphi}_{\rm run} + \widehat{H}_{\rm edge} \\
\downarrow & & & & \downarrow \\
Q_{\rm DM} & \longrightarrow & \text{superselection} & \longrightarrow & \Gamma_\partial(q) = q[\ln(q/q_\star) - 1] - \log Z_{\rm edge}(q) \\
\downarrow & & & & \downarrow \\
P_{\rm val}(D) & \longrightarrow & \text{valuation alphabet} & \longrightarrow & q_\star \longrightarrow \Lambda_\star \\
\downarrow & & & & \downarrow \\
S_{\rm obs} & \longrightarrow & \text{null tests/free spectrum} & \longrightarrow & \text{falsification.}
\end{array}$$

This diagram is the audit instrument for the manuscript: definitions, gates, theorems, and empirical tests occupy different vertices and cannot be interchanged.

The terminal statement is

$$\boxed{\text{cosmological constant} = \text{edge Hilbert-space eigenvalue.}}$$

Equivalently,

$$\boxed{\Lambda_\star = \frac{3\pi}{\ell_P^2 q_\star}, \qquad q_\star = \arg\min \Gamma_\partial.}$$

The hierarchy 10^{122} is therefore not a parameter of the local action. It is the output of the non-perturbative de Sitter horizon edge partition function.

LIII. CONCLUSION

The invariant sieve has reached its submission form. The trace-free field equation

$$R_{\mu\nu} - \frac{1}{4}Rg_{\mu\nu} = 8\pi G\left(T_{\mu\nu} - \frac{1}{4}Tg_{\mu\nu}\right)$$

annihilates constant vacuum-energy shifts. The Bianchi identity returns an integration mode. The capacity postulate fixes the global form of that mode,

$$\Lambda_{\rm cap} = \frac{3\pi}{\ell_P^2 \operatorname{Cap}(\mathcal{H}_\partial; \mathcal{A}_{\rm grav}, Q_{\rm DM}, P_{\rm val})}.$$

Thus Λ is neither a bare local coupling nor a fitted fluid. It is the eigenvalue of an admissible horizon edge Hilbert-space sector.

The final microscopic object is

$$\boxed{\widehat{H}_{\partial,q} = \widehat{H}^{\text{common kernel}}_{\text{adm},q} + \widehat{H}^{N_\varphi}_{\text{run},q} + \widehat{H}^{\rm HC}_{\text{edge},q}.}$$

The three edge gates determine the registry rank, the beta product and scheme factor, and the determinant-line dependence:

$$\boxed{N_\varphi = 292, \qquad b\, g_\partial^2(\ell_P^{-1}) = 0.559754586\ldots, \qquad -\log Z_{\rm edge}(q) = \frac{1}{2}}$$

Consequently

$$\boxed{\Gamma_\partial(q) = q[\ln(q/q_\star) - 1] - \log Z_{\rm edge}(q),}$$

with

$$\boxed{q_\star = \Xi_{\rm edge} \exp\left[\frac{16\pi^2}{b\, g_\partial^2(\ell_P^{-1})}\right] = 3.307251460713979 \times 10^{122}.}$$

The physical integer is

$$\boxed{q_{\mathrm{phys}} = \operatorname*{arg\,min}_{q\in\mathbb{Z}_{>0}\cap\mathrm{Adm}} \Gamma_\partial(q),}$$

and the selected amplitude is

$$\boxed{\Lambda_\star = \frac{3\pi}{\ell_P^2 q_{\mathrm{phys}}}.}$$

The role of observation is not to fit this amplitude back into the theory, but to falsify the same edge-character branch by BAO, CMB, supernovae, growth, lensing, curvature split, free-spectrum tests, and the $c_T = c$ condition. The preferred branch to test is the interacting valuation sector $(s_-, s_+) = (-1, +1)$, with admissibility priors only. The same three gates also define laboratory and analog tests: Fibonacci-registry processors test $N_\varphi = 292$ and Ξ_{edge}, spherical monopole traps test $\eta_q(0) = 0$ and $\nu = 1/2$, and Harish–Chandra edge correlators test the holographic determinant line.

This is the compression of the cosmological constant problem achieved here:

vacuum shift → trace-free kernel → integration mode → capacity sector → edge-character minimizer.

The hierarchy is not placed in the local action. It is the output of the physical de Sitter horizon edge partition function.

DATA AVAILABILITY

The analytic theorem layer uses no proprietary data. The illustrative cosmological checks use public survey products cited in the text, together with the approximations stated in the corresponding sections. The figures are included with the submission source. The falsification engine specified in Section LII consists of the finite-cutoff recursion evaluator, the matching ledger, the zeta/Harish–Chandra determinant evaluator, the joint likelihood, admissibility priors, null tests, free-spectrum reconstruction, curvature split, sampler settings, posterior model-order diagnostics, and evidence calculation. The authors will release the reference implementation, covariance-level likelihood interface, chains, and numerical ledgers in a public repository and will provide working files to editors and referees upon reasonable request.

[1] D. Lovelock, "The Einstein tensor and its generalizations," *J. Math. Phys.* **12**, 498–501 (1971), doi:10.1063/1.1665613.

[2] E. Curiel, "A Simple Proof of the Uniqueness of the Einstein Field Equation in All Dimensions," arXiv:1601.03032.

[3] T. Padmanabhan and D. Kothawala, "Lanczos–Lovelock models of gravity," *Phys. Rept.* **531**, 115–171 (2013), doi:10.1016/j.physrep.2013.05.007.

[4] J. L. Anderson and D. Finkelstein, "Cosmological constant and fundamental length," *Am. J. Phys.* **39**, 901–904 (1971), doi:10.1119/1.1986321.

[5] W. G. Unruh, "A Unimodular Theory of Canonical Quantum Gravity," *Phys. Rev. D* **40**, 1048–1052 (1989), doi:10.1103/PhysRevD.40.1048.

[6] M. Henneaux and C. Teitelboim, "The cosmological constant and general covariance," *Phys. Lett. B* **222**, 195–199 (1989), doi:10.1016/0370-2693(89)91251-3.

[7] S. Weinberg, "The cosmological constant problem," *Rev. Mod. Phys.* **61**, 1–23 (1989), doi:10.1103/RevModPhys.61.1.

[8] N. Kaloper and A. Padilla, "Sequestering the standard model vacuum energy," *Phys. Rev. Lett.* **112**, 091304 (2014), doi:10.1103/PhysRevLett.112.091304.

[9] N. Aghanim *et al.* (Planck Collaboration), "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641**, A6 (2020), doi:10.1051/0004-6361/201833910.

[10] J. D. Bekenstein, "Black holes and entropy," *Phys. Rev. D* **7**, 2333–2346 (1973), doi:10.1103/PhysRevD.7.2333.

[11] S. W. Hawking, "Particle creation by black holes," *Commun. Math. Phys.* **43**, 199–220 (1975), doi:10.1007/BF02345020.

[12] G. W. Gibbons and S. W. Hawking, "Cosmological event horizons, thermodynamics, and particle creation," *Phys. Rev. D* **15**, 2738–2751 (1977), doi:10.1103/PhysRevD.15.2738.

[13] R. Bousso, "The holographic principle," *Rev. Mod. Phys.* **74**, 825–874 (2002), doi:10.1103/RevModPhys.74.825.

[14] V. L. Ginzburg and L. D. Landau, "On the theory of superconductivity," *Zh. Eksp. Teor. Fiz.* **20**, 1064–1082 (1950).

[15] A. A. Abrikosov, "Nobel Lecture: Type-II superconductors and the vortex lattice," *Reviews of Modern Physics* **76**, 975–979 (2004), doi:10.1103/RevModPhys.76.975.

[16] M. Tinkham, *Introduction to Superconductivity*, 2nd ed., Dover Publications, Mineola, NY (2004).

[17] M. F. Atiyah and I. M. Singer, "The index of elliptic operators on compact manifolds," *Bull. Amer. Math. Soc.* **69**, 422–433 (1963), doi:10.1090/S0002-9904-1963-10957-X.

[18] A. A. Abrikosov Jr., "Dirac operator on the Riemann sphere," arXiv:hep-th/0212134.

[19] K. Hasebe, "Hopf maps, lowest Landau level, and fuzzy spheres," *SIGMA* **6**, 071 (2010), doi:10.3842/SIGMA.2010.071.

[20] V. P. Nair, "Landau-Hall states and Berezin–Toeplitz quantization of matrix algebras," *Phys. Rev. D* **102**, 025015 (2020), doi:10.1103/PhysRevD.102.025015.

[21] W. D. Kirwin, "Coherent states in geometric quantization," arXiv:math/0502026.

[22] P. A. M. Dirac, "The theory of gravitation in Hamiltonian form," *Proc. Roy. Soc. Lond. A* **246**, 333–343

(1958), doi:10.1098/rspa.1958.0142.
[23] R. Arnowitt, S. Deser, and C. W. Misner, "The dynamics of general relativity," in *Gravitation: An Introduction to Current Research*, ed. L. Witten (Wiley, New York, 1962), pp. 227–265, arXiv:gr-qc/0405109.
[24] M. Henneaux and C. Teitelboim, *Quantization of Gauge Systems*, Princeton University Press, Princeton (1992).
[25] H. Hadwiger, *Vorlesungen über Inhalt, Oberfläche und Isoperimetrie*, Springer, Berlin (1957).
[26] R. Schneider, *Convex Bodies: The Brunn–Minkowski Theory*, 2nd expanded ed., Cambridge University Press, Cambridge (2014).
[27] B. P. Abbott *et al.* (LIGO Scientific Collaboration and Virgo Collaboration), "GW170817: Observation of Gravitational Waves from a Binary Neutron Star Inspiral," *Phys. Rev. Lett.* **119**, 161101 (2017), doi:10.1103/PhysRevLett.119.161101.
[28] B. P. Abbott *et al.*, "Gravitational Waves and Gamma-rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A," *Astrophys. J. Lett.* **848**, L13 (2017), doi:10.3847/2041-8213/aa920c.
[29] D. Clowe, M. Bradac, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones, and D. Zaritsky, "A direct empirical proof of the existence of dark matter," *Astrophys. J. Lett.* **648**, L109–L113 (2006), doi:10.1086/508162.
[30] M. Markevitch, A. H. Gonzalez, D. Clowe, A. Vikhlinin, W. Forman, C. Jones, S. Murray, and W. Tucker, "Direct constraints on the dark matter self-interaction cross-section from the merging galaxy cluster 1E 0657–56," *Astrophys. J.* **606**, 819–824 (2004), doi:10.1086/383178.
[31] M. Abdul Karim *et al.* (DESI Collaboration), "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," *Phys. Rev. D* **112**, 083515 (2025), doi:10.1103/tr6y-kpc6, arXiv:2503.14738.
[32] K. Lodha *et al.* (DESI Collaboration), "Extended Dark Energy analysis using DESI DR2 BAO measurements," arXiv:2503.14743.
[33] D. Wang and D. Mota, "Did DESI DR2 truly reveal dynamical dark energy?" arXiv:2504.15222.
[34] NIST Reference on Constants, Units, and Uncertainty, "CODATA Value: Planck length," 2022 CODATA recommended values, physics.nist.gov/cgi-bin/cuu/Value?plkl=.
[35] K. V. Kuchař, "Geometry of hyperspace. I," *J. Math. Phys.* **17**, 777 (1976); see also C. Teitelboim, "How commutators of constraints reflect the spacetime structure," *Ann. Phys.* **79**, 542 (1973).
[36] P. A. M. Dirac, *Lectures on Quantum Mechanics*, Belfer Graduate School of Science, Yeshiva University, New York, 1964.
[37] R. Loll, "Quantum Gravity from Causal Dynamical Triangulations: A Review," *Class. Quantum Grav.* **37**, 013002 (2020), doi:10.1088/1361-6382/ab57c7.
[38] F. Saueressig, "The Functional Renormalization Group in Quantum Gravity," *Living Rev. Relativ.* **26**, 2 (2023), doi:10.1007/s41114-023-00042-3.
[39] C.-P. Ma and E. Bertschinger, "Cosmological perturbation theory in the synchronous and conformal Newtonian gauges," *Astrophys. J.* **455**, 7 (1995), doi:10.1086/176550, arXiv:astro-ph/9506072.
[40] CobayaSampler Collaboration, "bao_data: BAO data repository including DESI DR2," GitHub repository, github.com/CobayaSampler/bao_data.
[41] Pantheon+SH0ES Collaboration, "Pantheon-PlusSH0ES/DataRelease," GitHub repository, github.com/PantheonPlusSH0ES/DataRelease.
[42] A. Reeves, "planck_compressed: compressed Planck18 CMB TTTEEE+lensing likelihood," GitHub repository, github.com/alexreevesy/planck_compressed.
[43] R. Bernardo, "datasets: compiled cosmology data sets including growth-rate measurements," GitHub repository, github.com/reggiebernardo/datasets.
[44] C. Rovelli, "Black Hole Entropy from Loop Quantum Gravity," arXiv:gr-qc/9603063.
[45] A. Ashtekar, J. Baez, A. Corichi, and K. Krasnov, "Quantum Geometry and Black Hole Entropy," *Phys. Rev. Lett.* **80**, 904 (1998), doi:10.1103/PhysRevLett.80.904.
[46] T. Jacobson, "Renormalization and black hole entropy in Loop Quantum Gravity," arXiv:0707.4026.
[47] J. Diaz-Polo and D. Pranzetti, "Isolated Horizons and Black Hole Entropy in Loop Quantum Gravity," *SIGMA* **8**, 048 (2012), doi:10.3842/SIGMA.2012.048, arXiv:1112.0291.
[48] J. F. Barbero G. and D. Pranzetti, "Black Hole Entropy in Loop Quantum Gravity," arXiv:2212.13469.
[49] R. D. Sorkin, "Is the cosmological 'constant' a nonlocal quantum residue of discreteness of the causal set type?" arXiv:0710.1675.
[50] N. Zwane, N. Afshordi, and R. D. Sorkin, "Cosmological Tests of Everpresent Λ," *Class. Quantum Grav.* **35**, 194002 (2018), doi:10.1088/1361-6382/aad70a, arXiv:1703.06265.
[51] K. Hasebe, "Deformation of matrix geometry via Landau level evolution," *Phys. Rev. D* **111**, 045008 (2025), doi:10.1103/PhysRevD.111.045008.
[52] M. Li, "A model of holographic dark energy," *Phys. Lett. B* **603**, 1–5 (2004), doi:10.1016/j.physletb.2004.10.014, arXiv:hep-th/0403127.
[53] S. A. Tsilioukas, E. N. Saridakis, and C. Tzerefos, "Dark energy from topology change induced by microscopic Gauss–Bonnet wormholes," *Phys. Rev. D* **109**, 084010 (2024), doi:10.1103/PhysRevD.109.084010, arXiv:2312.07486.
[54] D. B. Ray and I. M. Singer, "R-torsion and the Laplacian on Riemannian manifolds," *Adv. Math.* **7**, 145–210 (1971), doi:10.1016/0001-8708(71)90045-4.
[55] C. Bär, "The Dirac operator on space forms of positive curvature," *J. Math. Soc. Japan* **48**, 69–83 (1996), doi:10.2969/jmsj/04810069.
[56] D. V. Vassilevich, "Heat kernel expansion: user's manual," *Phys. Rept.* **388**, 279–360 (2003), doi:10.1016/j.physrep.2003.09.002, arXiv:hep-th/0306138.
[57] M. B. Einhorn and D. R. T. Jones, "Naturalness and dimensional transmutation in classically scale-invariant gravity," arXiv:1410.8513.
[58] A. Salvio, "Dimensional transmutation in gravity and cosmology," arXiv:2012.11608.
[59] M. Geiller and P. Jai-akson, "Extended actions, dynamics of edge modes, and entanglement entropy," arXiv:1912.06025.
[60] L. Smolin, "Quantum gravity with a positive cosmological constant," arXiv:hep-th/0209079.
[61] N. Kent *et al.*, "Creation and observation of Hopfions in magnetic multilayer systems," *Nature Communications* **12**, 1562 (2021), doi:10.1038/s41467-021-21846-5, arXiv:2010.08674.

[62] F. Zheng *et al.*, "Hopfion rings in a cubic chiral magnet," *Nature* **623**, 718–723 (2023), doi:10.1038/s41586-023-06658-5.

[63] P. J. Ackerman, T. Boyle, and I. I. Smalyukh, "Self-assembly and electrostriction of arrays and chains of hopfion particles in chiral liquid crystals," *Nature Communications* **6**, 6012 (2015), doi:10.1038/ncomms7012.

[64] A. F. Rañada, "A topological theory of the electromagnetic field," *Letters in Mathematical Physics* **18**, 97–106 (1989), doi:10.1007/BF00401864.

[65] H. K. Moffatt, "The degree of knottedness of tangled vortex lines," *Journal of Fluid Mechanics* **35**, 117–129 (1969), doi:10.1017/S0022112069000991.

[66] H. K. Urbantke, "The Hopf fibration—seven times in physics," *Journal of Geometry and Physics* **46**, 125–150 (2003), doi:10.1016/S0393-0440(02)00121-3.

[67] A. F. Vakulenko and L. V. Kapitanskii, "Stability of solitons in S^2 in the nonlinear sigma model," *Soviet Physics Doklady* **24**, 433–434 (1979).

[68] D. Harland, "Topological energy bounds for the Skyrme and Faddeev models with massive pions," *Physics Letters B* **728**, 518–523 (2014), doi:10.1016/j.physletb.2013.11.062, arXiv:1311.2403.

[69] J. Schwinger, "On Gauge Invariance and Vacuum Polarization," *Physical Review* **82**, 664–679 (1951), doi:10.1103/PhysRev.82.664.

[70] G. V. Dunne, "New Strong-Field QED Effects at ELI: Nonperturbative Vacuum Pair Production," *European Physical Journal D* **55**, 327–340 (2009), doi:10.1140/epjd/e2009-00022-0, arXiv:0812.3163.

[71] E. B. Bogomolny, "The stability of classical solutions," *Soviet Journal of Nuclear Physics* **24**, 449–454 (1976).

[72] A. Jaffe and C. Taubes, *Vortices and Monopoles*, Birkhäuser, Boston, 1980.

[73] S. B. Bradlow, "Vortices in holomorphic line bundles over closed Kähler manifolds," *Communications in Mathematical Physics* **135**, 1–17 (1990), doi:10.1007/BF02097654.

[74] C. Adam, J. M. Speight, and A. Wereszczyński, "The volume of a vortex and the Bradlow bound," arXiv:1703.08848.

[75] J. Zhang, "Abrikosov lattice solutions to the Ginzburg–Landau equations," lecture notes, University of Copenhagen, online PDF.

[76] T. T. Wu and C. N. Yang, "Dirac Monopole Without Strings: Monopole Harmonics," *Nucl. Phys. B* **107**, 365–380 (1976), doi:10.1016/0550-3213(76)90143-7.

[77] B. P. Dolan, "Ground state wave functions for the quantum Hall effect on a sphere and the Atiyah–Singer index theorem," arXiv:2001.02208.

[78] M. F. Atiyah, V. K. Patodi, and I. M. Singer, "Spectral asymmetry and Riemannian geometry. I," *Mathematical Proceedings of the Cambridge Philosophical Society* **77**, 43–69 (1975), doi:10.1017/S0305004100049410.

[79] X. Dai and D. S. Freed, "Eta invariants and determinant lines," *Journal of Mathematical Physics* **35**, 5155–5194 (1994), doi:10.1063/1.530747.

[80] E. Witten, "Anomaly Inflow and the η-Invariant," *The Shoucheng Zhang Memorial Workshop*, arXiv:1909.08775, arXiv:1909.08775.

[81] S. Coleman and E. Weinberg, "Radiative Corrections as the Origin of Spontaneous Symmetry Breaking," *Phys. Rev. D* **7**, 1888–1910 (1973), doi:10.1103/PhysRevD.7.1888.

[82] J. D. Bekenstein, "Universal upper bound on the entropy-to-energy ratio for bounded systems," *Phys. Rev. D* **23**, 287–298 (1981), doi:10.1103/PhysRevD.23.287.

[83] L. Susskind, "The World as a hologram," *J. Math. Phys.* **36**, 6377–6396 (1995), arXiv:hep-th/9409089.

[84] S. C. van Lingen, *Relating Division Algebras to Hopf Fibrations using K-theory*, MSc thesis, Radboud University (2020), online PDF.

[85] H. Gluck, F. Warner, and W. Ziller, "The geometry of the Hopf fibrations," *Enseign. Math.* **32**, 173–198 (1986), online PDF.

[86] B. Heidenreich, J. McNamara, M. Montero, M. Reece, T. Rudelius, and I. Valenzuela, "Chern–Weil global symmetries and how quantum gravity avoids them," *JHEP* **11**, 053 (2021), doi:10.1007/JHEP11(2021)053.

[87] J. McNamara and C. Vafa, "Cobordism Classes and the Swampland," arXiv:1909.10355.

[88] E. Zeckendorf, "Représentation des nombres naturels par une somme de nombres de Fibonacci ou de nombres de Lucas," *Bulletin de la Société Royale des Sciences de Liège* **41**, 179–182 (1972).

[89] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., revised by D. R. Heath-Brown and J. H. Silverman, Oxford University Press, Oxford, 2008.

[90] M. Senechal, *Quasicrystals and Geometry*, Cambridge University Press, Cambridge, 1995.

[91] A. Jagannathan, "The Fibonacci quasicrystal: case study of hidden dimensions and multifractality," *Reviews of Modern Physics* **93**, 045001 (2021), doi:10.1103/RevModPhys.93.045001.

[92] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, "Non-Abelian anyons and topological quantum computation," *Reviews of Modern Physics* **80**, 1083–1159 (2008), doi:10.1103/RevModPhys.80.1083, arXiv:0707.1889.

[93] R. O. Bauer, "Analysis of the horizontal Laplacian for the Hopf fibration," *Forum Mathematicum* **17**, 903–920 (2005), doi:10.1515/form.2005.17.6.903, arXiv:math/0209127.

[94] R. J. Miatello and R. A. Podestá, "The spectrum of twisted Dirac operators on compact flat manifolds," arXiv:math/0312004.

[95] P. B. Gilkey, R. J. Miatello, and R. A. Podestá, "The eta invariant and equivariant bordism of flat manifolds with cyclic holonomy group of odd prime order," *Annals of Global Analysis and Geometry* **37**, 275–306 (2010), doi:10.1007/s10455-009-9185-5, arXiv:0910.0635.

[96] W. Donnelly and A. C. Wall, "Geometric entropy and edge modes of the electromagnetic field," *Phys. Rev. D* **94**, 104053 (2016), doi:10.1103/PhysRevD.94.104053, arXiv:1506.05792.

[97] J. R. David and J. Mukherjee, "Partition functions of p-forms from Harish–Chandra characters," *JHEP* **09**, 094 (2021), doi:10.1007/JHEP09(2021)094, arXiv:2105.03662.

[98] M. Grewal, Y. T. A. Law, and K. Parmentier, "Black hole horizon edge partition functions," *JHEP* **06**, 025 (2023), doi:10.1007/JHEP06(2023)025, arXiv:2211.16644.

[99] A. Ball, Y. T. A. Law, and G. Wong, "Dynamical edge modes and entanglement in Maxwell theory," arXiv:2403.14542.

[100] A. Ball and Y. T. A. Law, "Dynamical edge modes in p-form gauge theories," *JHEP* **02**, 182 (2025),

doi:10.1007/JHEP02(2025)182, arXiv:2411.02555.

[101] Y. T. A. Law, "De Sitter horizon edge partition functions," *Phys. Rev. D* (2026), arXiv:2501.17912.

[102] A. Dabholkar, E. Harris, and U. Moitra, "Edge modes on stringy horizons," *Phys. Rev. Lett.* (2026), arXiv:2601.13131.

[103] Y. T. A. Law and V. Lochab, "Horizon edge partition functions in $\Lambda > 0$ quantum gravity," arXiv:2603.20913.

[104] S. Xu *et al.*, "Non-Abelian braiding of Fibonacci anyons with a superconducting processor," arXiv:2404.00091.

[105] Z. K. Minev *et al.*, "Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials," *Nature Communications* **16**, 6225 (2025), arXiv:2406.12820.

[106] J. Torrado and A. Lewis, "Cobaya: Code for Bayesian Analysis of hierarchical physical models," *JCAP* **05**, 057 (2021), doi:10.1088/1475-7516/2021/05/057, arXiv:2005.05290.

[107] D. Brout *et al.*, "The Pantheon+ Analysis: Cosmological Constraints," *Astrophys. J.* **938**, 110 (2022), doi:10.3847/1538-4357/ac8e04, arXiv:2202.04077.

[108] Euclid Consortium, "Euclid data release: Q1," project release note (2026), online.

[109] Vera C. Rubin Observatory, "Legacy Survey of Space and Time," project description (2026), online.

[110] CMB-S4 Collaboration, "Revised CMB-S4 Project Plan," project report (2025), CMB-S4.